

20260129 Comments from Coauthors

So far I received responses from the following people:

1. Alessandro Boselli: moderate comments via pdf (attached in the end of this file).
2. Daniel Dale: minor comments via overleaf.
3. Sara Ellison: suggestion to add a reference.
4. Andrei Ristea: many comments via overleaf.
5. Vicente Villanueva: moderate comments via pdf (attached in the end of this file).
6. Christine Wilson: some questions via emails.
7. Andrew Battisti: moderate comments via overleaf.
8. Toby Brown: minor comments via overleaf.
9. Kristine Spekkens: moderate comments via overleaf.

Below is the email received from some of them.

Alessandro Boselli

Ciao Rongjun

I read with attention the draft, and I send you here a few comments that you might want to consider while preparing the revised version. I marked most of them on the pdf. Please feel free to consider them only if you think that they are useful.

As a general comment, the paper is interesting but a bit complex for a first reading for non experts. My only major comment is related to the time variations that you consider in your model since I have the impression that you do not consider the fact that these objects are Virgo cluster members, where gas infall is probably suppressed (although this is on global scales rather than local scales) and on the other hand stripping is occurring on timescales of ~ 100 Myr.

The perturbations induced by the interaction of galaxies with their surrounding environment might remove the gas, or mix it locally, changing gas density etc on relatively short timescales. N4383, which is an outlier here, confirms that environmental driven processes might be important. It would thus be interesting to consider this aspect in the analysis and discussion. A way of doing that would be to see whether there are systematic differences with molecular gas density and distribution, which might be altered during the interaction, or using a comparative analysis between the most and less perturbed objects of your sample.

In any case congratulation for the nice work

Ciao

Daniel Dale

Nice work, Rongjun! The analysis is very sound and you are a good writer, too. I added some comments on overleaf for your consideration. I'm not sure that alone merits co-authorship, but it was a pleasure to read the manuscript nonetheless!

Best,

Danny

Sara Ellison

Dear Rongjun,

I will endeavour to provide some useful comments before your deadline, but just in case I don't, may I humbly please make one request? In the introduction, when you introduce the FMR, the first paper to show this relation was actually Ellison et al 2008. Perhaps you could include this reference?

Thanks, and happy holidays,

Sara

Andrei Ristea

Hi Rongjun,

Congratulations on getting the paper draft ready, it seems to be a very robust body of work, and the draft is in very good shape. I have added lots of comments on the overleaf, a good number of them (I think) rather minor. My main comment is related to the exclusion of NGC 4383 from the sample; see my comment just ahead of Section 6 on Overleaf.

Congratulations again on putting together the paper, it's a very interesting body of work!

All the best,

Andrei

Vicente Villanueva

Hi Rongjun,

Hoping that you are well, I am contacting you for two reasons:

1. I checked out your manuscript "MAUVE–MUSE: When Metallicity Follows or Fights Star Formation—A Mass-Dependent Inversion in Virgo Galaxies".

First of all, congratulations for the draft! Is very well written and I almost didn't find any typos or grammar corrections on it.

In terms of my feedback, my main comments are related to include a more detailed comparison between the last results observed in field

galaxies and those included in your paper. I think they could be very helpful in terms of learning how environmental effects increase

the local variations of the physical conditions (compared with in principle more steady local conditions in field galaxies), particularly at

100 pc scales. Maybe this could be the main motivation for a follow-up study, but including a little bit more content in this regard

wouldn't hurt. I think a good reference to start with is Sanchez et al. (2020; <https://ui.adsabs.harvard.edu/abs/2020ARA%26A..58..99S/abstract>).

I am attaching a PDF file as well with some comments I found pertinent so you can use them as you wish. My affiliation and acknowledgements are

included at the end of this email.

2. In regards to your last email, I think it's better if you could send me the stellar and SFR maps directly. Let me know how I can help in this.

Are they heavy in terms of memory? How much memory do they weigh?

Do not hesitate to contact me in case you need further information.

Best regards,

Vicente V.

Christine Wilson

Hi, Rongjun,

I have a couple of high-level comments on your rMZR paper that I think are best sent by email (see below). Feel free to share my comments with anyone relevant e.g. your supervisor.

Either there are significant problems with your treatment of the data, or I don't understand the method enough to comment properly. Either way, please don't add me as a co-author at this time.

I have mostly looked at the figures and captions, and sections 2.1-2.3 on the data. It is confusing that the caption for Figure 3 seems to be wrong (someone else noted this already in the overleaf document) and that error bars are not shown in the lower panels.

For the Spearman rank correlation coefficients (e.g. Figure 5), is there a way to quantify correlations as e.g. strong/moderate/weak? For example, are correlation coefficients between -0.2 and 0.2 simply indicating uncorrelated data?

Cheers,

Chris

Comments

1. SFR measured on 100 pc resolution are going to have a huge amount of stochasticity. The conversion from $H\alpha$ to SFR depends on the age of the region and I believe can change by a factor of 10 or more. This isn't mentioned anywhere, and I wonder how this kind of scatter would affect your conclusions.

2. I was able to understand your description of the Voronoi binning and its application to stellar mass in sections 2.1 and 2.2. However, I got confused in section 2.3 describing the star formation rate surface density:

- it seems that you are working on a spaxel basis to determine the Balmer decrement and any extinction correction, as well as to classify the spaxels as HII (or not)

- you say that only 26% of valid spaxels meet your conservative HII selection.

- you don't say it explicitly in section 2.3, but I assume SFR and $\Sigma(\text{SFR})$ are calculated for the Voronoi grid?

- key point: when you calculate $\Sigma(\text{SFR})$, how do you account for the 74% of spaxels that are not classified as HII? for example, if you have a Voronoi grid with 40 spaxels, and 10 of them are valid, is the $\Sigma(\text{SFR})$ calculated by dividing by the area of 10 or 40 spaxels? this is important, because if you had another Voronoi grid of 40 spaxels that had (luckily) 20 valid spaxels, you would get quite different $\Sigma(\text{SFR})$

Hi Chris,

Thanks for your comments. While Luca is still on leave, I am happy to try to answer your questions.

1. For Figure 3, the error bars here are the standard error of the mean (i.e., $SEM = \text{std}/\sqrt{N}$) instead of scatter (i.e., std in Figure 2), and most of the SEM is less than 0.01 so they are not that visible in figure. And the caption is now fixed.
2. For Spearman rank in Figure 5, we use $|\rho| < 0.2$ to indicate a very weak association and we treat this range as ambiguous in the sense, so it does not provide strong evidence for either a correlated or anti-correlated regime.
3. For this extra SFR scatter caused by 100pc $H\alpha$ measurement, I think it is also captured in my derived model. That says, In equation 31, such stochasticity acts as additional contribution in $\Delta \log(\Sigma_{\text{SFR}})$ (i.e., the SFR-driven variability), so that it will (1) weaken the actual Z_{gas} -SFR relation if they are correlated; or (2) enhance if they are anti-correlated. Additionally, this may also affect the sign-change position in Figure 5: probably bias to higher stellar mass. I can add this clarification in the end of Section 4.
4. Actually, I think it is more like bin-basis analysis. For the spaxels in the same Voronoi bin, they share the exact same properties and so they will be classified as `HII` or `nonHII` consistently.

Please let me know if you have any questions.

Best,

Rongjun

Hi, Rongjun,

Thanks for your response

On 2026-01-13 12:53 a.m., Rongjun Huang wrote:

1. *For this extra SFR scatter caused by 100pc H α measurement, I think it is also captured in my derived model. That says, In equation 31, such stochasticity acts as additional contribution in $\Delta \log(\Sigma_{\text{SFR}})$ (i.e., the SFR-driven variability), so that it will (1) weaken the actual Z_{gas}-SFR relation if they are correlated; or (2) enhance if they are anti-correlated. Additionally, this may also affect the sign-change position in Figure 5: probably bias to higher stellar mass. I can add this clarification in the end of Section 4.*

OK, so clearly I have to read your model to understand this. I'm not sure I have time to do that anytime soon.

- 1.
2. *Actually, I think it is more like bin-basis analysis. For the spaxels in the same Voronoi bin, they share the exact same properties and so they will be classified as `HII` or `nonHII` consistently.*

So does this mean that everywhere you use the term "spaxel" in section 2.3, you actually mean "Voronoi bin"? If so, that is confusing and the language should be changed. (If not, and you actually apply e.g. the Balmer line detection criteria on each 0.2" spaxel, then you haven't answered my original question).

Cheers,

Chris

Hi Chris,

Sorry for the delayed reply, and thanks again — you're right to flag the ambiguity.

Please take your time to read, and feel free to send any further comments whenever convenient (by Feb 1st would be perfect but let me know if you need more time).

In Section 2.3, the independent measurements are performed on the Voronoi-binned spectra (i.e., per Voronoi bin). Each Voronoi bin comprises one or more native 0.2" spaxels whose spectra are co-added to form a single binned spectrum; measurements are performed per Voronoi bin and then mapped back to the native grid by assigning each constituent spaxel the bin's derived values. As a result, quantities such as Balmer-line detection/SNR cuts, H II vs non-H II classification, Balmer Decrement, extinction correction, and the derived Σ_{SFR} & Z_{gas} are all determined per Voronoi bin, not per the original unbinned 0.2" spaxels.

Where my wording was misleading is that we then reassign each bin's derived properties back onto the native 0.2" grid for map-making: every spaxel in a bin is assigned that bin's values. In the subsequent analyses (e.g., rMZR trends), we treat these "post-binning spaxels" (i.e., the 0.2" pixels carrying the binned-spectrum-derived quantities) as the data points; or equivalently, weighting each Voronoi bin by the number of spaxels it contains (in other word, by area it occupies).

So, when I used "spaxel" in Section 2.3, I meant spaxels after binning (post-binning spaxels), rather than the original unbinned spaxels. I will revise Section 2.3 to make this explicit and consistently distinguish "Voronoi bin" (measurement unit) from "post-binning spaxel" (map grid / analysis sampling).

Best,

Rongjun

MAUVE–MUSE: When Metallicity Follows or Fights Star Formation—A Mass-Dependent Inversion in Virgo Galaxies

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17 December 2025

ABSTRACT

We use 13 Virgo spiral galaxies from the MAUVE–MUSE survey to revisit the resolved mass–metallicity relation (rMZR) and its secondary dependence on star formation rate (SFR) surface density (Σ_{SFR}) at ~ 100 pc resolution. At this resolution, we construct co-spatial maps of stellar mass surface density (Σ_*), star formation rate surface density, gas-phase oxygen abundance and verify our results across several strong-line metallicity diagnostics. The MAUVE–MUSE galaxies follow a standard rMZR, but once binned in Σ_* we find a clear mass-dependent inversion in the $Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ relation: at fixed Σ_* , low- Σ_* regions show an anti-correlation between Z_{gas} and Σ_{SFR} , whereas high- Σ_* regions show a positive correlation, with a zero-crossing at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \simeq 7.5\text{--}8.0$. We find that both correlated and anti-correlated H II regions coexist within the same discs, and that the observed mass dependence emerges only as an ensemble trend. To interpret this behaviour, we develop a spatially-resolved gas–regulator model and show that the observed correlation and anti-correlation between Z_{gas} and Σ_{SFR} arise from the competition between star-formation-driven and gas-supply-driven variability: the sign of the local $Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ relation is set by which of these dominates. This framework can be naturally extrapolated to the integrated scenario, providing a unified explanation for both resolved and global $Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ relation. Finally, we find that the strength of the inversion is diagnostic-dependent and can weaken, or even disappear, for some metallicity prescriptions.

Key words: galaxies: abundances — galaxies: ISM — galaxies: star formation — galaxies: evolution — ISM: abundances

1 Introduction

The global gas-phase mass–metallicity relation (MZR), well-established from previous works (e.g. [Lequeux et al. 1979](#); [Tremonti et al. 2004](#); [Erb et al. 2006](#); [Maiolino et al. 2008](#); [Zahid et al. 2014](#); [Curti et al. 2020](#); [Sanders et al. 2021](#)), demonstrates that the gas-phase oxygen abundance increases with galaxy stellar mass (M_*), such that more massive galaxies host a more metal-enriched interstellar medium (ISM). While this global relation provides an essential benchmark for galaxy chemical evolution, the advent of wide-field integral-field spectroscopy (IFS) has shifted attention toward understanding how metallicity is regulated within galaxies. By resolving discs into individual star-forming spaxels, IFS surveys now enable direct measurement of local scaling relations between gas-phase oxygen abundance and local stellar and star-formation properties.

The Calar Alto Legacy Integral Field Area (CALIFA) survey ([Sánchez et al. 2013, 2017](#); [Cresci et al. 2019](#)), revealed that the global MZR is reflected locally as a tight correlation between metallicity and stellar mass surface density (Σ_*). This resolved mass–metallicity relation (rMZR) was subsequently confirmed and expanded by Mapping Nearby Galaxies at APO (MaNGA [Barrera-Ballesteros et al. 2016](#); [Baker et al. 2023](#)) and the Sydney-AAO Multi-object Integral-field spectrograph (SAMI [Sánchez et al. 2019](#)) surveys, establishing the rMZR as a ubiquitous feature of star-

forming discs on kpc scales in the local Universe. More recently, IFS surveys based on Very Large Telescope/Multi-Unit Spectroscopic Explorer (VLT/MUSE) have extended these galactic resolved scaling relations to intermediate redshift while retaining comparable spatial resolution. In particular, the MUSE–Wide survey demonstrated the existence of a rMZR at $z \sim 0.26$, with a shape similar to that at $z \sim 0$ but a modestly lower normalization, consistent with chemical evolution over cosmic time ([Yao et al. 2022](#)). Likewise, the Middle Ages Galaxy Properties with Integral Field Spectroscopy (MAGPI) survey confirmed a rMZR in galaxies at $z \sim 0.3$ on a similar (kpc-scale) spatial resolution ([Koller et al. 2024](#)). Taken together, these IFS studies indicate that gas-phase oxygen abundance correlates more tightly with local stellar mass surface density than with galactocentric radius alone: at fixed radius, spaxels with higher Σ_* tend to be more metal rich, and a similar $\Sigma_*\text{--}Z_{\text{gas}}$ relation is recovered in galaxies with very different sizes and radial profiles. In this sense, the rMZR appears to reflect genuinely local physics, even though it is closely linked to the underlying radial structure of galactic discs.

On global, galaxy-integrated scales, there is now broad consensus that gas-phase metallicity depends on both stellar mass and star formation rate (SFR): the classical MZR is usually interpreted as a projection of a more fundamental $M_*\text{--}Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ surface, often referred to as the Fundamental Metallicity Relation (FMR; [Mannucci et al. 2010](#); [Lara-López et al. 2010](#); [Yates et al. 2012](#); [Bothwell et al. 2013](#); [Peng & Maiolino 2014](#); [Bothwell et al. 2016](#); [Gao et al. 2018](#); [Cresci et al. 2019](#); [Curti et al. 2020](#); [Pallottini et al. 2025](#)). In most

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of these studies, galaxies with higher SFR at fixed M_* tend to be more metal-poor, i.e. Z_{gas} and SFR are anti-correlated, a behaviour often interpreted as the signature of metal-poor gas accretion diluting the ISM. This naturally raises the question of whether an analogous three-parameter relation exists on resolved scales, i.e. whether the rMZR carries a meaningful secondary dependence on local star formation rate surface density (Σ_{SFR}). Several works report that once the primary Σ_* dependence is removed, residual trends with Σ_{SFR} are weak or absent, suggesting that local metallicity primarily traces longer-term mass assembly and recycling (e.g. [Sánchez et al. 2013](#); [Yao et al. 2022](#)). However, recent studies find statistically significant $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (anti-)correlation and motivate a spatially-resolved analogue of the Fundamental Metallicity Relation (i.e., rFMR), wherein metallicity at fixed Σ_* depends on Σ_{SFR} ([Baker et al. 2023](#); [Koller et al. 2024](#)). Additionally, MAGPI data present a clear signal that the *sign* of this correlation can reverse with Σ_* : low- Σ_* regions show an anti-correlation between Z_{gas} and Σ_{SFR} , while high- Σ_* regions exhibit a flat or positive relation ([Koller et al. 2024](#)). Such global and resolved studies therefore suggest that any FMR- or rFMR-like behaviour might not be universal but instead depend on local physical regime, calibration choice, on whether variability in metallicity is dominated by star-formation-driven enrichment or by -driven dilution.

Interpreting such mass-dependent sign changes requires a framework that links metallicity variations to the gas–star formation cycle. A widely used description is provided by “gas–regulator” or “bathtub” models, in which the gas content and metal budget of a galaxy (or region) are governed by the balance between gas inflow, star formation, and outflows (e.g. [Davé et al. 2012](#); [Lilly et al. 2013](#); [Peng & Maiolino 2014](#); [Forbes et al. 2014](#); [Belfiore et al. 2019](#); [Sharda et al. 2021](#)). In their galaxy-integrated form, these models show that the gas-phase metallicity tends toward an effective equilibrium set by the metal yield, the metallicity of the accreted gas, and the interplay between star formation and inflow rate, naturally reproducing both the shape of the global MZR and its commonly observed anti-correlation with SFR. A more explicit connection between time-dependent star formation and metallicity has been established by [Wang & Lilly \(2021, hereafter WL21\)](#), who treated individual “gas–regulator units” (from galaxy-wide down to ~ 100 pc scales) as reservoirs subject to time-variable inflow rate, and star formation efficiency (SFE). Within this framework, the residual correlation between metallicity and SFR at fixed stellar mass (or potential) is set by which source of variability dominates: when fluctuations in SFE drive changes in the SFR at roughly fixed inflow rate, metallicity and SFR are predicted to vary in the same sense, whereas strongly time-variable inflow at nearly constant SFE produces metal dilution and thus an anti-correlation between metallicity and SFR. However, testing such analytical frameworks on local scales requires spatially-resolved data that simultaneously (i) resolve individual H II regions on ~ 100 pc scales and (ii) span a broad dynamic range in Σ_* , so that both SFR-driven and inflow-driven regimes can be probed within and across galaxies.

In this work, we use the  **first 14 Virgo-cluster spirals** observed as part of the Multiphase Astrophysics to Unveil the Virgo Environment (MAUVE) survey ([Catinella et al. 2025](#)) to revisit the rMZR and its secondary dependence on Σ_{SFR} at ~ 100 pc resolution.  **The cluster environment provides an ideal laboratory**: outer discs are susceptible to stripping and environmentally driven gas-supply perturbations, while inner discs probe dense, metal-rich regimes where enrichment may dominate. We show that the MAUVE–MUSE sample exhibits a robust mass-dependent inversion in both $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation and residual coupling ($\Delta Z_{\text{gas}}-\Delta \Sigma_{\text{SFR}}$), consistent with, but extending be-

yond, previous findings. To characterise the spatial heterogeneity of this coupling, we introduce a local bivariate Moran-like correlation index ([Moran 1950](#); [Anselin 1995](#)) that maps where star-formation and metallicity offsets are locally concordant or discordant. Finally, we develop a spatially-resolved regulator model and interpret these patterns in our framework. The results demonstrate that the observed mass-dependent sign flip in the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation is governed by the relative amplitude of local star-formation versus gas-supply variability.

This paper is organized as follows: [Section 2](#) introduces the data and methods used in this study, [Section 3](#) demonstrates our results, [Section 4](#) presents the spatially-resolved framework, its interpretation and the extrapolation to a global integrated scenario, [Section 5](#) discusses the robustness of our findings across different metallicity calibrations, star-forming spaxel selections, and the special case of the metallicity outlier NGC 4383, and [Section 6](#) outlines our conclusions. Throughout this work, we adopt a distance of 16.5 Mpc for Virgo Cluster members ([Mei et al. 2007](#)), aligning with previous and ongoing MAUVE studies.

2 Data & Methods

2.1 MAUVE–MUSE Data

The data analysed in this work are drawn from the internal release v2 (December 2024) of MAUVE–MUSE survey. Designed to investigate how the cluster environment shapes the gas–star formation cycle, MAUVE targets 40 late-type Virgo Cluster galaxies using a suite of multi-wavelength facilities ([Catinella et al. 2025](#)). For each target, we use the fully reduced MUSE datacubes produced with `pymusepipe` v2.28.2 and `esorex` 3.13.6, following closely the reduction, sky subtraction, and flux-calibration procedures described by [Emsellem et al. \(2022\)](#). Each galaxy is delivered as a science-ready, fully-reduced FITS file whose *DATA*, *STAT*, and *DQ* extensions contain the flux, variance, and data-quality information, respectively, on the native $0.2''$ pixel grid over the full 4750–9350 Å spectral range of VLT/MUSE.

All physical quantities used in this paper are derived from these cubes using the `nGIST` pipeline ([Fraser-McKelvie et al. 2025](#)), a `pPXF`-based ([Cappellari & Emsellem 2004](#); [Cappellari 2016, 2023](#)) wrapper optimised for MAUVE–MUSE data. Briefly, we first construct an adaptively binned representation of each cube by applying the Voronoi algorithm of [Cappellari & Emsellem \(2004\)](#) to the stellar continuum in the 4800–7000 Å window, targeting a minimum continuum signal-to-noise ratio of $S/N \geq 40$ per bin. The resulting Voronoi tessellation is then adopted as the common spatial grid for all `nGIST` products. This approach ensures strict one-to-one correspondence between all the following derived properties without any additional resampling or smoothing. The MAUVE–MUSE observations have a typical seeing full width at half maximum (FWHM) of $\sim 1''$, corresponding to a physical scale of ~ 80 pc, so that the effective angular resolution is already set by the point spread function (PSF) rather than by the MUSE sampling pixel scale of $0.2'' \text{ pixel}^{-1}$. The $S/N \geq 40$ Voronoi binning on the stellar continuum typically groups a few up to several tens of spaxels per bin with 95% containing fewer than 40 spaxels, yielding characteristic bin sizes comparable to the seeing footprint and hence also of order ~ 100 pc. Throughout this work, we therefore refer to our spatially-resolved measurements as probing ~ 100 pc scales.

On this common grid, the stellar continuum and kinematics in each bin are modelled with `pPXF` using `MILES` single-stellar-population

(SSP) templates (Vazdekis et al. 2010) convolved to match the MUSE line-spread function (LSF). Multiplicative Legendre polynomials are included to absorb residual calibration mismatches. After subtracting the best-fitting stellar model, Gaussian profiles are fitted simultaneously to the main nebular emission lines (e.g. $\text{H}\alpha$, $\text{H}\beta$, $[\text{O III}]$, $[\text{N II}]$, $[\text{S II}]$), yielding maps of observed line fluxes, velocities, and velocity dispersions with associated uncertainties.

While MAUVE–MUSE observations are still in progress, a total of 14 galaxies have been fully observed, reduced and therefore analyzed in this paper: IC 3392, NGC 4064, NGC 4192, NGC 4293, NGC 4298, NGC 4330, NGC 4383, NGC 4396, NGC 4419, NGC 4457, NGC 4501, NGC 4522, NGC 4694, and NGC 4698.

2.2 Stellar Mass Surface Density

We first derive spatially-resolved stellar mass surface densities from the MAUVE–MUSE datacubes and the associated nGIST value-added products. For each galaxy, we start from the nGIST star-formation-history (SFH) products, which provide, for every Voronoi bin, a vector of SFH weights and an accompanying grid of stellar population parameters sampled in stellar metallicity and age, $([\text{M}/\text{H}], \log_{10}(t))$. We then implement photometric properties for MILES-based SSPs (Vazdekis et al. 2010) on BASTI isochrone grids (Pietrinferni et al. 2004) with a Chabrier (2003, hereafter Chabrier) initial mass function (IMF). Restricting to the Chabrier-IMF entries, we construct a lookup table of R -band mass-to-light ratios $(M/L)_R$ as a function of SSP metallicity and age, $([\text{M}/\text{H}], \log_{10}(t))$, by rounding the age and metallicity of each grid point to a uniform grid and mapping to the corresponding $(M/L)_R$ in the table. The SFH weights for each Voronoi bin are then combined with this $(M/L)_R$ grid to compute a light-weighted R -band mass-to-light ratio,

$$(M/L)_R^{\text{bin}} = \sum_k w_k (M/L)_{R,k}, \quad (1)$$

where w_k and $(M/L)_{R,k}$ are the SFH weight and R -band mass-to-light ratio of the k -th SSP component in the bin. This procedure yields a map of $(M/L)_R^{\text{bin}}$ defined over all Voronoi zones.

To obtain the corresponding R -band luminosity per bin, for each spaxel in the DATA extension, we construct the observed spectral energy distribution F_λ and convolve it with the `bessell-R` filter response using the `specLite` package (Kirkby et al. 2023). This yields an AB R -band magnitude m_R and flux (in mJy) per spaxel. We then collapse these spaxel-level fluxes onto the Voronoi tessellation by averaging, for each BINID, only over spaxels assigned to that bin, while preserving the survey’s native masking of invalid pixels. The resulting bin-averaged AB magnitudes are corrected for Galactic extinction using the nGIST $E(B-V)$ map and an Calzetti et al. (2000, hereafter Calzetti) attenuation curve. Corrected apparent magnitudes (m_R^{corr}) are converted into absolute magnitudes via the distance modulus corresponding to the adopted Virgo-cluster distance of 16.5 Mpc, and hence into R -band luminosities (L_R) in solar units using the solar R -band absolute magnitude, $M_{R,\odot} = 4.61$ (Willmer 2018).

The stellar mass in each Voronoi bin is then obtained as

$$M_*^{\text{bin}} = L_R^{\text{bin}} \times (M/L)_R^{\text{bin}}. \quad (2)$$

Dividing M_*^{bin} by the corresponding projected bin area yields a map of projected stellar mass surface density, Σ_* (in $M_\odot \text{ kpc}^{-2}$).

Finally, we account for the inclination of each disc, adopting inclination angles from the Virgo Environment Traced in CO (VERTICO) Survey (Brown et al. 2021). Assuming an oblate disc geometry with intrinsic axial ratio $q_0 = 0.2$, we compute the inclination

correction factor for each galaxy:

$$\frac{b}{a} = \sqrt{(1 - q_0^2) \cos^2 i + q_0^2}, \quad (3)$$

which encodes the observed flattening of a disc of finite thickness. The stellar mass surface density and the star formation rate surface density in the following Section 2.3 are then multiplied by this b/a factor to correct from the projected to the intrinsic face-on properties.

is this a limit per pixel
or arcsec?

2.3 Star Formation Rate Surface Density

Nebular dust attenuation is quantified from the Balmer decrement measured on the same Voronoi grid as all other MAUVE–MUSE products. For each spaxel we start from the nGIST $\text{H}\alpha$ and $\text{H}\beta$ flux and error maps and impose a basic quality-control cut requiring $F_{\text{line}}/\sigma_{\text{line}} \geq \text{cut}$ with $\text{cut} = 3$, together with a minimum flux threshold of $F_{\text{line}} \geq 20$ in the native MAUVE–MUSE units of $10^{-20} \text{ erg s}^{-1} \text{ cm}^{-2}$. Spaxels that fail this detection criterion for either Balmer line are retained for the construction of SFR maps, but their fluxes are set to $\max(\text{threshold}, F_{\text{line}})$ (effectively assuming zero extinction) to prevent artifacts such as unphysical extinction corrections. These low-S/N spaxels will be excluded from our star-forming spaxel selection later, so this choice has negligible impact on our results. The Balmer decrement is defined as $\text{BD} = F_{\text{H}\alpha}/F_{\text{H}\beta}$ (using observed fluxes) and we impose a floor at the Case B recombination value at $\text{BD} = 2.86$, appropriate for $T_e = 10^4 \text{ K}$ and $n_e = 100 \text{ cm}^{-3}$, i.e. any measured $\text{BD} < 2.86$ is set to 2.86 to avoid unphysical negative colour excesses. The corresponding colour excess is obtained using the analytic Calzetti attenuation curve (Osterbrock & Ferland 2006).

$$E(B-V)_{\text{gas}} = \frac{2.5}{k(\text{H}\beta) - k(\text{H}\alpha)} \log_{10} \left(\frac{\text{BD}}{2.86} \right), \quad \text{This is not a Calzetti law, but just the definition of } E(B-V)$$

where $k(\lambda)$ is starburst reddening curve with the definition of $E(B-V)$ at V -band $R_V = 4.05$ and here k is evaluated at the wavelengths of $\text{H}\alpha$ and $\text{H}\beta$. All strong nebular lines entering the subsequent analysis are then dereddened with this $E(B-V)_{\text{gas}}$ via

$$F_\lambda^{\text{corr}} = F_\lambda 10^{0.4 E(B-V)_{\text{gas}} k(\lambda)}. \quad (5)$$

Using these dust-corrected fluxes we construct the standard $[\text{N II}]$ and $[\text{S II}]$ Baldwin et al. (1981, hereafter BPT) ratios, namely $F_{[\text{N II}]\lambda 6583}/F_{\text{H}\alpha}$, $F_{[\text{S II}](\lambda 6716+\lambda 6731)}/F_{\text{H}\alpha}$, and $F_{[\text{O III}]\lambda 5007}/F_{\text{H}\beta}$, propagating flux-to-error ratios into the corresponding line-ratio uncertainties. Each spaxel is classified on both the $[\text{N II}]$ and $[\text{S II}]$ BPT diagrams according to the Kewley et al. (2006) demarcation curves, and we require that the central value and its $\pm 1\sigma$ error bars in $(\log_{10}[\text{line}/\text{H}\beta], \log_{10}[\text{line}/\text{H}\alpha])$ space all remain within the same BPT region. Only spaxels that satisfy this criterion and the S/N cuts in all lines entering a given diagram are assigned a secure class.

In order to strictly define star-forming regions, we adopt a conservative H II-region selection that is used consistently throughout the paper. Specifically, we define “star-forming” spaxels (hereafter **HII** spaxels) as those that lie in the H II regime of both $[\text{N II}]$ and $[\text{S II}]$ BPT diagrams and whose $\pm 1\sigma$ error bars also remain entirely within the H II regime on both diagrams, with robust detections in all lines entering the classification. This yields a clean sample of classical H II regions purely dominated by O/B stars’ photoionization with minimal contamination from AGN or LINER-like excitation. All SFR and metallicity maps in this work are therefore computed exclusively on this **HII** mask, and all subsequent resolved analyses implicitly refer to these H II-classified star-forming regions. Across the sample, the

Line ratio	Definition
$N2$	$\frac{F_{[\text{N II}](\lambda 6548+\lambda 6584)}}{F_{\text{H}\beta}}$
$S2$	$\frac{F_{[\text{S II}](\lambda 6717+\lambda 6731)}}{F_{\text{H}\beta}}$
$R3$	$\frac{F_{[\text{O III}](\lambda 4959+\lambda 5007)}}{F_{\text{H}\beta}}$
$O3N2$	$\log_{10}\left(\frac{F_{[\text{O III}]\lambda 5007}/F_{\text{H}\beta}}{F_{[\text{N II}]\lambda 6583}/F_{\text{H}\alpha}}\right)$
$N2S2$	$\log_{10}\left(\frac{F_{[\text{N II}]\lambda 6583}}{F_{[\text{S II}](\lambda 6716+\lambda 6731)}}\right)$
y_{D16}	$N2S2 + 0.264 \log_{10}\left(\frac{F_{[\text{N II}]\lambda 6583}}{F_{\text{H}\alpha}}\right)$

Table 1. Definition of key emission-line ratios used in the extinction, SFR, and metallicity analysis, where all fluxes are dust-corrected. Here we adopt $F_{[\text{N II}](\lambda 6548+\lambda 6584)} = 1.34F_{[\text{N II}]\lambda 6584}$ and $F_{[\text{O III}](\lambda 4959+\lambda 5007)} = 1.33F_{[\text{O III}]\lambda 5007}$.

conservative **HII** selection retains only $\sim 26\%$ of the total valid spaxels used in this work (with substantial galaxy-to-galaxy variation). Additionally, in Section 5.2, we perform a robustness test by relaxing this conservative “star-forming” restriction and repeating the analysis with an expanded **HII+Comp** mask that includes spaxels classified as either **H II** and/or composite region on both **BPT** diagrams.

Star formation rate maps are constructed from the dust-corrected $\text{H}\alpha$ flux. After applying the Balmer-decrement attenuation correction described above, we then apply the Calzetti et al. (2007) calibration, rescaled to a Chabrier IMF by Kennicutt & Evans (2012), to obtain

$$\text{SFR} [M_{\odot} \text{ yr}^{-1}] = 4.98 \times 10^{-42} L_{\text{H}\alpha}^{\text{corr}} [\text{erg s}^{-1}]. \quad (6)$$

Similarly, we take the projected surface density of SFR and account for the inclination by Equation 3, obtaining corrected Σ_{SFR} .

2.4 Gas-phase Metallicity

Gas-phase oxygen abundances[†] are inferred from the same dust-corrected emission-line maps, restricted to the **HII** spaxels defined in Section 2.3. In practice, using only spaxels that meet the S/N and **BPT** requirements, we focus on a subset of four widely used strong-line prescriptions that are robustly applicable across the MAUVE–MUSE sample: the $N2S2$ – $\text{H}\alpha$ calibration of Dopita et al. (2016, hereafter **D16**), the S –calibration of Pilyugin & Grebel (2016, hereafter **Scal-PG16**), the $O3N2$ index calibration of Marino et al. (2013, hereafter **O3N2-M13**), and the $O3N2$ calibration of Curti et al. (2020, hereafter **O3N2-C20**). The basic line ratios entering these prescriptions are collected in Table 1. In practice, similar to Watts et al. (2024), we reconstruct the total $[\text{N II}] \lambda 6548, 6583$ and $[\text{O III}] \lambda 4959, 5007$ fluxes by scaling the measured $[\text{N II}] \lambda 6583$ and $[\text{O III}] \lambda 5007$ fluxes by fixed factors of 1.34 and 1.33, respectively. All strong-line calibrations are applied within the recommended range $7.63 \leq [\text{O}/\text{H}] \leq 9.23$ by Kewley et al. (2019).

The **D16** prescription combines the $N2S2$ and $[\text{N II}]/\text{H}\alpha$ ratios into a single abundance indicator. Using the shorthand in Table 1, we obtain the oxygen abundance as

$$[\text{O}/\text{H}]_{\text{D16}} = 8.77 + y_{\text{D16}} + 0.45(y_{\text{D16}} + 0.30)^5. \quad (7)$$

The **Scal-PG16** calibration uses the $N2$, $S2$, and $R3$ ratios listed in

Table 1. We evaluate the branch-dependent polynomial expressions given in Pilyugin & Grebel (2016):

$$[\text{O}/\text{H}]_{\text{PG16}} = \begin{cases} 8.424 + 0.030 \log_{10}(R3/S2) + 0.751 \log_{10}(N2) \\ \quad + [-0.349 + 0.182 \log_{10}(R3/S2) \\ \quad + 0.508 \log_{10}(N2)] \log_{10}(S2), \\ \log_{10}(N2) \geq -0.6; \\ 8.072 + 0.789 \log_{10}(R3/S2) + 0.726 \log_{10}(N2) \\ \quad + [1.069 - 0.170 \log_{10}(R3/S2) \\ \quad + 0.022 \log_{10}(N2)] \log_{10}(S2), \\ \log_{10}(N2) < -0.6. \end{cases} \quad (8)$$

For the **O3N2-M13** calibration, we adopt the $O3N2$ index defined in Table 1 and use the corresponding abundance relation:

$$[\text{O}/\text{H}]_{\text{O3N2-M13}} = 8.533 - 0.214 O3N2, \quad (9)$$

again restricted to the validity interval tabulated in that work.

Finally, we also adopt the $O3N2$ -based calibration of Curti et al. (2020), which uses the same $O3N2$ index but fits a higher-order polynomial to stacked Sloan Digital Sky Survey (SDSS) spectra (Abazajian et al. 2009). We evaluate the recommended **O3N2-C20** functional form spaxel-by-spaxel over the valid range:

$$O3N2 = 0.281 - 4.765x - 2.268x^2, \quad (10)$$

where $x \equiv [\text{O}/\text{H}]_{\text{O3N2-C20}} - 8.69$. In the course of this work, we also implemented the “combined” strategy advocated by Curti et al. (2020), in which all 6 actionable indicators in MUSE wavelength are selected on a spaxel-by-spaxel basis according to χ^2 minimization. While this approach yields self-consistent estimation with minimum uncertainties among available line diagnostics, we found that it can introduce indicator-dependent discontinuities within individual galaxies. To ensure internal consistency and straightforward interpretation of spatial trends, we therefore adopt the **O3N2-C20** calibration alone instead of this proposed “combined” approach.

All four metallicity prescriptions are evaluated exclusively on the conservative **HII** star-forming mask, on the same Voronoi grid as the stellar-mass and SFR surface-density maps. This guarantees that Σ_* , Σ_{SFR} , and $12 + \log_{10}(\text{O}/\text{H})$ are strictly co-spatial for every bin used in the resolved mass–metallicity–SFR analysis presented in Section 3. Throughout the analysis of this paper, to be consistent, we adopt **O3N2-M13** as the standard prescription. However, we acknowledge the different results caused by different calibrations, and we will therefore discuss this effect in Section 5.1.

2.5 Local Moran-like Correlation Index (I_i)

To quantify how two spatially-resolved fields co-vary in a genuinely local sense, we construct a “local Moran-like” product that combines the value of one property at a given position with the neighbourhood behaviour of a second property. This construction is inspired by the classical Moran’s I statistic for spatial autocorrelation (Moran 1950) and its generalization: local indicators of spatial association (Anselin 1995). Consider two generic maps $A(i)$ and $B(i)$ defined on the same set of valid spaxels i . We first compute the mean and standard deviation of each field over all valid locations, denoted (μ_A, σ_A) and (μ_B, σ_B) , and form standardised residuals

$$\begin{cases} z_A(i) = \frac{A(i) - \mu_A}{\sigma_A}, \\ z_B(i) = \frac{B(i) - \mu_B}{\sigma_B}. \end{cases} \quad (11)$$

[†] Throughout this paper, we use $[\text{O}/\text{H}] \equiv 12 + \log_{10}(\text{O}/\text{H})$ to denote the gas-phase oxygen abundance.

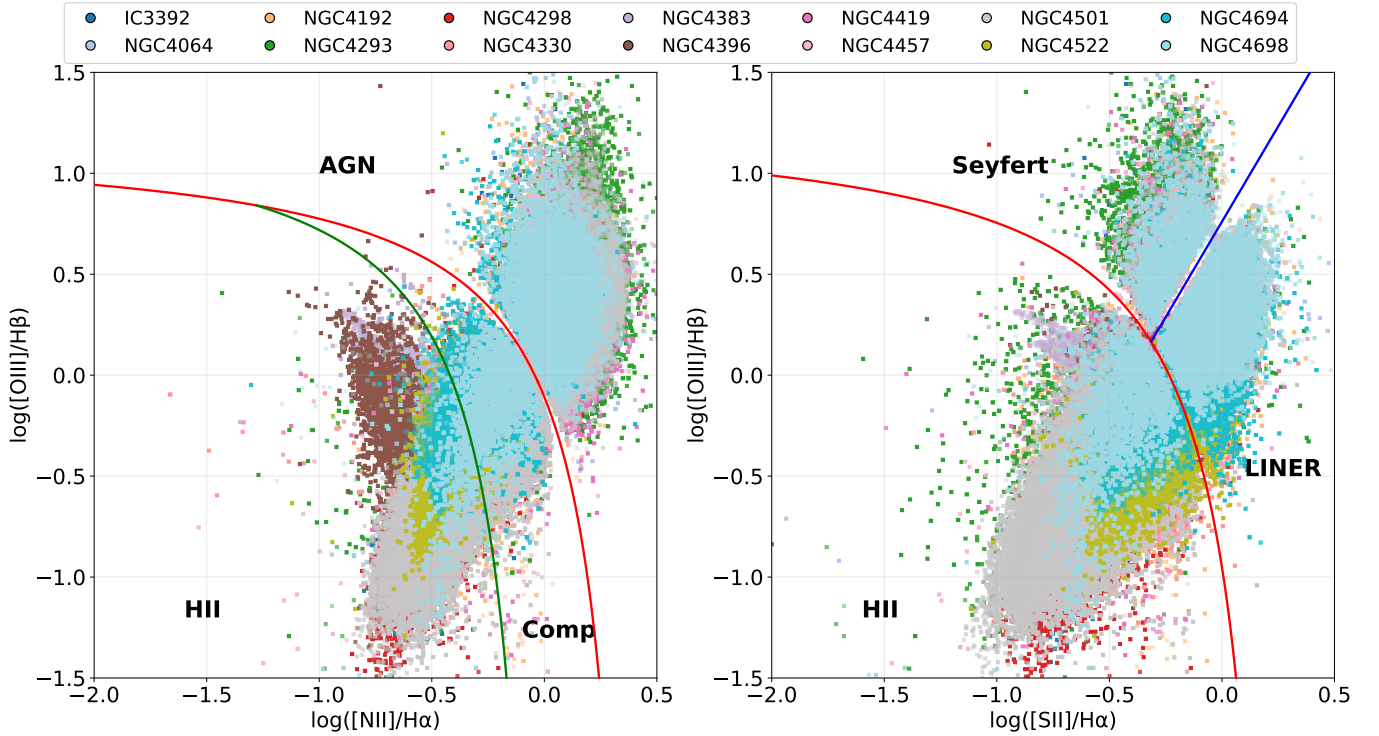


Figure 1. [N II] (left) and [S II] (right) BPT diagrams for “classified” spaxels of all 14 MAUVE–MUSE galaxies. Each point corresponds to a Voronoi spaxel classified using demarcation lines within 1σ confidence described in Section 2.3. Red and green curves show the Kewley et al. (2001) and Kauffmann et al. (2003) demarcation lines in the [N II] diagram; while red and blue curves indicate the Kewley et al. (2001) and Kewley et al. (2006) divisions in the [S II] diagram. Spaxels classified as star-forming (H II spaxels) require identification as H II regions on both BPT diagrams.

The offsets $z_A(i)$ and $z_B(i)$ measure the local departure of each spaxel from the global behaviour of the sample in units of the intrinsic scatter, after subtracting the azimuthally averaged radial profile (or the other secondary dependence) of each field. Positive values indicate that a given location lies above the mean in that field, while negative values indicate a deficit relative to the mean.

To embed this information in its spatial context, we next characterise the local environment of B around each spaxel i by averaging the z_B values of its neighbouring spaxels. In the present work, we adopt a simple eight-connected neighbourhood, so that the neighbour set $N(i)$ consists of the immediately adjacent spaxels in the two-dimensional grid and $N_i = 8$ for interior pixels. The local “spatial lag” of B at position i is then defined as

$$z_{\text{lag},B}(i) \equiv \frac{1}{N_i} \sum_{j \in N(i)} z_B(j). \quad (12)$$

This quantity is a dimensionless measure of whether the environment surrounding i is systematically enhanced or suppressed in B relative to the global mean, again in units of the intrinsic dispersion: $z_{\text{lag},B}(i) > 0$ implies that the neighbours tend to be above the mean in B , while $z_{\text{lag},B}(i) < 0$ indicates a locally depressed environment.

The final local Moran-like product at each position is then defined as

$$I_i \equiv z_A(i) z_{\text{lag},B}(i). \quad (13)$$

This scalar encapsulates both the sign and the strength of the local correspondence between the value of A at i and the surrounding behaviour of B . When $z_A(i)$ and $z_{\text{lag},B}(i)$ share the same sign, $I_i > 0$ and the spaxel is “in agreement” with its environment: locations with $z_A(i) > 0$ and $z_{\text{lag},B}(i) > 0$ correspond to regions that are elevated

in A and embedded in an environment that is likewise elevated in B (“high–high”), whereas $z_A(i) < 0$ and $z_{\text{lag},B}(i) < 0$ identify “low–low” regions that are suppressed with respect to both fields. Conversely, if $z_A(i)$ and $z_{\text{lag},B}(i)$ have opposite signs, then $I_i < 0$ and the spaxel is locally “inverted” with respect to its surroundings: a positive $z_A(i)$ combined with a negative $z_{\text{lag},B}(i)$ signals a high value of A embedded in a depressed environment in B (“high–low”), while the opposite configuration corresponds to “low–high” structure. The magnitude $|I_i|$ quantifies the statistical significance of this local pattern in units of the product of the two scatters. The resulting I_i map therefore provides a compact diagnostic of where, and how strongly, the field A is locally correlated or anti-correlated with the spatial environment of the field B , enabling a spatially-resolved assessment of concordant versus discordant structures in any pair of astrophysical quantities.

3 Results

3.1 The rMZR Relation

Before analysing the ensemble resolved relations derived from stacking all MAUVE–MUSE galaxies, we note that one of the 14 target galaxies, NGC 4383, is identified as a metallicity outlier relative to the rest of the sample. Across all four strong-line prescriptions adopted in this work (D16, Scal-PG16, O3N2-M13, and O3N2-C20), the oxygen abundances of star-forming spaxels in NGC 4383, especially the high- Σ_{SFR} spaxels (star-forming disc), are systematically offset low by at least 0.2 dex compared to the locus defined by the other 13 galaxies. This phenomenon is robust to the choice of calibration and to the adopted H II masking, and is therefore unlikely to

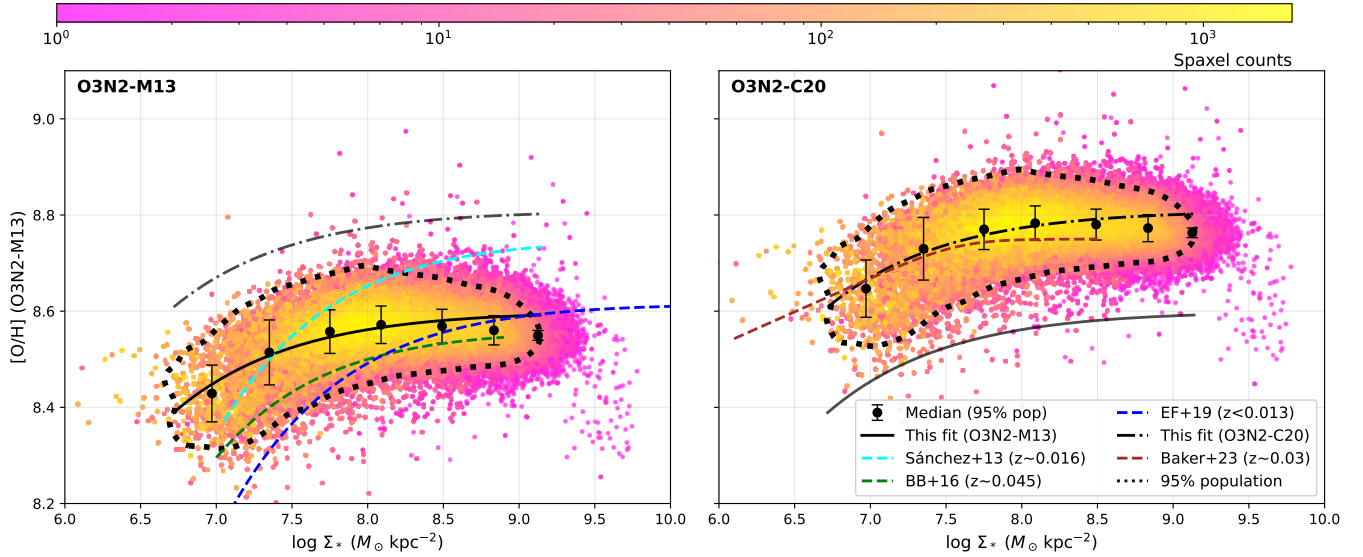


Figure 2. The resolved mass–metallicity relation (rMZR) for 13 MAUVE–MUSE galaxies (excluding NGC 4383). The left panel shows the rMZR derived using the **O3N2-M13** calibration, while the right panel shows the rMZR using the **O3N2-C20** calibration. Individual spaxels are shown as colored points, with the color scale indicating the number of spaxel counts. The dotted contours indicate the 95% population bounds, and the black points with error bars show the median metallicity within the contours. The solid and dot-dashed black curves represent the best-fitting rMZR to our data using Equation 14 under **O3N2-M13** and **O3N2-C20** prescriptions, respectively. For comparison, we show literature relations: the blue dashed curve from MAD data ($z < 0.013$; Erroz-Ferrer et al. 2019), the cyan dot-dashed curve from CALIFA data ($z \sim 0.016$; Sánchez et al. 2013), the green dot-dashed curve from MaNGA data ($z \sim 0.045$; Barrera-Ballesteros et al. 2016) in the left panel, and the brown dashed curve from MaNGA data ($z \sim 0.03$; Baker et al. 2023) in the right panel. Note that Baker et al. (2023) adopted the “combined” prescription from Curti et al. (2020).

The cyan curve seems dashed, not dot dashed, this is also the case for the green curve

be driven by methodological systematics. Because this single galaxy would otherwise dominate the low-metallicity tail and bias the ensemble trends, we temporarily exclude NGC 4383 from the stacked analyses below. We will return to discuss its properties in Section 5.3.

The stacked spaxel population of **13 MAUVE–MUSE** galaxies follows the familiar rMZR shape reported in previous IFS surveys (e.g., Sánchez et al. 2013; Barrera-Ballesteros et al. 2016; Sánchez et al. 2019; Erroz-Ferrer et al. 2019; Yao et al. 2022; Baker et al. 2023; Koller et al. 2024): Z_{gas} increases steeply with Σ_* at the low-mass end and gradually transitions to a shallower, saturation-like behaviour toward high Σ_* . In Figure 2, we visualise this locus using the 95% population contour and its median trend (black points), shown for both the **O3N2-M13** (left) and **O3N2-C20** (right) calibrations. Within the 95% contour, we perform a nonlinear least-square fit by adopting the functional form from Sánchez et al. (2013) to describe the trend of rMZR:

$$[\text{O}/\text{H}] = a + b (\log_{10} \Sigma_* - c) e^{-(\log_{10} \Sigma_* - c)}, \quad (14)$$

The best-fitting parameters for our 13 MAUVE–MUSE galaxies are $a = 8.597$, $b = 0.0024$, $c = 10$ for **O3N2-M13** (left) and $a = 8.807$, $b = 0.0023$, $c = 10$ **O3N2-C20** (right) prescription, respectively.

In the left panel of Figure 2, we compare our best-fitting rMZR obtained with the **O3N2-M13** calibration to three other spatially-resolved local-universe surveys that adopt the same metallicity prescription. Overall, the MAUVE–MUSE relation shows the familiar saturating rMZR behaviour but exhibits the shallowest low- Σ_* rise among the plotted studies. Relative to the MAD relation of Erroz-Ferrer et al. (2019) at comparable (~ 100 pc) resolution, we recover a similar high- Σ_* plateau at $12 + \log(\text{O}/\text{H}) \approx 8.6$, whereas the MAD curve declines more steeply toward low Σ_* , reaching ~ 0.3 dex lower metallicity around $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.0$. Compared to the MaNGA-based relation of Barrera-Ballesteros et al. (2016), our rMZR is systematically higher by ~ 0.05 – 0.1 dex over the entire dy-

namic range of $\log(\Sigma_*)$. The comparison to CALIFA (Sánchez et al. 2013) is non-monotonic: in the intermediate- and high-mass regimes ($7.5 \leq \log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \leq 9.0$) our curve lies below Sánchez et al. (2013), but at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \leq 7.5$ it becomes higher, consistent with a comparatively shallow low- Σ_* metallicity decline in MAUVE–MUSE data.

The right panel shows that adopting **O3N2-C20** shifts the best-fitting rMZR upward by ~ 0.2 dex relative to **O3N2-M13** at fixed Σ_* , underscoring that the absolute normalisation is calibration-dependent (see further discussion in Section 5.1). In this representation, the “combined” prescription used by Baker et al. (2023) yields a best-fitting relation that is highly consistent with our **O3N2-C20** fit, particularly at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \lesssim 7.5$. Taken together, these comparisons suggest that the comparatively metal-rich appearance of our **O3N2-M13** rMZR (relative to some local references) is more plausibly driven by differences in sample selection and calibration systematics than by redshift evolution. Indeed, although not shown here, our **O3N2-M13** relation shares a similar curvature to higher-redshift studies (e.g., Yao et al. 2022; Koller et al. 2024) but with a higher normalisation; as discussed by Koller et al. (2024), such offsets are unlikely to be attributable to redshift evolution of the rMZR alone.

3.2 rMZR’s Secondary Dependence of rMZR on Σ_{SFR}

Colour-coding the rMZR by Σ_{SFR} reveals a clear secondary dependence on local star formation intensity (see the left column of Figure 3). At fixed Σ_* , high- Σ_{SFR} spaxels tend to lie at higher Z_{gas} in the high- Σ_* regime, whereas the ordering reverses at low Σ_* , where lower- Σ_{SFR} spaxels are preferentially more metal-rich. The transition between these regimes occurs at $\log_{10}(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.5$ – 8.0 . While the exact reversal point is imprecise due to limited statistics

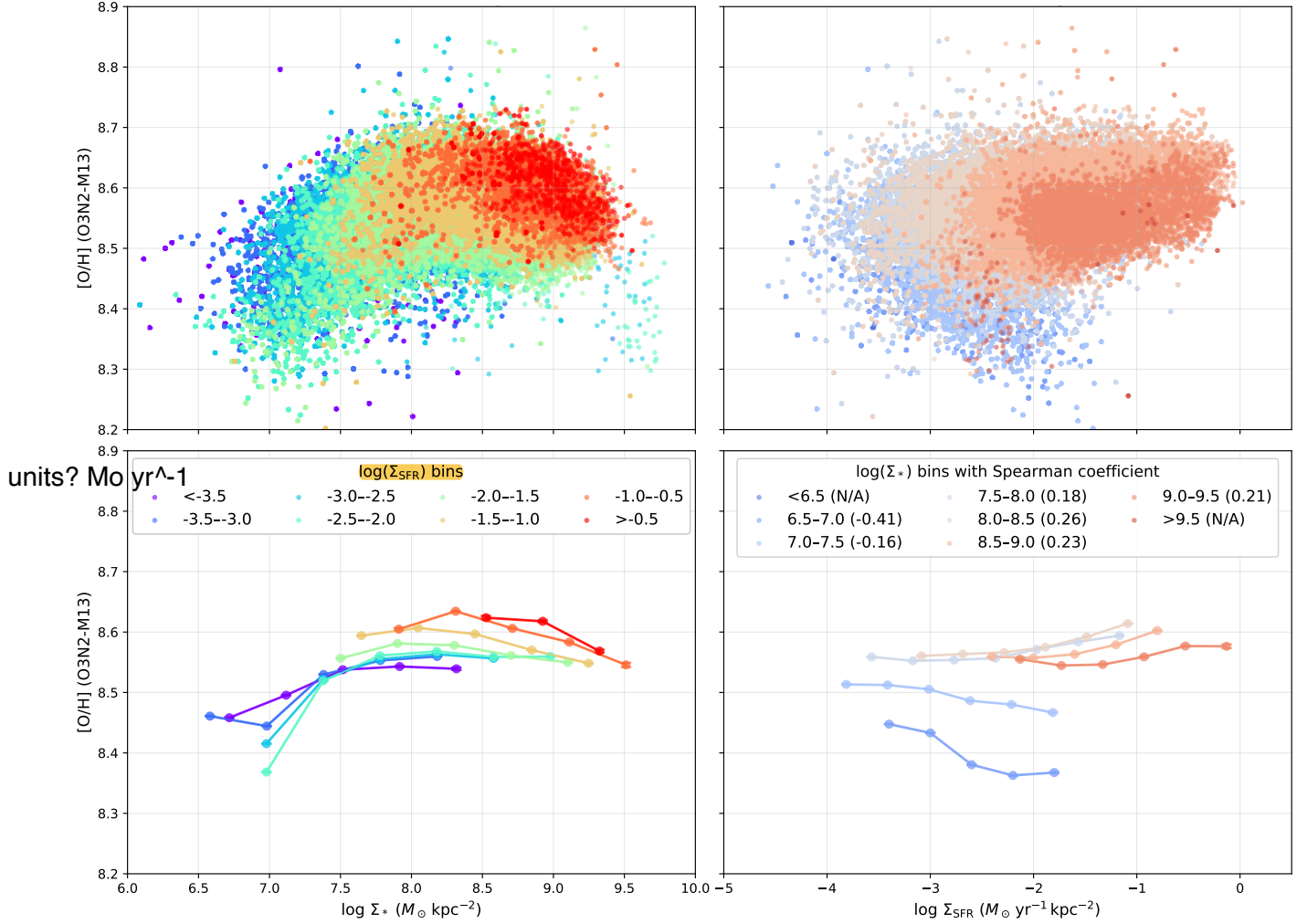


Figure 3. The left column shows the rMZR colour-coded by Σ_{SFR} for 13 MAUVE-MUSE galaxies (excluding NGC 4383), while the right column shows the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation colour-coded by Σ_* . The upper panels display the scatter plots, while the lower panels show the corresponding median trends of each scatter plot, with error bars indicating the standard deviation. In the top left panel, we use the black contour to enclose the 95% population of **HII** spaxels. Within this contour, we present the median trends with errorbars indicating the standard error of the median (typical value at $\lesssim 0.01$), and perform the best-fitting rMZR as a black curve using Equation 14. Additionally, the brown dash curve and dot-dash grey curve are the rMZR fits from MAGPI (Koller et al. 2024) and MUSE-Wide (Yao et al. 2022), respectively. Note that due to insufficient statistics, the median trends of the lowest and highest Σ_* bins in the bottom right panel are not shown.

i do not see errorbars
in the lower plots

in the narrow overlap region, its location is consistent with the inversion reported from MAGPI by Koller et al. (2024), who found a clear sign change near $\log_{10}(\Sigma_*) \sim 7.8$ using O3N2-M13 prescription as well. The MAUVE-MUSE data therefore support a mass-dependent inversion in the resolved $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling, implying that the physical driver of the secondary dependence may differ between low- and high- Σ_* environments. Similarly, although not a strong signal, under the prescriptions of Curti et al. (2020), MaNGA data also seem to suggest a flipped correlation in the spatially-resolved scenario (Baker et al. 2023). Furthermore, adopting both D16 and Scal-PG16 calibrations, WL21 reported a predominantly positive local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation using the MUSE Atlas of Disks (MAD) data at ~ 100 pc resolution, whereas with MaNGA data on global galactic scales they found an overall negative correlation.

To visualise this dependence more directly, the right column of Figure 3 shows Z_{gas} as a function of Σ_{SFR} in bins of Σ_* . For each Σ_* bin we also compute the Spearman rank correlation coefficient ρ between Z_{gas} and Σ_{SFR} after removing the 3σ outliers. The lowest- Σ_* bins ($\log \Sigma_*$ at 6.5 - 7.0) exhibit the strongest anti-correlation at

Spearman coefficient $\rho = -0.41$, with metallicity decreasing as Σ_{SFR} rises. With increasing Σ_* , the median trends flatten progressively, and ultimately the relation becomes positively correlated, such that regions of elevated star formation are also metal-enhanced. This systematic, mass-ordered sequence from negative through flat to positive slopes encapsulates the rMZR's second dependence on Σ_{SFR} and motivates the offset and spatial-correlation analyses developed in the next section.

3.3 Offset Relation and Local Moran-like Correlation Map

To isolate the secondary coupling between metallicity and star formation from their primary dependence on stellar mass surface density, we construct mass-normalised offset maps for each galaxy. Specifically, within each galaxy we divide all valid **HII** spaxels into 12 equal-width bins in $\log_{10} \Sigma_*$ (illustrated for one of MAUVE-MUSE targets, NGC 4192, in the bottom-right panel of Figure 4). For every bin m , we compute the mean spaxel values $\langle \log_{10} \Sigma_{\text{SFR}} \rangle_m$ and

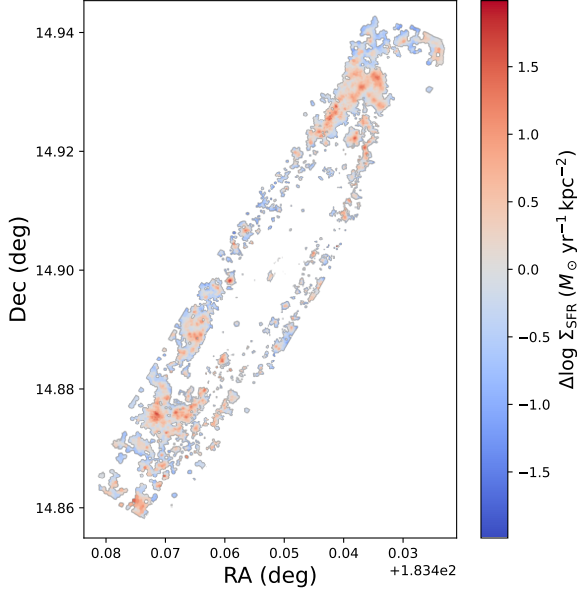
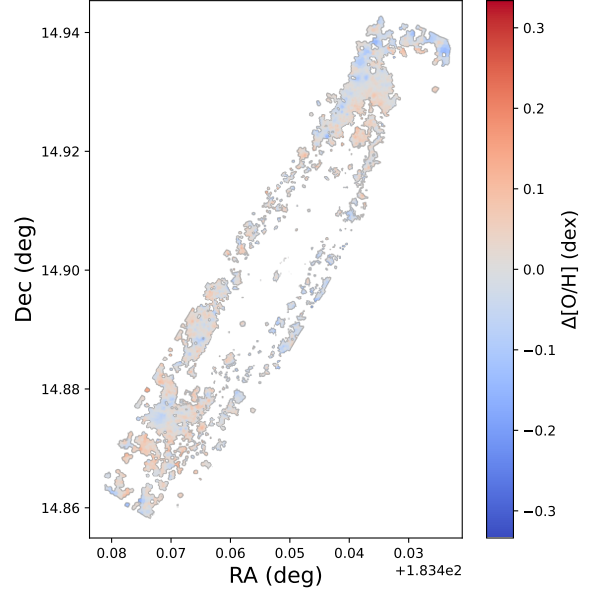
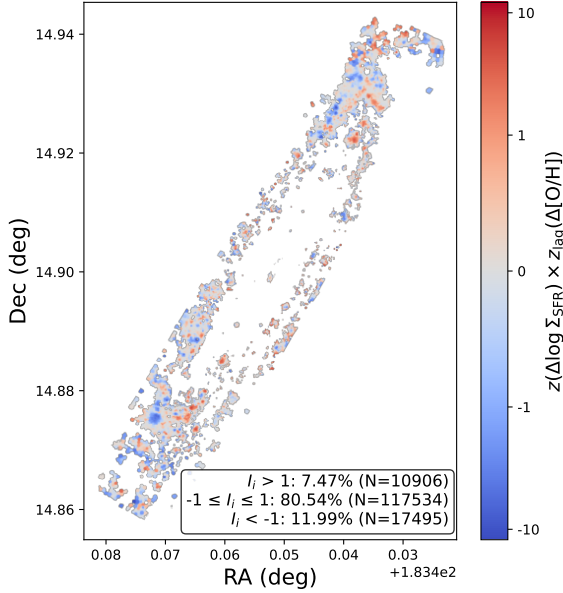
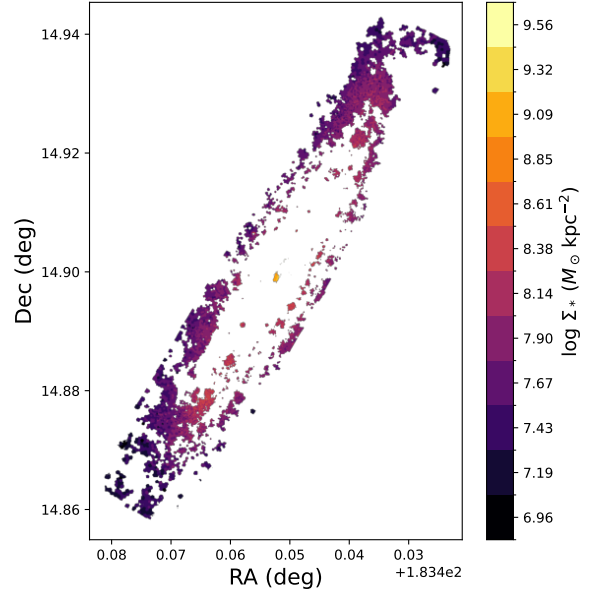
NGC4192: $\Delta \log \Sigma_{\text{SFR}}$ (offset from bin mean)NGC4192: $\Delta[\text{O}/\text{H}]_{\text{O3N2-M13}}$ (offset from bin mean)NGC4192: local bivariate Moran-like I_i NGC4192: $\log \Sigma_*$ (12 discrete bins)

Figure 4. Example offset and local Moran-like correlation maps for **HII** spaxels of NGC 4192. The top panels show the offsets $\Delta \log_{10} \Sigma_{\text{SFR}}$ (left) and $\Delta[\text{O}/\text{H}]_{\text{O3N2-M13}}$ (right) after removing the mean trends with Σ_* (see text). The bottom-left panel presents the local bivariate Moran-like product I_i constructed from these two offset fields, highlighting spatially local correlation (red) and anti-correlation (blue). The bottom-right panel shows $\log_{10} \Sigma_*$ for the **HII** spaxels used in the other panels, divided into 12 equal-width bins.

$\langle [\text{O}/\text{H}]_{\text{O3N2-M13}} \rangle_m$, and define spaxel-level offsets

$$\begin{cases} \Delta \log_{10} \Sigma_{\text{SFR}}(i) \equiv \log_{10} \Sigma_{\text{SFR}}(i) - \langle \log_{10} \Sigma_{\text{SFR}} \rangle_m, \\ \Delta[\text{O}/\text{H}](i) \equiv [\text{O}/\text{H}](i) - \langle [\text{O}/\text{H}] \rangle_m, \end{cases} \quad (15)$$

where spaxel i belongs to bin m . By construction, these offsets remove the mean rMZR and resolve star forming main sequence (rSFMS) trends within each Σ_* interval, leaving only the relative fluctuations of Σ_{SFR} and Z_{gas} at fixed local stellar mass density. The resulting maps therefore trace spatial variations that are not driven by the underlying Σ_* distribution and its associated large-scale gradients (e.g., see top panels of Figure 4).

Using these mass-normalised quantities, we also measure the residual $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling by computing, in each Σ_* bin of each galaxy, the Spearman rank correlation coefficient between $\Delta[\text{O}/\text{H}]_{\text{O3N2-M13}}$ and $\Delta \log_{10} \Sigma_{\text{SFR}}$ after removing the 3σ outliers. The resulting $\rho(\Sigma_*)$ trends are shown in Figure 5. For the stacked sample (black curve), ρ increases smoothly from negative values at low Σ_* to positive values at high Σ_* : rising from $\rho \approx -0.35$ at $\log_{10} \Sigma_* \sim 7$ to $\rho \approx +0.3$ at $\log_{10} \Sigma_* \gtrsim 8$, with a zero-crossing within $\log_{10}(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.5-8.0$. Individual galaxies follow the same qualitative ordering: systems that predominantly sample the low- Σ_* regime (e.g. NGC 4396, NGC 4457, and part of NGC 4192)

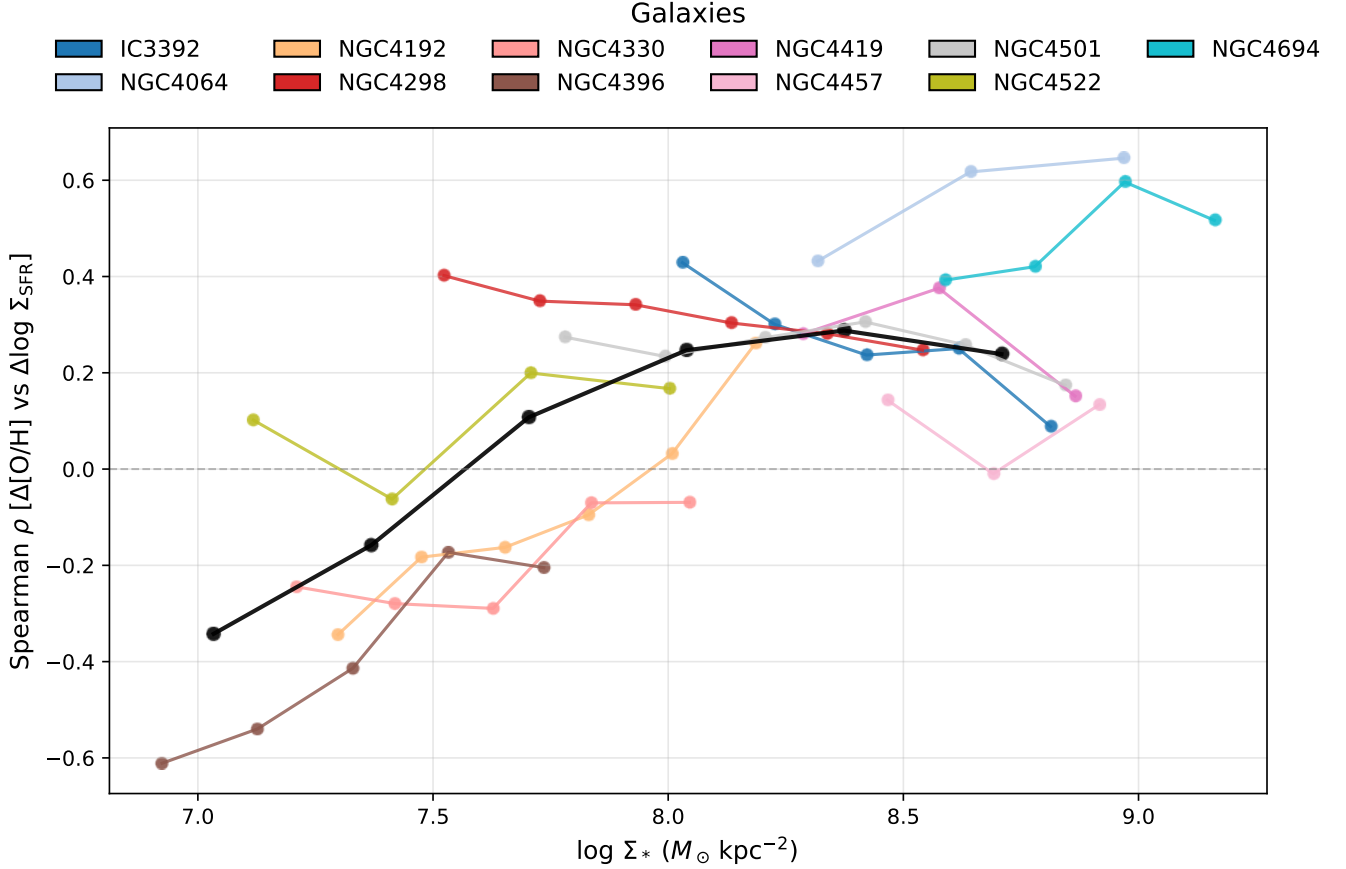


Figure 5. Spearman rank correlation coefficient ρ between $\Delta[\text{O}/\text{H}]_{\text{O3N2-M13}}$ and $\Delta\log_{10} \Sigma_{\text{SFR}}$ as a function of $\log_{10} \Sigma_*$, measured in discrete Σ_* bins for 13 MAUVE–MUSE galaxies (excluding NGC 4383). Each coloured curve corresponds to one galaxy (colour scheme as in Figure 1); the Spearman trends of NGC 4293 and NGC 4698 are omitted due to insufficient statistics. The black curve shows the stacked result obtained by combining all **HII** spaxels from the 13 galaxies.

contribute to the anti-correlated branch, whereas galaxies whose **HII** spaxels lie mainly at high Σ_* populate the positively correlated regime. Notably, NGC 4192 spans a broad Σ_* range that straddles the inversion point and exhibits a clear internal sign reversal, indicating that the transition is intrinsic to the subtle physics within galaxies rather than an artefact of combining different galaxy populations.

We further assess the spatial locality of this residual coupling by constructing the local bivariate Moran-like product I_i defined in Section 2.5. As an illustrative example, the bottom-left panel of Figure 4 shows the I_i map for NGC 4192, derived from the corresponding $\Delta\log_{10} \Sigma_{\text{SFR}}$ and $\Delta[\text{O}/\text{H}]$ fields. As indicated by Equation 15, these two fields are defined as residuals at fixed Σ_* , i.e. after subtracting, in each Σ_* bin, the trends of mean rMZR and rSFMS. In practice, this removes the dominant radial (mass-driven) profiles, so that the I_i map traces purely local spatial correspondence between star formation and metallicity fluctuations. By construction, I_i is the product of two standardised quantities, $z_{\Delta\log_{10} \Sigma_{\text{SFR}}}(i)$ and $z_{\text{lag}, \Delta[\text{O}/\text{H}]}(i)$, each expressed in units of each Σ_* bin’s standard deviation of the corresponding field. Thus $|I_i| \sim 1$ corresponds to the joint product of two $\sim 1\sigma$ deviations (one at the spaxel position, one in its local environment), whereas $|I_i| \ll 1$ indicates that at least one of the two is close to the global mean. Therefore, while the choice is inherently arbitrary, we adopt $|I_i| = 1$ as a convenient, dimensionless threshold to

distinguish weak or typical patterns from stronger local correspondence.

For NGC 4192, most spaxels in Figure 4 have $|I_i| \leq 1$ (grey), implying weak or ambiguous local correspondence, while a non-negligible fraction show significant local correlation ($I_i > 1$, red: higher (lower) Σ_{SFR} with higher (lower) Z_{gas}) or anti-correlation ($I_i < -1$, blue: higher (lower) Σ_{SFR} with lower (higher) Z_{gas}). These two tails are of comparable magnitude, with 7.47% of spaxels having $I_i > 1$ and 11.99% having $I_i < -1$, while the majority (80.54%; $N=117534$) remain in the weak/ambiguous regime. This near-balance between correlated and anti-correlated **HII** spaxels is consistent with NGC 4192 being the clearest “inversion” case in our sample: depending on Σ_* , the net $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling can change sign in Figure 5. Interpreting high- Σ_{SFR} peaks as tracing the cores of individual **HII** regions, the coexistence of red and blue structures within the same Σ_* bins suggests that the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling is spatially heterogeneous: once mass trends of these two properties are removed, neighbouring regions of comparable Σ_* can display either concordant or discordant metallicity responses to star formation. Moreover, the highest $|I_i|$ values are spatially coincident with bright **HII**/high- Σ_{SFR} clumps that satisfy our **HII** BPT selection, whereas diffuse low- Σ_{SFR} regions tend to show $|I_i| \lesssim 1$, supporting that the observed signal is associated with genuine **HII** regions rather than being driven by diffuse ionised gas (DIG) contamination. These

maps show that 1) the sign of the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling is set by spatially varying local physics, so correlated and anti-correlated H II regions can coexist even at fixed Σ_* ; and 2) the observed mass-dependent behaviour arises only as an emergent population trend when one aggregates enough spaxels to sample both sides of the transition near $\log_{10}(\Sigma_*/M_\odot, \text{kpc}^{-2}) \sim 7.5-8.0$.

4 Discussion: The Origin of $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (Anti-)Correlation in a Gas-regulator Framework

To interpret the varying sign of the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation, we adopt a spatially-resolved gas-regulator (“bathtub”) framework formulated entirely in spaxel-by-spaxel variations. Throughout this section, all quantities are written as surface densities to reflect the fact that our observational constraints are local (~ 100 pc resolution) rather than global. The model is intended to describe the evolution of the gas regulator in the local Universe (present day), over timescales that are short compared to the ~ 10 Gyr past history of star formation and chemical enrichment. In practice, the variations that matter for our observed $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (anti-)correlation are closer to the lifetimes of individual H II regions and local dynamical or mixing timescales (i.e. a few-tens of Myr), as suggested by the fact that the strongest signal is associated with bright H α /high- Σ_{SFR} clumps. Over such intervals, the accumulated growth of the stellar mass surface density Σ_* is negligible compared to its present-day value (or at most comparable to observational uncertainties), so we neglect the time evolution of Σ_* and treat it as a fixed control parameter. We therefore do not attempt to reconstruct the full past star formation or metal enrichment history of each region. Instead, we assume that different H II spaxels within a galaxy began their recent evolution from broadly similar initial conditions and have since evolved under the same set of governing equations.

Within this framework, the time variable, t , plays the role of an evolution parameter that controls the instantaneous state of the local regulator (gas mass, SFR, metallicity, gas supply properties) at a given location. Mathematically, even though we observe different spaxels at the same cosmic time, each with its own set of properties, P (here $P \in \{\Sigma_*, \Sigma_{\text{SFR}}, Z_{\text{gas}}, \text{etc.}\}$), it is useful to think of these spatial variations as sampling different points along a common family of time-dependent solutions. Equivalently, given any pair of spaxels within the same Σ_* bin, the differences in their local properties can be interpreted as the result of having evolved for different effective “phases” or having experienced different recent perturbations under the same functional form of the regulator equations. In this sense, the time coordinate, t , may be regarded as a parameter that indexes the instantaneous value of each property and allows us to trace how fluctuations in one property, P , induce corresponding responses in another. What follows is therefore a local, time-dependent description that we use as a physical language for organizing the spatially-resolved fluctuations measured in the MAUVE–MUSE data.

We first write the star-formation rate surface density as

$$\Sigma_{\text{SFR}}(t) \equiv \frac{\Sigma_g(t)}{\tau_{\text{dep}}(t)}, \quad (16)$$

where $\Sigma_g(t)$ is the gas surface density and $\tau_{\text{dep}}(t)$ the local depletion time. Then, we use the following two differential equations to describe our spatially-resolved framework: the gas mass surface density obeys

$$\frac{d\Sigma_g(t)}{dt} = \Sigma_\Phi(t) - [1 - R + \lambda] \Sigma_{\text{SFR}}(t), \quad (17)$$

R should be t dependent? this is the case if t is long

where $\Sigma_\Phi(t)$ is the *gas supply rate (GSR) surface density*[‡], i.e. the rate at which the region gains fresh gas per unit area, **R is the returned fraction**, and λ is the mass-loading factor; while the surface density of metals (Σ_Z) follows

$$\frac{d\Sigma_Z(t)}{dt} = y(1 - R) \Sigma_{\text{SFR}}(t) - [1 - R + \lambda] \Sigma_{\text{SFR}}(t) Z(t) + \Sigma_\Phi(t) Z_\Phi, \quad (18)$$

with metal yield per stellar, y , and metallicity of the supplied gas, Z_Φ (i.e., background metallicity).

Defining the gas-phase metallicity as

$$Z(t) \equiv \frac{\Sigma_Z(t)}{\Sigma_g(t)}, \quad (19)$$

we obtain

$$\frac{dZ(t)}{dt} = \frac{1}{\Sigma_g(t)} [y(1 - R) \Sigma_{\text{SFR}}(t) + (Z_\Phi - Z(t)) \Sigma_\Phi(t)], \quad (20)$$

so that at any location, the metallicity responds to enrichment from ongoing star formation and dilution by the supplied gas.

It is convenient to introduce the gas supply timescale

$$\tau_\Phi(t) \equiv \frac{\Sigma_g(t)}{\Sigma_\Phi(t)} \leq \tau_{\text{dep}}(t), \quad (21)$$

which measures how quickly the local gas reservoir would be replenished by the current GSR. Using $\Sigma_{\text{SFR}} = \Sigma_g/\tau_{\text{dep}}$ and $\Sigma_\Phi = \Sigma_g/\tau_\Phi$ in Equation 20 yields

$$\frac{dZ(t)}{dt} = \frac{1}{\tau_\Phi(t)} \left[Z_\Phi + y(1 - R) \frac{\Sigma_{\text{SFR}}(t)}{\Sigma_\Phi(t)} - Z(t) \right]. \quad (22)$$

This motivates the definition of the *instantaneous target metallicity*

$$Z^\dagger(t) \equiv Z_\Phi + y(1 - R) \frac{\Sigma_{\text{SFR}}(t)}{\Sigma_\Phi(t)}, \quad (23)$$

so that Equation 22 becomes

$$\frac{dZ(t)}{dt} = \frac{1}{\tau_\Phi(t)} [Z^\dagger(t) - Z(t)] \iff Z^\dagger(t) = Z(t) + \frac{dZ(t)}{dt} \tau_\Phi(t). \quad (24)$$

This explicitly shows that $Z^\dagger(t)$ acts as the *driver*: it is the metallicity that the system would reach if the current rate of change dZ/dt were sustained over one gas supply timescale. Only when $Z^\dagger(t) = Z(t)$ does Equation 24 imply $dZ/dt = 0$, corresponding to a *local minimum or maximum* of the real-time metallicity $Z(t)$ rather than a long-lived equilibrium.

With these definitions, the observed spaxels can be separated conceptually into two complementary cases: either reach (Section 4.1) or evolve toward/away from (Section 4.2) the local local extremum stages.

4.1 Case I: Correlation at $Z = Z^\dagger$

For spaxels observed at a series of local extremum time $t = t_m$, the real-time metallicity satisfies $dZ/dt|_{t_m} = 0$, hence $Z(t_m) = Z^\dagger(t_m)$,

[‡] We avoid the term “inflow rate surface density” because inflow usually denotes pristine, galaxy-scale accretion from circumgalactic medium (CGM), whereas on local scales it is more appropriate to describe a **region’s** or giant molecular cloud (GMC)’s ability to gain gas supply from its ambient ISM; moreover, for our observed Virgo members, **the global inflow is likely suppressed or truncated**. (starvation: Larson et al. 1980)

with

$$Z^\dagger(t_m) = Z_\Phi + \gamma(1-R) \frac{\Sigma_{\text{SFR}}(t_m)}{\Sigma_\Phi(t_m)}. \quad (25)$$

Taking a first-order variation of Equation 25 yields

$$\delta Z^\dagger(t_m) = [Z^\dagger(t_m) - Z_\Phi] [\delta \log_{10} \Sigma_{\text{SFR}}(t_m) - \delta \log_{10} \Sigma_\Phi(t_m)], \quad (26)$$

so the sign of the metallicity fluctuation at extrema is controlled by the competition between spatial variations in SFR and GSR. If Σ_{SFR} varies more strongly than Σ_Φ across neighbouring spaxels, then $\delta Z^\dagger$ follows $\delta \log_{10} \Sigma_{\text{SFR}}$ and a positive $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation is expected. If instead GSR variations dominate while Σ_{SFR} and Σ_Φ co-vary, dilution drives $\delta Z^\dagger$ opposite to $\delta \log_{10} \Sigma_{\text{SFR}}$, yielding an anti-correlation. Where the two contributions are comparable, $\delta Z^\dagger$ is small and little correlation is produced.

Using $\Sigma_{\text{SFR}} \equiv \Sigma_g/\tau_{\text{dep}}$ and $\tau_\Phi \equiv \Sigma_g/\Sigma_\Phi$, Equation 26 is equivalently

$$\delta Z^\dagger(t_m) \propto \delta \log_{10} \tau_\Phi(t_m) - \delta \log_{10} \tau_{\text{dep}}(t_m), \quad (27)$$

highlighting that, at metallicity extrema, the correlation sign is set by whether variations in the gas supply timescale outpace those in depletion time. Furthermore, if we generalize (i.e., substitute) t_m with t , which in fact describes the change of instantaneous target metallicity rather than the real-time metallicity at extrema, we still have the same functional forms of Equation 26 and Equation 27. This tells us a broader context that the change of instantaneous target metallicity is also determined by the comparison of the relative change magnitude between the local SFR and GSR. We will return to this relation later in our discussion.

4.2 Case II: Correlation at $Z \neq Z^\dagger$

In reality, most spaxels are not observed exactly at a local metallicity extremum. To relate their real-time metallicity to the driver $Z^\dagger$, we expand around the nearest extremum time t_m by rewriting $t = t_m + \zeta$ and, for any quantity P ($P \in \{\Sigma_{\text{SFR}}, Z, Z^\dagger, \text{etc.}\}$), defining

$$\Delta_P(\zeta) \equiv P(t_m + \zeta) - P(t_m). \quad (28)$$

Linearising Equation 24 about t_m yields a first-order response,

$$\frac{d\Delta_Z(\zeta)}{d\zeta} + \frac{\Delta_Z(\zeta)}{\tau_\Phi(t_m)} = \frac{\Delta_{Z^\dagger}(\zeta)}{\tau_\Phi(t_m)}, \quad (29)$$

whose solution is

$$\Delta_Z(\zeta) = \frac{1}{\tau_\Phi(t_m)} \int_0^\zeta \exp\left[-\frac{\zeta - \nu}{\tau_\Phi(t_m)}\right] \Delta_{Z^\dagger}(\nu) d\nu. \quad (30)$$

Equation 30 shows that $\Delta_Z(\zeta)$ is a low-pass-filtered version of the driver $\Delta_{Z^\dagger}(\zeta)$: the real-time metallicity responds as an exponential moving average of the target metallicity, smoothed over the gas supply timescale $\tau_\Phi(t_m)$. Physically, this “memory” arises because $Z(t)$ cannot respond instantaneously to changes in $Z^\dagger(t)$, due to finite delays in metal injection from recent star formation and in turbulent mixing (metal diffusion) through the ISM. In practice, because of detection procedures and measurement uncertainties, our observations predominantly probe the regime $\zeta \gtrsim \tau_\Phi(t_m)$, i.e. the time-averaged response rather than an instantaneous snapshot of $Z^\dagger$.

For such observed spaxels, $\delta Z(t)$ follows the recent behaviour of $\delta Z^\dagger(t)$ and therefore inherits the same sign set by star formation versus gas supply variability. Combining Equation 23 with Equation 30 gives the consistent prediction as Equation 26 and Equation 27

$$\begin{aligned} \delta Z(t) &\propto \delta \log_{10}(\Sigma_{\text{SFR}}(t)) - \delta \log_{10}(\Sigma_\Phi(t)) \\ &\propto \delta \log_{10}(\tau_\Phi(t)) - \delta \log_{10}(\tau_{\text{dep}}(t)). \end{aligned} \quad (31)$$

if this was the case, you should see systematic differences in cluster galaxies where gas supply isn't lacking (starvation), and you rather have gas stripping. Is it the case?

Thus, away from extrema the observed local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (anti-)correlation is still governed by whether spatial variations are driven primarily by changes in star-formation efficiency (through τ_{dep}) or by changes in the gas supply rate (through τ_Φ), with the finite response time encoded in τ_Φ introducing additional scatter and spatially varying phase lags.

if this is the case you should see correlations with the molecular gas distribution/surface density, is it the case?

4.3 Interpretation of the Observed (Anti-)Correlation

Within this framework, our resolved MAUVE–MUSE trends in Figure 5 can be interpreted as the outcome of a competition between GSR-driven and SFR-driven variability. Observationally, in high- Σ_* regions (typically inner discs), dense gas and short depletion times make spatial variations in Σ_{SFR} or τ_{dep} relatively strong, while the ability to gain fresh gas (i.e., the GSR) is comparatively steady from spaxel to spaxel. In this regime, Equation 31 favours a positive $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation: fluctuations in SFR dominate over variations in GSR, and oxygen abundance increases with Σ_{SFR} at fixed Σ_* .

On the other hand, at low Σ_* (outer discs), especially in the Virgo environment where the local gas supply can fluctuate strongly due to environmental processes, spatial variations in τ_Φ become more important. When GSR fluctuations dominate while Σ_{SFR} and Σ_Φ co-vary, the dilution associated with enhanced supply tends to lower Z in regions of relatively higher SFR, driving the correlation toward zero or negative values. In between these extremes, where variations in τ_Φ and τ_{dep} are comparable, the net effect on metallicity is small and the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation becomes weak or flat.

WL21 also explain this (anti-)correlation from a theoretical point of view. Their model attributes the positive case (in ~ 100 pc scales) to time-variable SFE acting against a relatively steady gas inflow rate, and the negative case (in integrated galactic scales) to strongly time-variable inflow acting against nearly constant SFE. Our MAUVE–MUSE galaxies, however, show the coexistence of both positive and negative correlations already at ~ 100 pc scales, and our regulator analysis indicates that the sign is set locally by the relative amplitude of variations in Σ_{SFR} and the gas-supply-rate surface density Σ_Φ , rather than by scale alone. If the spaxel-to-spaxel changes in Σ_{SFR} are interpreted as being driven primarily by fluctuations in the SFE of individual GMCs, while Σ_Φ is taken as the local analogue of an inflow rate, then our expression in Equation 31 is fully consistent with the WL21 framework: positive (negative) correlations emerge where SFE- or SFR-driven (GSR-driven) variability dominates, and both regimes can naturally coexist within the same galaxy at fixed spatial resolution.

4.4 From Spatially-resolved to Integrated Relations

Although this framework is written entirely in terms of surface densities, the underlying regulator equations have exactly the same functional form if one replaces $(\Sigma_g, \Sigma_*, \Sigma_{\text{SFR}}, \Sigma_\Phi)$ by their galaxy-integrated counterparts $(M_g, M_*, \text{SFR}, \Phi)$ and the associated timescales by effective global depletion and inflow timescales. In this sense, Equation 31 can be read as a scale-free statement: at fixed M_* , the sign of the $Z_{\text{gas}}-\text{SFR}$ (anti-)correlation is determined by the comparison of the relative fluctuation amplitudes in SFR and in the inflow rate, or equivalently by whether variations in depletion or inflow timescale dominate. If global fluctuations in SFR (or SFE) are larger than those in inflow, one expects a positive $Z_{\text{gas}}-\text{SFR}$ correlation at fixed M_* ; if stochasticity in inflow dominates, dilution produces an anti-correlation. This picture provides a natural physical interpretation for the integrated FMR found in large samples, where the SFR

dependence of metallicity is negative at low stellar mass and commonly observed but weakens and may even change sign at the high-mass end (e.g. Mannucci et al. 2010; Yates et al. 2012; Curti et al. 2020). In our view, these global trends reflect the same basic competition encapsulated in Equation 31, with low-mass galaxies residing in a regime where inflow-driven variability is more significant, and massive systems shifting toward a regime where SFR- or SFE-driven variability becomes comparatively stronger.

Within this picture, the apparent prevalence of anti-correlations in both resolved and integrated studies can be understood, at least in part, as a consequence of how different surveys sample the M_* and/or Σ_* parameter space. Global FMR analyses are dominated by galaxies with $M_* \lesssim 10^{10} M_\odot$, while more massive systems with $M_* \gtrsim 10^{10.5} M_\odot$ are rarer and span a narrower range of specific SFR, so the low-mass, inflow-dominated regime contributes a larger fraction of the statistical weight. An analogous effect arises in spatially-resolved work: at kpc-scale resolution, many surveys preferentially sample spaxels with $\log(\Sigma_*/M_\odot, \text{kpc}^{-2}) \lesssim 7.8$ and therefore see predominantly anti-correlated or flat $Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ behaviour (e.g. Baker et al. 2023; Koller et al. 2024), whereas at ~ 100 pc resolution our MAUVE–MUSE, and possibly MAD data in WL21, maps contain a much larger fraction of spaxels with $\log(\Sigma_*/M_\odot, \text{kpc}^{-2}) \gtrsim 8$, so the positive branch dominates the stacked trends. In both the global and resolved cases, the sign flip emerges only when one samples across the transition region (around $\log \Sigma_* \sim 7.5\text{--}8.0$ locally and $\log M_* \sim 10.0\text{--}10.5$ globally), and the balance between correlated and anti-correlated regimes is then controlled by the relative number of galaxies/spaxels on either side of this inversion. Furthermore, the more frequent detection of anti-correlation, as well as the weaker and less clear (anti-)correlation signal on global scales than on local scales, may simply raise from an averaging effect, whereby integrating over many spaxels blends regions together with opposite local behaviours. However, what sets the precise location of these transition scales, whether the local (Σ_*) and global (M_*) inversions are causally linked or merely coincident, and which physical conditions change across them (e.g. gas fraction, disc stability, outflow efficiency, or environmental processes), remain open questions that we defer to future work.

5 Robustness of Our Findings

In this section, we access how robust our observed $Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ (anti-)correlation is to three key sources of systematics with Figure 6 as a compact visual guide. We first examine the effect of adopting different strong-line abundance prescriptions in Section 5.1, by comparing the four rows of columns 1 and 3 of Figure 6 (D16, Scal-PG16, O3N2-M13, and O3N2-C20). We then test the sensitivity to star-forming spaxel selection in Section 5.2, by contrasting the conservative **HII** results (columns 1 and 3) with the more inclusive **HII+Comp** results (columns 2 and 4). Finally, we investigate the special case of the metallicity outlier NGC 4383 in Section 5.3, highlighted by dotted contours or curves in each panel of Figure 6 relative to the solid loci of the remaining galaxies, noting that the stacked Spearman trend in black continues to exclude this object. The broader purpose of these checks is to emphasise that, although calibration choice, excitation mixing, environment, and mass-dependent sampling can modulate the apparent scatter and the detailed slope in individual regimes, they do not alter the underlying physical interpretation implied by Equation 31, namely that the sign of the local residual correlation is set by the competition between the relative fluctuations

of star formation and gas supply, $\delta \log_{10} \Sigma_{\text{SFR}}$ versus $\delta \log_{10} \Sigma_\Phi$, or equivalently by the balance between $\delta \log_{10} \tau_\Phi$ and $\delta \log_{10} \tau_{\text{dep}}$.

5.1 Sensitivity to Metallicity Calibration

A well-known limitation of any resolved metallicity analysis is that strong-line prescriptions are not anchored to a unique absolute scale. Different calibrations rely on different reference samples and methodologies—direct electron-temperature (T_e) measurements, photoionization model grids, or hybrid/stacked approaches—and they respond differently to variations in excitation, ionization parameter, diffuse ionized gas (DIG), and N/O abundance. Consequently, systematic offsets of order $\sim 0.1\text{--}0.3$ dex (and in extreme cases larger up to ~ 0.5 dex) between calibrations applied to the same spectra are well documented, and can alter both the normalization and the detailed slopes of resolved relations (e.g. Kewley & Ellison 2008; Kewley et al. 2019; Maiolino & Mannucci 2019). These long-standing uncertainties motivate our use of four widely adopted diagnostics spanning empirical T_e -calibrated and model-based families: D16, Scal-PG16, O3N2-M13, and O3N2-C20.

The first column of Figure 6 shows that, despite calibration-dependent zero-points, the overall rMZR morphology is broadly stable across all prescriptions: Z_{gas} increases steeply with Σ_* at low surface densities and flattens toward a saturation-like regime at high Σ_* , in line with previous resolved surveys. However, the secondary ordering with Σ_{SFR} is not equally coherent for every diagnostic. For both O3N2-based calibrations, the Σ_{SFR} -stratified contours display a clear and systematic reversal with Σ_* : at low Σ_* , higher Σ_{SFR} preferentially occupies lower Z_{gas} , while at high Σ_* the ordering flips such that higher Σ_{SFR} corresponds to higher Z_{gas} . This behaviour is reflected in the stacked Spearman trends (third column), which show a coherent transition from $\rho < 0$ at low Σ_* to $\rho > 0$ at intermediate/high Σ_* for both O3N2 prescriptions (with galaxy-to-galaxy scatter but a consistent global trend). By contrast, the D16 and Scal-PG16 prescriptions exhibit substantially weaker and more heterogeneous Σ_{SFR} stratification in the H II-only sample. The D16 stacked ρ trend remains mildly positive over most of the Σ_* range; while Scal-PG16 shows a stronger dependence on Σ_* in the stacked statistic, but with opposite-sign behaviour compared to O3N2-M13 and O3N2-C20. Therefore, while the rMZR curvature itself is stable, the inferred $Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ coupling (and any apparent sign change) is diagnostic-dependent.

The comparison also reveals systematic differences between calibration families. In our sample, indicators tied more closely to T_e -based anchors (e.g. Scal-PG16 and the O3N2 calibrations) show tighter internal sequences with metallicity dispersions of $\sim 0.1\text{--}0.15$ dex at fixed Σ_* and Σ_{SFR} , whereas the model-based D16 prescription produces visibly larger scatter ($\sim 0.2\text{--}0.25$ dex) and spans a broader metallicity range ($\sim 0.4\text{--}0.5$ dex wider) across the same parameter space. This behaviour is consistent with PHANGS–MUSE (Kreckel et al. 2019), which shows that calibrations tied to photoionization models or to a small number of line ratios are more susceptible to local variations in ionization parameter and abundance ratios, leading to increased spaxel-to-spaxel dispersion and systematic offsets relative to T_e -anchored scales. Within the O3N2 family itself, the two prescriptions are highly consistent in their trends; the main difference is a nearly uniform offset, with O3N2-C20 yielding metallicities systematically higher than O3N2-M13 by $\sim 0.1\text{--}0.2$ dex.

While all calibrations recover an inversion, they do not yield identical qualitative behaviour in detail. The D16 Spearman trends are weighted toward positive ρ over most of the Σ_* range, mirroring the predominantly positive resolved correlation reported by WL21 on ~ 100 pc scales. This suggests that the *choice of calibration*

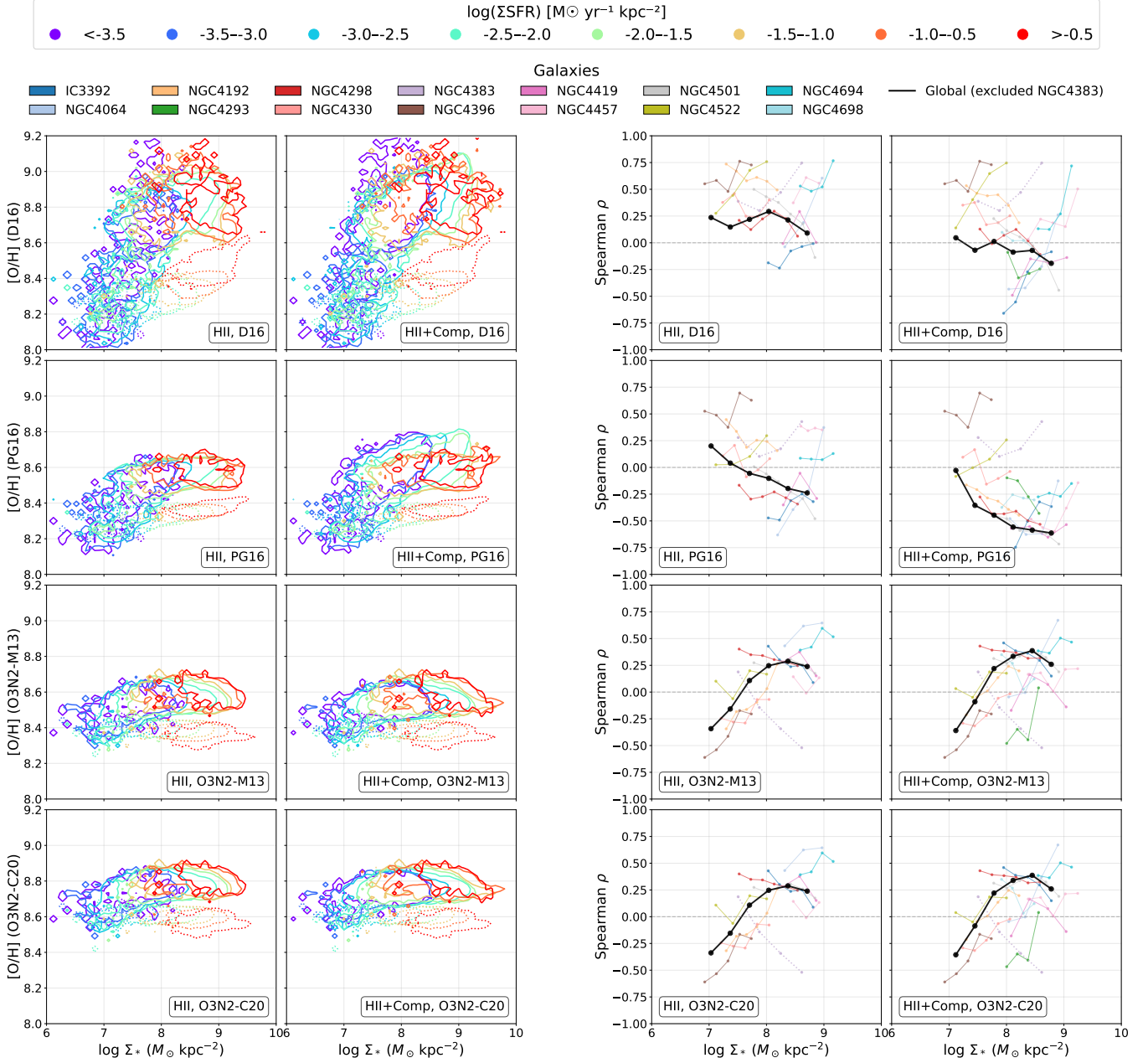


Figure 6. Comparison of the rMZR (left two columns) and the mass-dependent Spearman rank correlation coefficient ρ between $\Delta[\text{O}/\text{H}]$ and $\Delta \log_{10} \Sigma_{\text{SFR}}$ (right two columns) for four gas-phase metallicity calibrations: D16 (top row), Scal-PG16 (second row), O3N2-M13 (third row), and O3N2-C20 (bottom row). Columns 1 and 3 use only **HII**-classified star-forming spaxels, whereas columns 2 and 4 use the more inclusive **HII+Comp** selection (see the BPT demarcation in Figure 1). The rMZR panels are colour-coded by $\log_{10} \Sigma_{\text{SFR}}$ as in Figure 3. The Spearman panels show $\rho(\Sigma_*)$ measured in discrete Σ_* bins for individual galaxies (coloured curves) and for the stacked sample (black curve; excluding NGC 4383), following Figure 5. The parameter spaces occupied by NGC 4383 are indicated by dotted contours/curves in each corresponding panel.

can influence whether a given data set appears dominated by correlated or anti-correlated regimes, even when the underlying spaxel population is the same. Moreover, at the high- Σ_* end the O3N2 prescriptions favour a positive residual coupling, whereas D16 and Scal-PG16 lean toward weaker or even negative behaviour. Similar calibration-dependent qualitative differences have been noted in other MUSE-based surveys, where O3N2- and N2S2-based prescriptions can yield different resolved slopes from S- or N2S2-based prescriptions (e.g. Kreckel et al. 2019). We therefore caution that the detailed

mass dependence of the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation can diminish or even reverse under different metallicity calibrations.

We adopt the O3N2 indicator as our primary metallicity diagnostic for several reasons. First, both the O3N2-M13 and O3N2-C20 calibrations are derived from large IFS datasets (CALIFA and MaNGA, respectively), making them particularly well-suited for spatially-resolved studies like ours. Second, these calibrations are anchored to T_e -based abundance measurements, providing a more direct physical foundation. Third, O3N2 prescriptions show tighter internal sequences in our sample compared to some alternative di-

agnostics (see Figure 6). Importantly, results by O3N2 indicators clearly delineates a key observational feature: a negative $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation at the low-mass end transitioning to a positive correlation at the high-mass end. In our resolved sample, this inversion occurs at $\log_{10}(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.5 - 8.0$. This resolved trend mirrors the global, integrated picture established by Yates et al. (2012), whose observations derived by both strong line diagnostics and grids of photoionization models (see their Figure 1) reveal a similar reversal, negative at low masses and positive at high masses, with a transition point around a characteristic mass of $\log_{10}(M_*/M_\odot) \approx 10.0 - 10.5$. Similarly, Curti et al. (2020) also present such reversal of $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation at the high mass end in their Figure 6 and 7.

5.2 Sensitivity to the Star-forming Spaxel Definition

Our baseline analysis relies on a conservative **HII**-only mask, in which spaxels are required to fall within the **HII** locus of both the [N II] and [S II] BPT diagrams with robust line detections and uncertainties that remain fully inside the same excitation class. This stringent selection is designed to minimise contamination from shocks, AGN-like excitation, and diffuse ionised gas (DIG), all of which can bias strong-line metallicity indicators and $H\alpha$ -based Σ_{SFR} estimates. However, MAUVE-MUSE also provides a more inclusive “star-forming” mask that merges **HII** and composite regions (**HII+Comp**), motivated by the expectation that composite spaxels are still largely powered by O/B stars’ photonization but may include a non-negligible contribution from DIG or harder radiation fields. We therefore repeat the full rMZR and offset-correlation analysis using **HII+Comp** spaxels to assess the sensitivity of our conclusions to excitation masking.

The comparison is summarised in Figure 6 as well, where columns 1 and 3 use the conservative **HII** selection, and columns 2 and 4 adopt **HII+Comp**. Including composite spaxels expands the sampled parameter space, especially toward high Σ_* where inner discs and circumnuclear regions are more likely to host mixed excitation and more complex physics. For the two O3N2-based prescriptions (O3N2-M13 and O3N2-C20), both the rMZR colour ordering and the mass-dependent Spearman trends are essentially unchanged when composites are added. The stacked $\rho(\Sigma_*)$ curves retain the same negative-to-positive transition at $\log_{10}(\Sigma_*/M_\odot \text{ kpc}^{-2}) \sim 7.5-8.0$, and the scatter around the media rMZR locus increases only marginally. This stability indicates that our main result is not driven by a narrow excitation-class definition, and that O3N2 indicators are comparatively resilient to moderate DIG/composite admixture in the MAUVE-MUSE data.

By contrast, the N2S2- and S-calibrations show a stronger response to including composites. In particular, the Scal-PG16 rMZR exhibits a noticeable increase in scatter at the high- Σ_* end under the **HII+Comp** mask, consistent with the known sensitivity of S2- and N2-based ratios to harder ionizing spectra and DIG-enhanced low-ionization lines. The corresponding $\rho(\Sigma_*)$ trends also shift: for D16 the stacked Spearman curve becomes dominated by weak or negative correlations over a broader Σ_* range, and some individual galaxies show reduced significance or a delayed zero-crossing; while for Scal-PG16 the stacked Spearman trend become completely negative without no inversion point anymore. These changes underline that the N2S2- and S-calibrations (i.e., D16 and Scal-PG16 here) leveraging [N II] and [S II] can be more susceptible to excitation mixing, and they reinforce the need to treat excitation masking and calibration choice as coupled systematics when interpreting resolved $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relations.

Crucially, despite these quantitative shifts, the qualitative inversion of the residual $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling persists in the stacked sample for

both O3N2-M13 and O3N2-C20. The low- Σ_* anti-correlation branch and the high- Σ_* positive branch remain present, demonstrating that the sign flip is not an artefact of excluding composite spaxels when using O3N2 prescriptions.

5.3 NGC4383 – the Metallicity Outlier

NGC 4383 is a clear metallicity outlier within the MAUVE-MUSE sample. To illustrate its behaviour without allowing it to bias ensemble trends, we reintroduce its spaxels in Figure 6 as dotted contours (rMZR panels) and dotted light-mauve curves (Spearman panels), while keeping the stacked Spearman trend (black curve) computed from the other 13 galaxies only. From the rMZR comparison (first two columns), NGC 4383 remains systematically offset to lower Z_{gas} at fixed Σ_* by at least 0.2 dex under all four metallicity calibrations. The persistence of this depression across prescriptions and excitation selections indicates that it is not produced by calibration-specific systematics or by the choice of star-forming mask, motivating our designation of NGC 4383 as a metallicity outlier.

The Spearman trend of NGC 4383 adds a second, more striking peculiarity. Unlike the other galaxies, NGC 4383 shows a mass-dependent correlation pattern that is nearly the mirror image of the stacked relation: wherever the combined MAUVE-MUSE behaviour is positive (negative), the NGC 4383 curve tends to be negative (positive), especially in O3N2 indicators. In other words, the sign of the residual $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling in NGC 4383 appears to reverse relative to the dominant local regime traced by the rest of the sample. If included in the stack, this anti-aligned behaviour would disproportionately populate the low- Z_{gas} tail and partially cancel the mass-ordered trend, artificially steepening the low- Σ_* rMZR branch and shifting $\rho(\Sigma_*)$ toward the non-correlation case. We therefore exclude NGC 4383 from all stacked rMZR and correlation measurements in the main text, and investigate it separately here.

A plausible physical explanation for both its depressed metallicity scale and its opposite-sign residual coupling is unusually efficient removal of enriched gas. NGC 4383 has been identified as a system with strong feedback activity in the MAUVE-MUSE context, and intense metal-rich outflows could lower Z_{gas} at fixed Σ_* while also perturbing the local balance between enrichment and gas supply that sets the sign of the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation. Alternatively (or in addition), the anomaly could reflect a recent gas-accretion or re-accretion event: Chung et al. (2009) reported kinematic anomalies on the eastern side of the H I disc and suggested recent accretion. In our framework, the inflow of relatively metal-poor gas would naturally dilute Z_{gas} at fixed Σ_* and can drive an opposite-sign residual coupling by enhancing gas-supply variability relative to enrichment. In this sense, NGC 4383 provides a valuable pathological case for testing the limits of the resolved regulator picture and for isolating environment- or feedback-driven departures from the population-average behaviour. A detailed discussion of its ionized-gas properties and environmental context is presented by Watts et al. (2024), and we defer a full physical interpretation to future work.

6 Conclusions

In this paper, after removing the metallicity outlier (NGC 4383), we have used 13 Virgo spiral galaxies from the MAUVE-MUSE internal release to revisit the resolved mass-metallicity relation and its secondary dependence on star formation rate surface density on ~ 100 pc scales. By combining high-quality IFS data with a spatially-resolved regulator framework, we have characterised where and why

the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation (and also its residual coupling) changes sign. Our main conclusions are as follows.

(i) Our MAUVE–MUSE galaxies follow a well-defined resolved mass–metallicity relation that is consistent with previous IFS surveys. Using the [O3N2-M13](#) calibration as our fiducial metallicity scale, the rMZR rises steeply at low Σ_* and flattens toward high Σ_* , in line with earlier work on kpc-scale data. Colour-coding by Σ_{SFR} reveals a clear secondary dependence: at fixed Σ_* , low- Σ_* regions tend to show lower Z_{gas} at higher Σ_{SFR} , whereas high- Σ_* regions show the opposite ordering (see [Figure 3](#)).

(ii) After removing the primary rMZR and rSFMS trends, the residual coupling between metallicity and SFR surface density exhibits a robust mass-dependent inversion. The Spearman rank correlation coefficient between $\Delta[\text{O}/\text{H}]$ and $\Delta \log_{10} \Sigma_{\text{SFR}}$ transitions smoothly from $\rho < 0$ at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \lesssim 7.5$ to $\rho > 0$ at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \gtrsim 8.0$, with a zero-crossing in between (see [Figure 5](#)). Notably, NGC 4192 spans a broad Σ_* range that straddles the inversion point and exhibits a clear internal sign reversal, rather than simply populating one branch or the other. This suggests that the transition is driven by an intrinsic physical mechanism operating within individual galaxy, rather than arising as an artifact of stacking different galaxy populations. Local Moran-like correlation maps further show that, once Σ_* trends are removed, correlated and anti-correlated H II regions can coexist within the same galaxy, and the familiar mass-dependent behaviour emerges only as an ensemble property.

(iii) To interpret these patterns, we formulated a spatially-resolved gas–regulator model in surface densities and identified an “instantaneous target metallicity” $Z^\dagger$ that drives the evolution of the real-time metallicity Z . Analysis around metallicity extrema and toward/away from them leads to a simple condition (see [Equation 31](#)): at fixed Σ_* , the sign of the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (anti-)correlation is controlled by the relative fluctuation amplitudes of star formation and gas supply. Positive correlations arise where SFR- or SFE-driven variability dominates (short, strongly varying depletion times at nearly steady gas supply), while negative correlations arise where gas-supply variability dominates and dilution drives Z_{gas} opposite to Σ_{SFR} . Our MAUVE–MUSE trends are therefore naturally understood as the outcome of a competition between SFR-driven and GSR-driven fluctuations that varies systematically with Σ_* .

(iv) We argued that the same regulator physics can be extrapolated from spatially-resolved to integrated galaxy-wide relations ([Section 4.4](#)). Replacing surface densities by their global analogues, [Equation 31](#) implies that the sign of the integrated $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ dependence at fixed M_* is likewise set by whether fluctuations in SFR (or SFE) outweigh those in the inflow rate. In this view, the negative SFR dependence of metallicity commonly seen at low stellar masses and the weaker or reversed dependence at the high-mass end in global FMR analyses are two manifestations of the same underlying competition between depletion and inflow timescales. The apparent prevalence of anti-correlation, and the weaker and less clear direct correlation signal on global scales than on local scales, can be understood as a consequence of parameter-space sampling and spatial averaging: many surveys predominantly probe galaxies or spaxels on the low-mass, inflow-dominated side of the inversion and then integrate over regions with mixed local behaviour.

(v) Finally, we tested the robustness of these conclusions against three key systematics using [Figure 6](#). First, adopting four widely used metallicity diagnostics ([D16](#), [Scal-PG16](#), [O3N2-M13](#), and [O3N2-C20](#)), we find that the overall rMZR curvature is stable across all calibrations. However, the mass-dependent sign flip in the residual $Z_{\text{gas}}-$

Σ_{SFR} coupling remains tension according to the choice of diagnostic. The O3N2-based indicators ([O3N2–O3N2-M13](#) and [O3N2–O3N2-C20](#)) show the clearest and most internally consistent transition from anti-correlation at low Σ_* to correlation at high Σ_* . By contrast, the [Scal-PG16](#) prescription exhibits a completely opposite trend, while the [D16](#) calibration remains predominantly positively correlated over the full Σ_* range. We therefore conclude that, although the resolved metallicity scaling relations are qualitatively similar across diagnostics, the presence and sign of the inferred $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ inversion are calibration-dependent. Second, relaxing the star-forming mask from a conservative **HII**-only selection to an inclusive **HII+Comp** mask increases the scatter and modulates the detailed $\rho(\Sigma_*)$ trends, with the strongest changes occurring for the N2S2- and S-based diagnostics ([D16](#) and [Scal-PG16](#)). For the O3N2-based calibrations ([O3N2-M13](#) and [O3N2-C20](#)), the mass-dependent inversion remains identifiable. In contrast, under the [Scal-PG16](#) prescription the coupling becomes predominantly negative across the full Σ_* range, while [D16](#) shows no clear monotonic correlation, with the stacking Spearman trend fluctuating around zero. Third, we showed that NGC 4383 is a genuine metallicity outlier: it lies ~ 0.1 – 0.2 dex below the rMZR locus and exhibits an almost mirror-image residual correlation pattern relative to the stack ([Section 5.3](#)).

In general, these results suggest that the spatially-resolved $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation is not governed by a single universal trend, but by a balance between star-formation-driven and gas-supply-driven variability that depends on local mass surface density, environment, and feedback. Our MAUVE–MUSE data show that both correlated and anti-correlated regimes naturally coexist within the same discs at ~ 100 pc resolution, and that the familiar mass-dependent inversion emerges as a population-level imprint of this underlying competition. Future work combining MAUVE–MUSE data with resolved H I and CO observations, as well as higher-redshift IFS surveys will be crucial for constraining Σ_ϕ and its timescale more directly, and for testing whether the inversion scales identified here are universal or themselves evolve with redshift, environment, and galaxy assembly history.

Software

This research made use of [astropy](#) ([Astropy Collaboration et al. 2022](#), <https://www.astropy.org/>), [numpy](#) ([Harris et al. 2020](#), <https://numpy.org/>), [matplotlib](#) ([Hunter 2007](#), <https://matplotlib.org/>), [scipy](#) ([Virtanen et al. 2020](#), <https://scipy.org/>), [speclite](#) ([Kirkby et al. 2023](#), <https://speclite.readthedocs.io/en/latest/>) and [nGIST](#) ([Fraser-McKelvie et al. 2025](#), <https://geckos-survey.github.io/gist-documentation/>).

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Data Availability

The full set of MAUVE–MUSE datacubes and associated nGIST value-added products will be released as part of the main public data release once the survey is completed (anticipated in 2027). Requests for access prior to the public release should be directed to the corresponding author. Readers interested in joining the MAUVE collaboration and accessing the full dataset are encouraged to consult the [team](#) page on the MAUVE survey website. The analysis pipeline used in this work is publicly available at https://github.com/Rongjun-ANU/MAUVE_MUSE_Zgas_SigmaSFR.

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MAUVE–MUSE: When Metallicity Follows or Fights Star Formation—A Mass-Dependent Inversion in Virgo Galaxies

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ABSTRACT

We use 13 Virgo spiral galaxies from the MAUVE–MUSE survey to revisit the resolved mass–metallicity relation (rMZR) and its secondary dependence on star formation rate (SFR) surface density (Σ_{SFR}) at ~ 100 pc resolution. At this resolution, we construct co-spatial maps of stellar mass surface density (Σ_*), star formation rate surface density, gas-phase oxygen abundance and verify our results across several strong-line metallicity diagnostics. The MAUVE–MUSE galaxies follow a standard rMZR, but once binned in Σ_* we find a clear mass-dependent inversion in the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation: at fixed Σ_* , low- Σ_* regions show an anti-correlation between Z_{gas} and Σ_{SFR} , whereas high- Σ_* regions show a positive correlation, with a zero-crossing at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \simeq 7.5\text{--}8.0$. We find that both correlated and anti-correlated H II regions coexist within the same discs, and that the observed mass dependence emerges only as an ensemble trend. To interpret this behaviour, we develop a spatially-resolved gas–regulator model and show that the observed correlation and anti-correlation between Z_{gas} and Σ_{SFR} arise from the competition between star-formation-driven and gas-supply-driven variability: the sign of the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation is set by which of these dominates. This framework can be naturally extrapolated to the integrated scenario, providing a unified explanation for both resolved and global $Z_{\text{gas}}-\text{SFR}$ relation. Finally, we find that the strength of the inversion is diagnostic-dependent and can weaken, or even disappear, for some metallicity prescriptions.

Key words: galaxies: abundances — galaxies: ISM — galaxies: star formation — galaxies: evolution — ISM: abundances

1 Introduction

The global gas-phase mass–metallicity relation (MZR), well-established from previous works (e.g. [Lequeux et al. 1979](#); [Tremonti et al. 2004](#); [Erb et al. 2006](#); [Maiolino et al. 2008](#); [Zahid et al. 2014](#); [Curti et al. 2020](#); [Sanders et al. 2021](#)), demonstrates that the gas-phase oxygen abundance increases with galaxy stellar mass (M_*), such that more massive galaxies host a more metal-enriched interstellar medium (ISM). While this global relation provides an essential benchmark for galaxy chemical evolution, the advent of wide-field integral-field spectroscopy (IFS) has shifted attention toward understanding how metallicity is regulated within galaxies. By resolving discs into individual star-forming spaxels, IFS surveys now enable direct measurement of local scaling relations between gas-phase oxygen abundance and local stellar and star-formation properties.

The Calar Alto Legacy Integral Field Area (CALIFA) survey ([Sánchez et al. 2013, 2017](#); [Cresci et al. 2019](#)), revealed that the global MZR is reflected locally as a tight correlation between metallicity and stellar mass surface density (Σ_*). This resolved mass–metallicity relation (rMZR) was subsequently confirmed and expanded by Mapping Nearby Galaxies at APO (MaNGA [Barrera-Ballesteros et al. 2016](#); [Baker et al. 2023](#)) and the Sydney-AAO Multi-object Integral-field spectrograph (SAMI [Sánchez et al. 2019](#)) surveys, establishing the rMZR as a ubiquitous feature of star-

forming discs on kpc scales in the local Universe. More recently, IFS surveys based on Very Large Telescope/Multi-Unit Spectroscopic Explorer (VLT/MUSE) have extended these galactic resolved scaling relations to intermediate redshift while retaining comparable spatial resolution. In particular, the MUSE–Wide survey demonstrated the existence of a rMZR at $z \sim 0.26$, with a shape similar to that at $z \sim 0$ but a modestly lower normalization, consistent with chemical evolution over cosmic time ([Yao et al. 2022](#)). Likewise, the Middle Ages Galaxy Properties with Integral Field Spectroscopy (MAGPI) survey confirmed a rMZR in galaxies at $z \sim 0.3$ on a similar (kpc-scale) spatial resolution ([Koller et al. 2024](#)). Taken together, these IFS studies indicate that gas-phase oxygen abundance correlates more tightly with local stellar mass surface density than with galactocentric radius alone: at fixed radius, spaxels with higher Σ_* tend to be more metal rich, and a similar Σ_*-Z_{gas} relation is recovered in galaxies with very different sizes and radial profiles. In this sense, the rMZR appears to reflect genuinely local physics, even though it is closely linked to the underlying radial structure of galactic discs.

On global, galaxy-integrated scales, there is now broad consensus that gas-phase metallicity depends on both stellar mass and star formation rate (SFR): the classical MZR is usually interpreted as a projection of a more fundamental $M_*-Z_{\text{gas}}-\text{SFR}$ surface, often referred to as the Fundamental Metallicity Relation (FMR; e.g., [Ellison et al. 2008](#); [Mannucci et al. 2010](#); [Lara-López et al. 2010](#); [Yates et al. 2012](#); [Bothwell et al. 2013](#); [Peng & Maiolino 2014](#); [Bothwell et al. 2016](#); [Gao et al. 2018](#); [Cresci et al. 2019](#); [Curti et al. 2020](#); [Pallot-](#)

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ini et al. 2025). In most of these studies, galaxies with higher SFR at fixed M_* tend to be more metal-poor, i.e. Z_{gas} and SFR are anti-correlated, a behaviour often interpreted as the signature of metal-poor gas accretion diluting the ISM. This naturally raises the question of whether an analogous three-parameter relation exists on resolved scales, i.e. whether the rMZR carries a meaningful secondary dependence on local star formation rate surface density (Σ_{SFR}). Several works report that once the primary Σ_* dependence is removed, residual trends with Σ_{SFR} are weak or absent, suggesting that local metallicity primarily traces longer-term mass assembly and recycling (e.g. Sánchez et al. 2013; Yao et al. 2022). However, recent studies find statistically significant $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (anti-)correlation and motivate a spatially-resolved analogue of the Fundamental Metallicity Relation (i.e., rFMR), wherein metallicity at fixed Σ_* depends on Σ_{SFR} (Baker et al. 2023; Koller et al. 2024). Additionally, MAGPI data present a clear signal that the *sign* of this correlation can reverse with Σ_* : low- Σ_* regions show an anti-correlation between Z_{gas} and Σ_{SFR} , while high- Σ_* regions exhibit a flat or positive relation (Koller et al. 2024). Such global and resolved studies therefore suggest that any FMR- or rFMR-like behaviour might not be universal but instead depend on local physical regime, calibration choice, on whether variability in metallicity is dominated by star-formation-driven enrichment or by gas-supply-driven dilution.

Interpreting such mass-dependent sign changes requires a framework that links metallicity variations to the gas–star formation cycle. A widely used description is provided by “gas–regulator” or “bathtub” models, in which the gas content and metal budget of a galaxy (or region) are governed by the balance between gas inflow, star formation, and outflows (e.g. Davé et al. 2012; Lilly et al. 2013; Peng & Maiolino 2014; Forbes et al. 2014; Belfiore et al. 2019; Sharda et al. 2021). In their galaxy-integrated form, these models show that the gas-phase metallicity tends toward an effective equilibrium set by the metal yield, the metallicity of the accreted gas, and the interplay between star formation and inflow rate, naturally reproducing both the shape of the global MZR and its commonly observed anti-correlation with SFR. A more explicit connection between time-dependent star formation and metallicity has been established by Wang & Lilly (2021, hereafter WL21), who treated individual “gas–regulator units” (from galaxy-wide down to ~ 100 pc scales) as reservoirs subject to time-variable inflow rate, and star formation efficiency (SFE). Within this framework, the residual correlation between metallicity and SFR at fixed stellar mass (or potential) is set by which source of variability dominates: when fluctuations in SFE drive changes in the SFR at roughly fixed inflow rate, metallicity and SFR are predicted to vary in the same sense, whereas strongly time-variable inflow at nearly constant SFE produces metal dilution and thus an anti-correlation between metallicity and SFR. However, testing such analytical frameworks on local scales requires spatially-resolved data that simultaneously (i) resolve individual H II regions on ~ 100 pc scales and (ii) span a broad dynamic range in Σ_* , so that both SFR-driven and inflow-driven regimes can be probed within and across galaxies.

In this work, we use the first 14 Virgo-cluster spirals observed as part of the Multiphase Astrophysics to Unveil the Virgo Environment (MAUVE) survey (Catinella et al. 2025) to revisit the rMZR and its secondary dependence on Σ_{SFR} at ~ 100 pc resolution. The cluster environment provides an ideal laboratory: outer discs are susceptible to stripping and environmentally driven gas-supply perturbations, while inner discs probe dense, metal-rich regimes where enrichment may dominate. We show that the MAUVE–MUSE sample exhibits a robust mass-dependent inversion in both $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation and residual coupling ($\Delta Z_{\text{gas}}-\Delta \Sigma_{\text{SFR}}$), consistent with, but extending be-

yond, previous findings. To characterise the spatial heterogeneity of this coupling, we introduce a local bivariate Moran-like correlation index (Moran 1950; Anselin 1995) that maps where star-formation and metallicity offsets are locally concordant or discordant. Finally, we develop a spatially-resolved regulator model and interpret these patterns in our framework. The results demonstrate that the observed mass-dependent sign flip in the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation is governed by the relative amplitude of local star-formation versus gas-supply variability.

This paper is organized as follows: Section 2 introduces the data and methods used in this study, Section 3 demonstrates our results, Section 4 presents the spatially-resolved framework, its interpretation and the extrapolation to a global integrated scenario, Section 5 discusses the robustness of our findings across different metallicity calibrations, star-forming spaxel selections, and the special case of the metallicity outlier NGC 4383, and Section 6 outlines our conclusions. Throughout this work, we adopt a distance of 16.5 Mpc for Virgo Cluster members (Mei et al. 2007), aligning with previous and ongoing MAUVE studies.

2 Data & Methods

2.1 MAUVE–MUSE Data

The data analysed in this work are drawn from the internal release v2 (December 2024) of MAUVE–MUSE survey. Designed to investigate how the cluster environment shapes the gas–star formation cycle, MAUVE targets 40 late-type Virgo Cluster galaxies using a suite of multi-wavelength facilities (Catinella et al. 2025). For each target, we use the fully reduced MUSE datacubes produced with pymusepipe v2.28.2 and esorex 3.13.6, following closely the reduction, sky subtraction, and flux-calibration procedures described by Emsellem et al. (2022). Each galaxy is delivered as a science-ready, fully-reduced FITS file whose *DATA*, *STAT*, and *DQ* extensions contain the flux, variance, and data-quality information, respectively, on the native $0''.2$ pixel grid over the full 4750–9350 Å spectral range of VLT/MUSE.

All physical quantities used in this paper are derived from these cubes using the nGIST pipeline (Fraser-McKelvie et al. 2025), a pPXF-based (Cappellari & Emsellem 2004; Cappellari 2016, 2023) wrapper optimised for MAUVE–MUSE data. Briefly, we first construct an adaptively binned representation of each cube by applying the Voronoi algorithm of Cappellari & Emsellem (2004) to the stellar continuum in the 4800–7000 Å window, targeting a minimum continuum signal-to-noise ratio of $S/N \geq 40$ per bin. The resulting Voronoi tessellation is then adopted as the common spatial grid for all nGIST products. This approach ensures strict one-to-one correspondence between all the following derived properties without any additional resampling or smoothing. The MAUVE–MUSE observations have a typical seeing full width at half maximum (FWHM) of $\sim 1''$, corresponding to a physical scale of ~ 80 pc, so that the effective angular resolution is already set by the point spread function (PSF) rather than by the MUSE sampling pixel scale of $0''.2 \text{ pixel}^{-1}$. The $S/N \geq 40$ Voronoi binning on the stellar continuum typically groups a few up to several tens of spaxels per bin with 95% containing fewer than 40 spaxels, yielding characteristic bin sizes comparable to the seeing footprint and hence also of order ~ 100 pc. Throughout this work, we therefore refer to our spatially-resolved measurements as probing ~ 100 pc scales.

On this common grid, the stellar continuum and kinematics in each bin are modelled with pPXF using MILES single-stellar-population

(SSP) templates (Vazdekis et al. 2010) convolved to match the MUSE line-spread function (LSF). Multiplicative Legendre polynomials are included to absorb residual calibration mismatches. After subtracting the best-fitting stellar model, Gaussian profiles are fitted simultaneously to the main nebular emission lines (e.g. H α , H β , [O III], [N II], [S II]), yielding maps of observed line fluxes, velocities, and velocity dispersions with associated uncertainties.

While MAUVE–MUSE observations are still in progress, a total of 14 galaxies have been fully observed, reduced and therefore analyzed in this paper: IC 3392, NGC 4064, NGC 4192, NGC 4293, NGC 4298, NGC 4330, NGC 4383, NGC 4396, NGC 4419, NGC 4457, NGC 4501, NGC 4522, NGC 4694, and NGC 4698.

2.2 Stellar Mass Surface Density

We first derive spatially-resolved stellar mass surface densities from the MAUVE–MUSE datacubes and the associated nGIST value-added products. For each galaxy, we start from the nGIST star-formation-history (SFH) products, which provide, for every Voronoi bin, a vector of SFH weights and an accompanying grid of stellar population parameters sampled in stellar metallicity and age, ([M/H], $\log_{10}(t)$). We then implement photometric properties for MILES-based SSPs (Vazdekis et al. 2010) on BASTI isochrone grids (Pietrinferni et al. 2004) with a Chabrier (2003, hereafter Chabrier) initial mass function (IMF). Restricting to the Chabrier-IMF entries, we construct a lookup table of R -band mass-to-light ratios $(M/L)_R$ as a function of SSP metallicity and age, ([M/H], $\log_{10}(t)$), by rounding the age and metallicity of each grid point to a uniform grid and mapping to the corresponding $(M/L)_R$ in the table. The SFH weights for each Voronoi bin are then combined with this $(M/L)_R$ grid to compute a light-weighted R -band mass-to-light ratio,

$$(M/L)_R^{\text{bin}} = \sum_k w_k (M/L)_{R,k}, \quad (1)$$

where w_k and $(M/L)_{R,k}$ are the SFH weight and R -band mass-to-light ratio of the k -th SSP component in the bin. This procedure yields a map of $(M/L)_R^{\text{bin}}$ defined over all Voronoi zones.

To obtain the corresponding R -band luminosity per bin, for each spaxel in the DATA extension, we construct the observed spectral energy distribution F_λ and convolve it with the `bessell-R` filter response using the `specLite` package (Kirkby et al. 2023). This yields an AB R -band magnitude m_R and flux (in mJy) per spaxel. We then collapse these spaxel-level fluxes onto the Voronoi tessellation by averaging, for each BINID, only over spaxels assigned to that bin, while preserving the survey’s native masking of invalid pixels. The resulting bin-averaged AB magnitudes are corrected for Galactic extinction using the nGIST $E(B - V)$ map and a Calzetti et al. (2000, hereafter Calzetti) attenuation curve. Corrected apparent magnitudes (m_R^{corr}) are converted into absolute magnitudes via the distance modulus corresponding to the adopted Virgo-cluster distance of 16.5 Mpc, and hence into R -band luminosities (L_R) in solar units using the solar R -band absolute magnitude, $M_{R,\odot} = 4.61$ (Willmer 2018).

The stellar mass in each Voronoi bin is then obtained as

$$M_*^{\text{bin}} = L_R^{\text{bin}} \times (M/L)_R^{\text{bin}}. \quad (2)$$

Dividing M_*^{bin} by the corresponding projected bin area yields a map of projected stellar mass surface density, Σ_* (in $M_\odot \text{ kpc}^{-2}$).

Finally, we account for the inclination of each disc, adopting inclination angles from the Virgo Environment Traced in CO (VERTICO) Survey (Brown et al. 2021). Assuming an oblate disc geometry with intrinsic axial ratio $q_0 = 0.2$, we compute the inclination

correction factor for each galaxy:

$$\frac{b}{a} = \sqrt{(1 - q_0^2) \cos^2 i + q_0^2}, \quad (3)$$

which encodes the observed flattening of a disc of finite thickness. The stellar mass surface density and the star formation rate surface density in the following Section 2.3 are then multiplied by this b/a factor to correct from the projected to the intrinsic face-on properties.

2.3 Star Formation Rate Surface Density

Nebular dust attenuation is quantified from the Balmer decrement measured on the same Voronoi grid as all other MAUVE–MUSE products. For each spaxel we start from the nGIST H α and H β flux and error maps and impose a basic quality-control cut requiring $F_{\text{line}}/\sigma_{\text{line}} \geq \text{cut}$ with $\text{cut} = 3$, together with a minimum flux threshold of $F_{\text{line}} \geq 20$ in the native MAUVE–MUSE units of $10^{-20} \text{ erg s}^{-1} \text{ cm}^{-2}$. Spaxels that fail this detection criterion for either Balmer line are retained for the construction of SFR maps, but their fluxes are set to $\max(\text{threshold}, F_{\text{line}})$ (effectively assuming zero extinction) to prevent artifacts such as unphysical extinction corrections. These low-S/N spaxels will be excluded from our star-forming spaxel selection later, so this choice has negligible impact on our results. The Balmer decrement is defined as $\text{BD} = F_{\text{H}\alpha}/F_{\text{H}\beta}$ (using observed fluxes) and we impose a floor at the Case B recombination value at $\text{BD} = 2.86$, appropriate for $T_e = 10^4 \text{ K}$ and $n_e = 100 \text{ cm}^{-3}$, i.e. any measured $\text{BD} < 2.86$ is set to 2.86 to avoid unphysical negative colour excesses. The corresponding colour excess is obtained using the analytic Calzetti attenuation curve:

$$E(B - V)_{\text{gas}} = \frac{2.5}{k(\text{H}\beta) - k(\text{H}\alpha)} \log_{10} \left(\frac{\text{BD}}{2.86} \right), \quad (4)$$

where $k(\lambda)$ is starburst reddening curve with the effective obscuration at V -band $R_V = 4.05$ and here k is evaluated at the wavelengths of H α and H β . All strong nebular lines entering the subsequent analysis are then dereddened with this $E(B - V)_{\text{gas}}$ via

$$F_\lambda^{\text{corr}} = F_\lambda 10^{0.4 E(B - V)_{\text{gas}} k(\lambda)}. \quad (5)$$

Using these dust-corrected fluxes we construct the standard [N II] and [S II] Baldwin et al. (1981, hereafter BPT) ratios, namely $F_{[\text{N II}]\lambda 6583}/F_{\text{H}\alpha}$, $F_{[\text{S II}]\lambda 6716+\lambda 6731}/F_{\text{H}\alpha}$, and $F_{[\text{O III}]\lambda 5007}/F_{\text{H}\beta}$, propagating flux-to-error ratios into the corresponding line-ratio uncertainties. Each spaxel is classified on both the [N II] and [S II] BPT diagrams according to the Kewley et al. (2006) demarcation curves, and we require that the central value and its $\pm 1\sigma$ error bars in $(\log_{10}[\text{line}/\text{H}\beta], \log_{10}[\text{line}/\text{H}\alpha])$ space all remain within the same BPT region. Only spaxels that satisfy this criterion and the S/N cuts in all lines entering a given diagram are assigned a secure class.

In order to strictly define star-forming regions, we adopt a conservative H II-region selection that is used consistently throughout the paper. Specifically, we define “star-forming” spaxels (hereafter **HII** spaxels) as those that lie in the H II regime of both [N II] and [S II] BPT diagrams and whose $\pm 1\sigma$ error bars also remain entirely within the H II regime on both diagrams, with robust detections in all lines entering the classification. This yields a clean sample of classical H II regions purely dominated by O/B stars’ photonization with **minimal contamination from AGN or LINER-like excitation**. All SFR and metallicity maps in this work are therefore computed exclusively on this **HII** mask, and all subsequent resolved analyses implicitly refer to these H II-classified star-forming regions. Across the sample, the

Line ratio	Definition
$N2$	$\frac{F_{[\text{N II}](\lambda 6548+\lambda 6584)}}{F_{\text{H}\beta}}$
$S2$	$\frac{F_{[\text{S II}](\lambda 6717+\lambda 6731)}}{F_{\text{H}\beta}}$
$R3$	$\frac{F_{[\text{O III}](\lambda 4959+\lambda 5007)}}{F_{\text{H}\beta}}$
$O3N2$	$\log_{10}\left(\frac{F_{[\text{O III}](\lambda 5007)/F_{\text{H}\beta}}}{F_{[\text{N II}](\lambda 6583)/F_{\text{H}\alpha}}}\right)$
$N2S2$	$\log_{10}\left(\frac{F_{[\text{N II}](\lambda 6583)}}{F_{[\text{S II}](\lambda 6716+\lambda 6731)}}\right)$
y_{D16}	$N2S2 + 0.264 \log_{10}\left(\frac{F_{[\text{N II}](\lambda 6583)}}{F_{\text{H}\alpha}}\right)$

Table 1. Definition of key emission-line ratios used in the extinction, SFR, and metallicity analysis, where all fluxes are dust-corrected. Here we adopt $F_{[\text{N II}](\lambda 6548+\lambda 6584)} = 1.34F_{[\text{N II}](\lambda 6584)}$ and $F_{[\text{O III}](\lambda 4959+\lambda 5007)} = 1.33F_{[\text{O III}](\lambda 5007)}$.

conservative **HII** selection retains only $\sim 26\%$ of the total valid spaxels used in this work (with substantial galaxy-to-galaxy variation). Additionally, in Section 5.2, we perform a robustness test by relaxing this conservative “star-forming” restriction and repeating the analysis with an expanded **HII+Comp** mask that includes spaxels classified as either **H II** and/or composite region on both **BPT** diagrams.

Star formation rate maps are constructed from the dust-corrected $\text{H}\alpha$ flux. After applying the Balmer-decrement attenuation correction described above, we then apply the Calzetti et al. (2007) calibration, rescaled to a Chabrier IMF by Kennicutt & Evans (2012), to obtain

$$\text{SFR} [M_{\odot} \text{ yr}^{-1}] = 4.98 \times 10^{-42} L_{\text{H}\alpha}^{\text{corr}} [\text{erg s}^{-1}]. \quad (6)$$

Similarly, we take the projected surface density of SFR and account for the inclination by Equation 3, obtaining corrected Σ_{SFR} .

2.4 Gas-phase Metallicity

Gas-phase oxygen abundances[†] are inferred from the same dust-corrected emission-line maps, restricted to the **HII** spaxels defined in Section 2.3. In practice, using only spaxels that meet the S/N and **BPT** requirements, we focus on a subset of four widely used strong-line prescriptions that are robustly applicable across the MAUVE–MUSE sample: the $N2S2$ – $\text{H}\alpha$ calibration of Dopita et al. (2016, hereafter **D16**), the S –calibration of Pilyugin & Grebel (2016, hereafter **Scal-PG16**), the $O3N2$ index calibration of Marino et al. (2013, hereafter **O3N2-M13**), and the $O3N2$ calibration of Curti et al. (2020, hereafter **O3N2-C20**). The basic line ratios entering these prescriptions are collected in Table 1. In practice, similar to Watts et al. (2024), we reconstruct the total $[\text{N II}] \lambda 6548, 6583$ and $[\text{O III}] \lambda 4959, 5007$ fluxes by scaling the measured $[\text{N II}] \lambda 6583$ and $[\text{O III}] \lambda 5007$ fluxes by fixed factors of 1.34 and 1.33, respectively. All strong-line calibrations are applied within the recommended range $7.63 \leq [\text{O}/\text{H}] \leq 9.23$ by Kewley et al. (2019).

The **D16** prescription combines the $N2S2$ and $[\text{N II}]/\text{H}\alpha$ ratios into a single abundance indicator. Using the shorthand in Table 1, we obtain the oxygen abundance as

$$[\text{O}/\text{H}]_{\text{D16}} = 8.77 + y_{\text{D16}} + 0.45(y_{\text{D16}} + 0.30)^5. \quad (7)$$

The **Scal-PG16** calibration uses the $N2$, $S2$, and $R3$ ratios listed in

Table 1. We evaluate the branch-dependent polynomial expressions given in Pilyugin & Grebel (2016):

$$[\text{O}/\text{H}]_{\text{PG16}} = \begin{cases} 8.424 + 0.030 \log_{10}(R3/S2) + 0.751 \log_{10}(N2) \\ \quad + [-0.349 + 0.182 \log_{10}(R3/S2) \\ \quad + 0.508 \log_{10}(N2)] \log_{10}(S2), \\ \log_{10}(N2) \geq -0.6; \\ 8.072 + 0.789 \log_{10}(R3/S2) + 0.726 \log_{10}(N2) \\ \quad + [1.069 - 0.170 \log_{10}(R3/S2) \\ \quad + 0.022 \log_{10}(N2)] \log_{10}(S2), \\ \log_{10}(N2) < -0.6. \end{cases} \quad (8)$$

For the **O3N2-M13** calibration, we adopt the $O3N2$ index defined in Table 1 and use the corresponding abundance relation:

$$[\text{O}/\text{H}]_{\text{O3N2-M13}} = 8.533 - 0.214 O3N2, \quad (9)$$

again restricted to the validity interval tabulated in that work.

Finally, we also adopt the $O3N2$ -based calibration of Curti et al. (2020), which uses the same $O3N2$ index but fits a higher-order polynomial to stacked Sloan Digital Sky Survey (SDSS) spectra (Abazajian et al. 2009). We evaluate the recommended **O3N2-C20** functional form spaxel-by-spaxel over the valid range:

$$O3N2 = 0.281 - 4.765x - 2.268x^2, \quad (10)$$

where $x \equiv [\text{O}/\text{H}]_{\text{O3N2-C20}} - 8.69$. In the course of this work, we also implemented the “combined” strategy advocated by Curti et al. (2020), in which all 6 actionable indicators in MUSE wavelength are selected on a spaxel-by-spaxel basis according to χ^2 minimization. While this approach yields self-consistent estimation with minimum uncertainties among available line diagnostics, we found that it can introduce indicator-dependent discontinuities within individual galaxies. To ensure internal consistency and straightforward interpretation of spatial trends, we therefore adopt the **O3N2-C20** calibration alone instead of this proposed “combined” approach.

All four metallicity prescriptions are evaluated exclusively on the conservative **HII** star-forming mask, on the same Voronoi grid as the stellar-mass and SFR surface-density maps. This guarantees that Σ_* , Σ_{SFR} , and $12 + \log_{10}(\text{O}/\text{H})$ are strictly co-spatial for every bin used in the resolved mass–metallicity–SFR analysis presented in Section 3. Throughout the analysis of this paper, to be consistent, we adopt **O3N2-M13** as the standard prescription. However, we acknowledge the different results caused by different calibrations, and we will therefore discuss this effect in Section 5.1.

2.5 Local Moran-like Correlation Index (I_i)

To quantify how two spatially-resolved fields co-vary in a genuinely local sense, we construct a “local Moran-like” product that combines the value of one property at a given position with the neighbourhood behaviour of a second property. This construction is inspired by the classical Moran’s I statistic for spatial autocorrelation (Moran 1950) and its generalization: local indicators of spatial association (Anselin 1995). Consider two generic maps $A(i)$ and $B(i)$ defined on the same set of valid spaxels i . We first compute the mean and standard deviation of each field over all valid locations, denoted (μ_A, σ_A) and (μ_B, σ_B) , and form standardised residuals

$$\begin{cases} z_A(i) = \frac{A(i) - \mu_A}{\sigma_A}, \\ z_B(i) = \frac{B(i) - \mu_B}{\sigma_B}. \end{cases} \quad (11)$$

[†] Throughout this paper, we use $[\text{O}/\text{H}] \equiv 12 + \log_{10}(\text{O}/\text{H})$ to denote the gas-phase oxygen abundance.

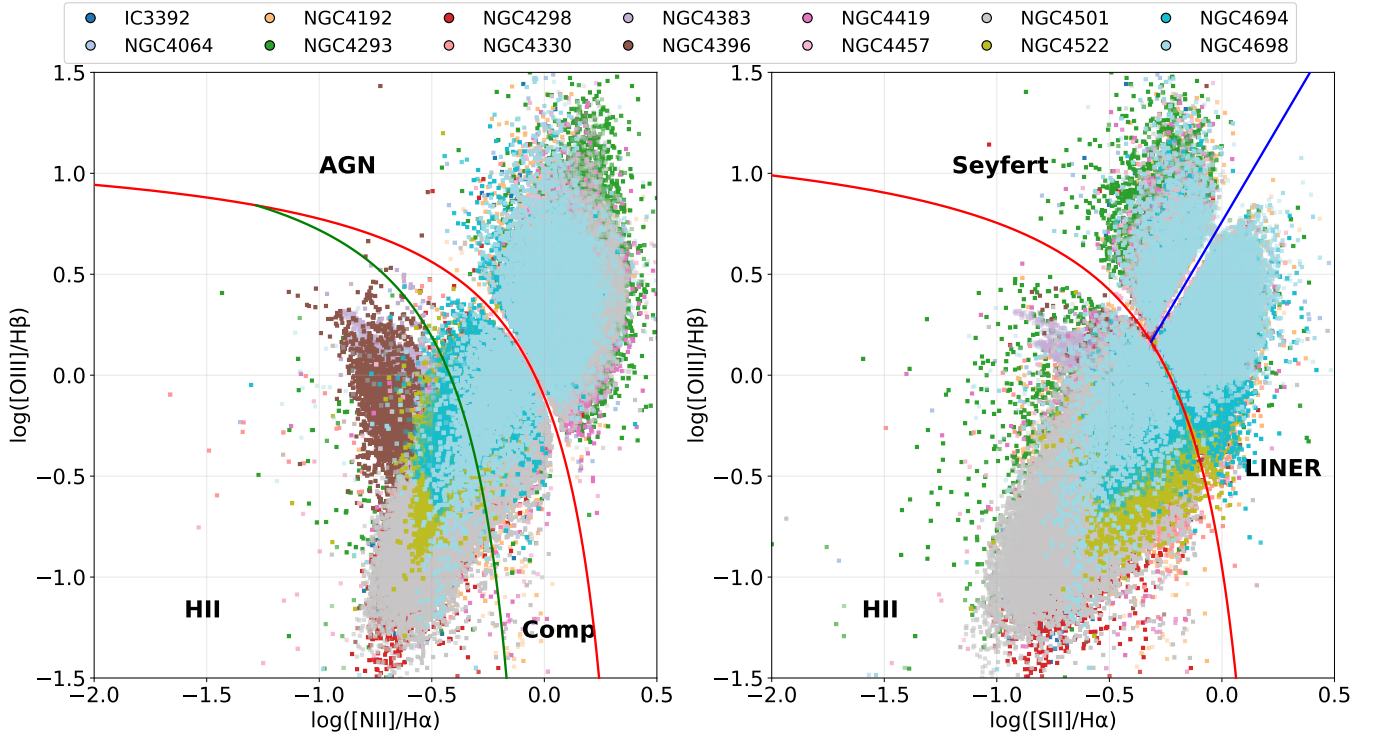


Figure 1. [N II] (left) and [S II] (right) BPT diagrams for “classified” spaxels of all 14 MAUVE–MUSE galaxies. Each point corresponds to a Voronoi spaxel classified using demarcation lines within 1σ confidence described in Section 2.3. Red and green curves show the Kewley et al. (2001) and Kauffmann et al. (2003) demarcation lines in the [N II] diagram; while red and blue curves indicate the Kewley et al. (2001) and Kewley et al. (2006) divisions in the [S II] diagram. Spaxels classified as star-forming (H II spaxels) require identification as H II regions on both BPT diagrams.

The offsets $z_A(i)$ and $z_B(i)$ measure the local departure of each spaxel from the global behaviour of the sample in units of the intrinsic scatter, after subtracting the azimuthally averaged radial profile (or the other secondary dependence) of each field. Positive values indicate that a given location lies above the mean in that field, while negative values indicate a deficit relative to the mean.

To embed this information in its spatial context, we next characterise the local environment of B around each spaxel i by averaging the z_B values of its neighbouring spaxels. In the present work, we adopt a simple eight-connected neighbourhood, so that the neighbour set $N(i)$ consists of the immediately adjacent spaxels in the two-dimensional grid and $N_i = 8$ for interior pixels. The local “spatial lag” of B at position i is then defined as

$$z_{\text{lag},B}(i) \equiv \frac{1}{N_i} \sum_{j \in N(i)} z_B(j). \quad (12)$$

This quantity is a dimensionless measure of whether the environment surrounding i is systematically enhanced or suppressed in B relative to the global mean, again in units of the intrinsic dispersion: $z_{\text{lag},B}(i) > 0$ implies that the neighbours tend to be above the mean in B , while $z_{\text{lag},B}(i) < 0$ indicates a locally depressed environment.

The final local Moran-like product at each position is then defined as

$$I_i \equiv z_A(i) z_{\text{lag},B}(i). \quad (13)$$

This scalar encapsulates both the sign and the strength of the local correspondence between the value of A at i and the surrounding behaviour of B . When $z_A(i)$ and $z_{\text{lag},B}(i)$ share the same sign, $I_i > 0$ and the spaxel is “in agreement” with its environment: locations with $z_A(i) > 0$ and $z_{\text{lag},B}(i) > 0$ correspond to regions that are elevated

in A and embedded in an environment that is likewise elevated in B (“high–high”), whereas $z_A(i) < 0$ and $z_{\text{lag},B}(i) < 0$ identify “low–low” regions that are suppressed with respect to both fields. Conversely, if $z_A(i)$ and $z_{\text{lag},B}(i)$ have opposite signs, then $I_i < 0$ and the spaxel is locally “inverted” with respect to its surroundings: a positive $z_A(i)$ combined with a negative $z_{\text{lag},B}(i)$ signals a high value of A embedded in a depressed environment in B (“high–low”), while the opposite configuration corresponds to “low–high” structure. The magnitude $|I_i|$ quantifies the statistical significance of this local pattern in units of the product of the two scatters. The resulting I_i map therefore provides a compact diagnostic of where, and how strongly, the field A is locally correlated or anti-correlated with the spatial environment of the field B , enabling a spatially resolved assessment of concordant versus discordant structures in any pair of astrophysical quantities.

3 Results

3.1 The rMZR Relation

Before analysing the ensemble resolved relations derived from stacking all MAUVE–MUSE galaxies, we note that one of the 14 target galaxies, NGC 4383, is identified as a metallicity outlier relative to the rest of the sample. Across all four strong-line prescriptions adopted in this work (D16, Scal-PG16, O3N2-M13, and O3N2-C20), the oxygen abundances of star-forming spaxels in NGC 4383, especially the high- Σ_{SFR} spaxels (star-forming disc), are systematically offset low by at least 0.2 dex compared to the locus defined by the other 13 galaxies. This phenomenon is robust to the choice of calibration and to the adopted H II masking, and is therefore unlikely to

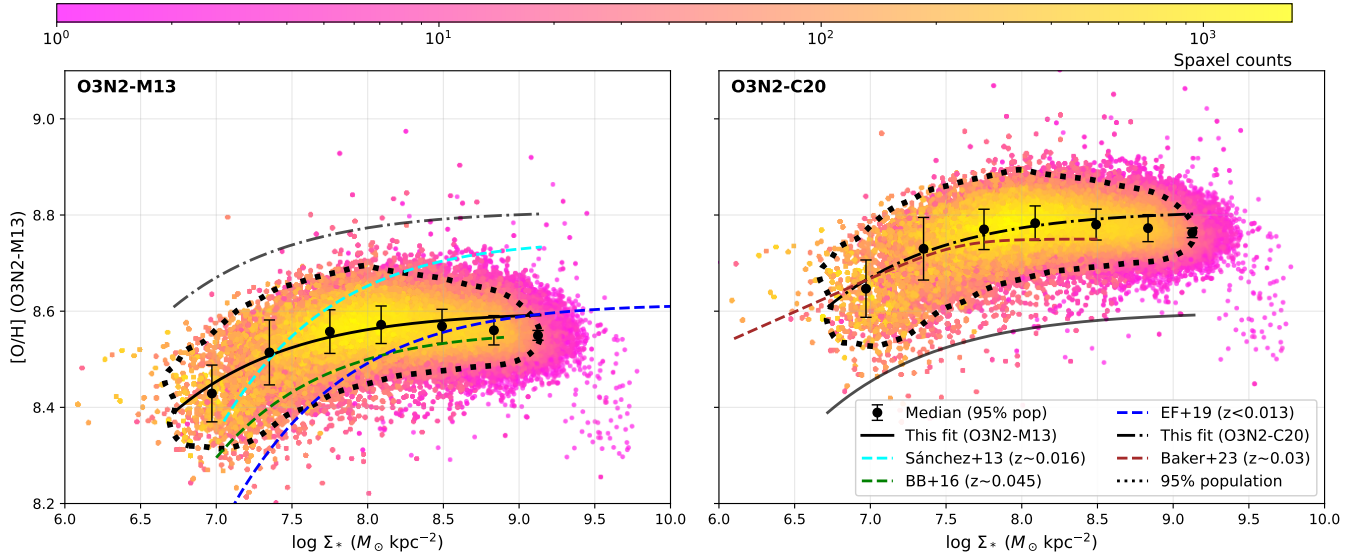


Figure 2. The resolved mass–metallicity relation (rMZR) for 13 MAUVE–MUSE galaxies (excluding NGC 4383). The left panel shows the rMZR derived using the **O3N2-M13** calibration, while the right panel shows the rMZR using the **O3N2-C20** calibration. Individual spaxels are shown as colored points, with the color scale indicating the number of spaxel counts. The dotted contours indicate the 95% population bounds, and the black points with error bars show the median metallicity within the contours. The solid and dot-dashed black curves represent the best-fitting rMZR to our data using Equation 14 under **O3N2-M13** and **O3N2-C20** prescriptions, respectively. For comparison, we show literature relations: the blue dashed curve from MAD data ($z < 0.013$; Erroz-Ferrer et al. 2019), the cyan dot-dashed curve from CALIFA data ($z \sim 0.016$; Sánchez et al. 2013), the green dot-dashed curve from MaNGA data ($z \sim 0.045$; Barrera-Ballesteros et al. 2016) in the left panel, and the brown dashed curve from MaNGA data ($z \sim 0.03$; Baker et al. 2023) in the right panel. Note that Baker et al. (2023) adopted the “combined” prescription from Curti et al. (2020).

be driven by methodological systematics. Because this single galaxy would otherwise dominate the low-metallicity tail and bias the ensemble trends, we temporarily exclude NGC 4383 from the stacked analyses below. We will return to discuss its properties in Section 5.3.

The stacked spaxel population of 13 MAUVE–MUSE galaxies follows the familiar rMZR shape reported in previous IFS surveys (e.g., Sánchez et al. 2013; Barrera-Ballesteros et al. 2016; Sánchez et al. 2019; Erroz-Ferrer et al. 2019; Yao et al. 2022; Baker et al. 2023; Koller et al. 2024): Z_{gas} increases steeply with Σ_* at the low-mass end and gradually transitions to a shallower, saturation-like behaviour toward high Σ_* . In Figure 2, we visualise this locus using the 95% population contour and its median trend (black points), shown for both the **O3N2-M13** (left) and **O3N2-C20** (right) calibrations. Within the 95% contour, we perform a nonlinear least-square fit by adopting the functional form from Sánchez et al. (2013) to describe the trend of rMZR:

$$[\text{O}/\text{H}] = a + b (\log_{10} \Sigma_* - c) e^{-(\log_{10} \Sigma_* - c)}, \quad (14)$$

The best-fitting parameters for our 13 MAUVE–MUSE galaxies are $a = 8.597$, $b = 0.0024$, $c = 10$ for **O3N2-M13** (left) and $a = 8.807$, $b = 0.0023$, $c = 10$ **O3N2-C20** (right) prescription, respectively.

In the left panel of Figure 2, we compare our best-fitting rMZR obtained with the **O3N2-M13** calibration to three other spatially-resolved local-universe surveys that adopt the same metallicity prescription. Overall, the MAUVE–MUSE relation shows the familiar saturating rMZR behaviour but exhibits the shallowest low- Σ_* rise among the plotted studies. Relative to the MAD relation of Erroz-Ferrer et al. (2019) at comparable (~ 100 pc) resolution, we recover a similar high- Σ_* plateau at $12 + \log(\text{O}/\text{H}) \approx 8.6$, whereas the MAD curve declines more steeply toward low Σ_* , reaching ~ 0.3 dex lower metallicity around $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.0$. Compared to the MaNGA-based relation of Barrera-Ballesteros et al. (2016), our rMZR is systematically higher by ~ 0.05 – 0.1 dex over the entire dy-

namic range of $\log(\Sigma_*)$. The comparison to CALIFA (Sánchez et al. 2013) is non-monotonic: in the intermediate- and high-mass regimes ($7.5 \leq \log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \leq 9.0$) our curve lies below Sánchez et al. (2013), but at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \lesssim 7.5$ it becomes higher, consistent with a comparatively shallow low- Σ_* metallicity decline in MAUVE–MUSE data.

The right panel shows that adopting **O3N2-C20** shifts the best-fitting rMZR upward by ~ 0.2 dex relative to **O3N2-M13** at fixed Σ_* , underscoring that the absolute normalisation is calibration-dependent (see further discussion in Section 5.1). In this representation, the “combined” prescription used by Baker et al. (2023) yields a best-fitting relation that is highly consistent with our **O3N2-C20** fit, particularly at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \lesssim 7.5$. Taken together, these comparisons suggest that the comparatively metal-rich appearance of our **O3N2-M13** rMZR (relative to some local references) is more plausibly driven by differences in sample selection and calibration systematics than by redshift evolution. Indeed, although not shown here, our **O3N2-M13** relation shares a similar curvature to higher-redshift studies (e.g., Yao et al. 2022; Koller et al. 2024) but with a higher normalisation; as discussed by Koller et al. (2024), such offsets are unlikely to be attributable to redshift evolution of the rMZR alone.

3.2 rMZR’s Secondary Dependence of rMZR on Σ_{SFR}

Colour-coding the rMZR by Σ_{SFR} reveals a clear secondary dependence on local star formation intensity (see the left column of Figure 3). At fixed Σ_* , high- Σ_{SFR} spaxels tend to lie at higher Z_{gas} in the high- Σ_* regime, whereas the ordering reverses at low Σ_* , where lower- Σ_{SFR} spaxels are preferentially more metal-rich. The transition between these regimes occurs at $\log_{10}(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.5$ – 8.0 . While the exact reversal point is imprecise due to limited statistics

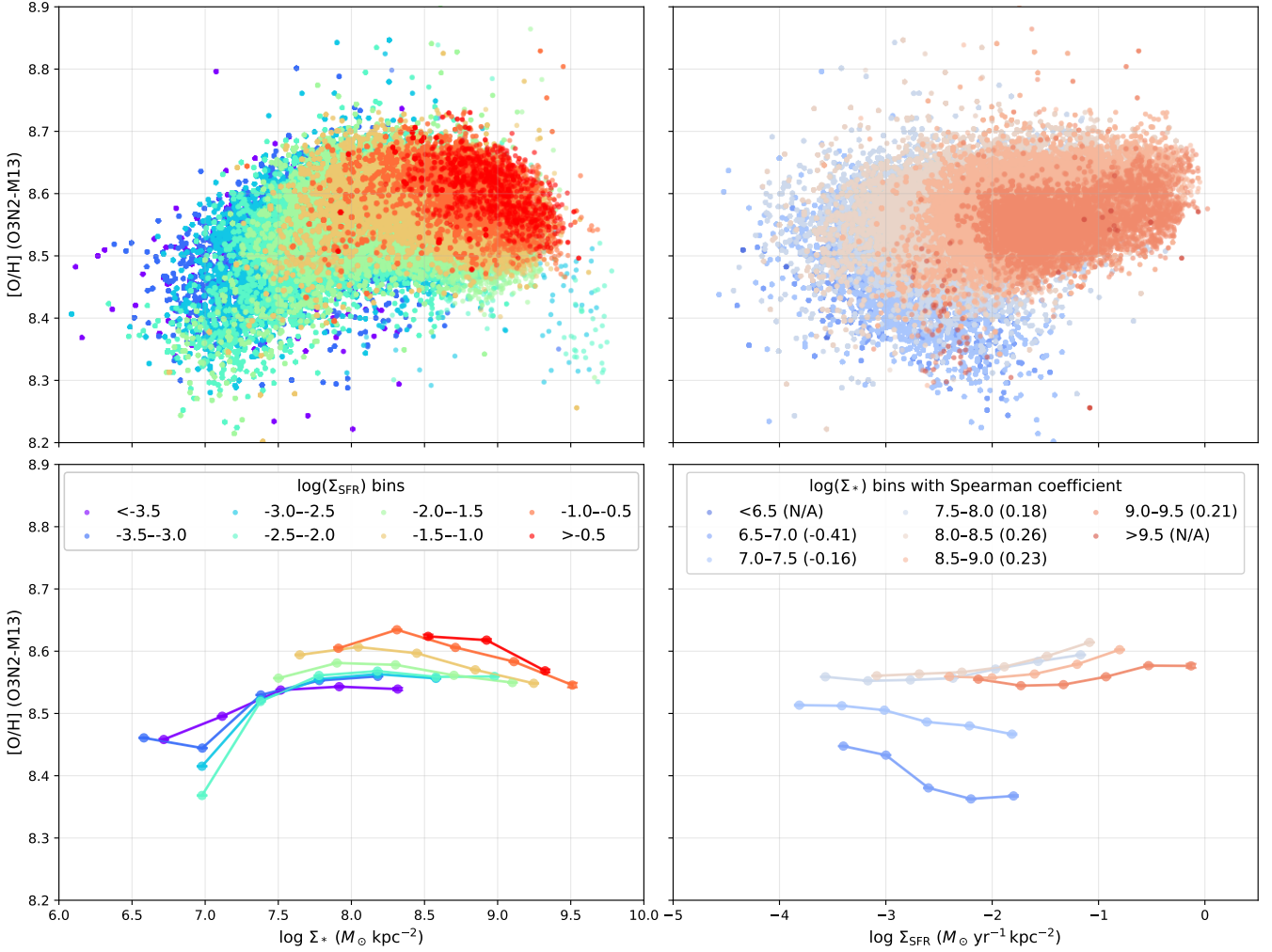


Figure 3. The left column shows the rMZR colour-coded by Σ_{SFR} for 13 MAUVE–MUSE galaxies (excluding NGC 4383), while the right column shows the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation colour-coded by Σ_* . The upper panels display the scatter plots, while the lower panels show the corresponding median trends of each scatter plot, with error bars indicating the standard deviation. In the top left panel, we use the black contour to enclose the 95% population of **HII** spaxels. Within this contour, we present the median trends with errorbars indicating the standard error of the median (typical value at ≤ 0.01), and perform the best-fitting rMZR as a black curve using Equation 14. Additionally, the brown dash curve and dot-dash grey curve are the rMZR fits from MAGPI (Koller et al. 2024) and MUSE–Wide (Yao et al. 2022), respectively. Note that due to insufficient statistics, the median trends of the lowest and highest Σ_* bins in the bottom right panel are not shown.

in the narrow overlap region, its location is consistent with the inversion reported from MAGPI by Koller et al. (2024), who found a clear sign change near $\log_{10}(\Sigma_*) \sim 7.8$ using O3N2–M13 prescription as well. The MAUVE–MUSE data therefore support a mass-dependent inversion in the resolved $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling, implying that the physical driver of the secondary dependence may differ between low- and high- Σ_* environments. Similarly, although not a strong signal, under the prescriptions of Curti et al. (2020), MaNGA data also seem to suggest a flipped correlation in the spatially-resolved scenario (Baker et al. 2023). Furthermore, adopting both D16 and Scal-PG16 calibrations, WL21 reported a predominantly positive local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation using the MUSE Atlas of Disks (MAD) data at ~ 100 pc resolution, whereas with MaNGA data on global galactic scales they found an overall negative correlation.

To visualise this dependence more directly, the right column of Figure 3 shows Z_{gas} as a function of Σ_{SFR} in bins of Σ_* . For each Σ_* bin we also compute the Spearman rank correlation coefficient ρ between Z_{gas} and Σ_{SFR} after removing the 3σ outliers. The lowest- Σ_* bins ($\log \Sigma_*$ at 6.5 - 7.0) exhibit the strongest anti-correlation at

Spearman coefficient $\rho = -0.41$, with metallicity decreasing as Σ_{SFR} rises. With increasing Σ_* , the median trends flatten progressively, and ultimately the relation becomes positively correlated, such that regions of elevated star formation are also metal-enhanced. This systematic, mass-ordered sequence from negative through flat to positive slopes encapsulates the rMZR’s second dependence on Σ_{SFR} and motivates the offset and spatial-correlation analyses developed in the next section.

3.3 Offset Relation and Local Moran-like Correlation Map

To isolate the secondary coupling between metallicity and star formation from their primary dependence on stellar mass surface density, we construct mass-normalised offset maps for each galaxy. Specifically, within each galaxy we divide all valid **HII** spaxels into 12 equal-width bins in $\log_{10} \Sigma_*$ (illustrated for one of MAUVE–MUSE targets, NGC 4192, in the bottom-right panel of Figure 4). For every bin m , we compute the mean spaxel values $\langle \log_{10} \Sigma_{\text{SFR}} \rangle_m$ and

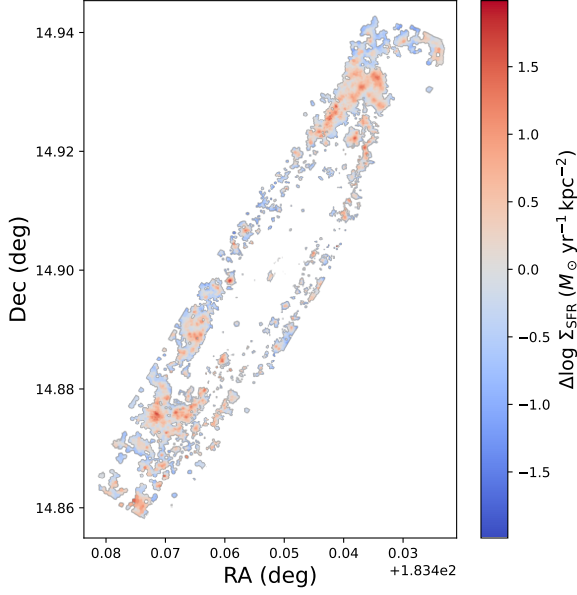
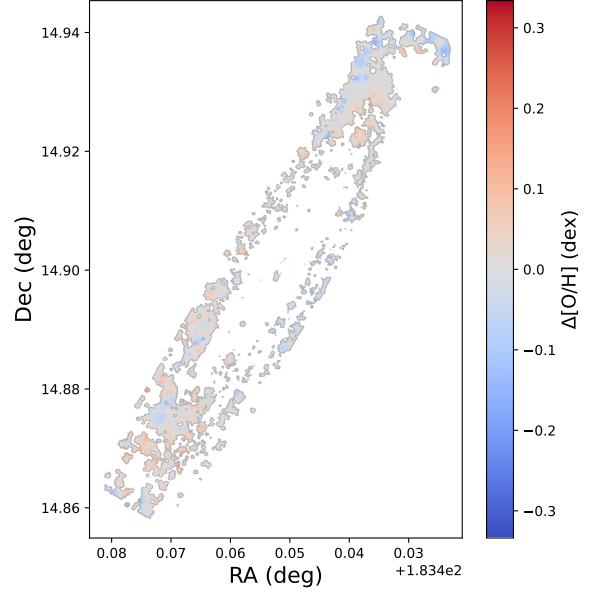
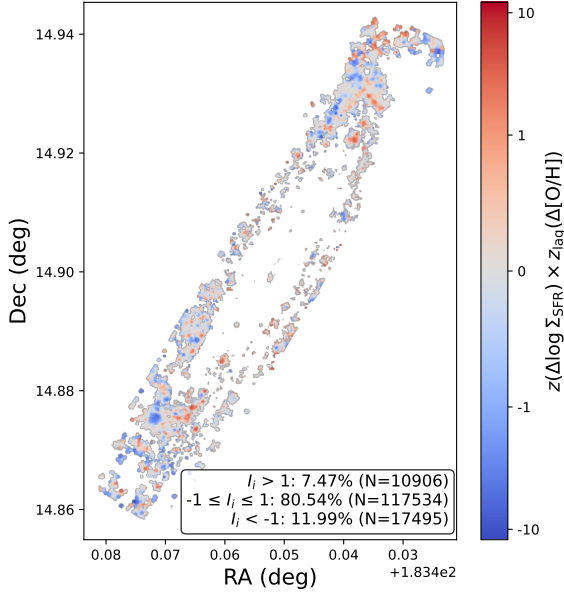
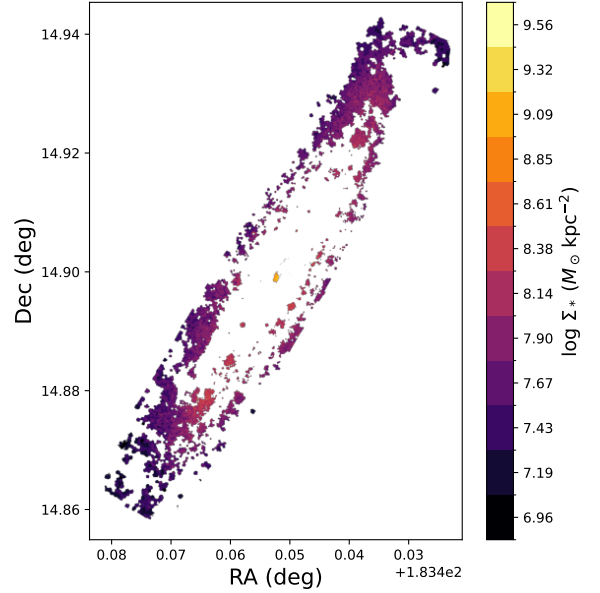
NGC4192: $\Delta \log \Sigma_{\text{SFR}}$ (offset from bin mean)NGC4192: $\Delta[\text{O}/\text{H}]_{\text{O3N2-M13}}$ (offset from bin mean)NGC4192: local bivariate Moran-like I_i NGC4192: $\log \Sigma_*$ (12 discrete bins)

Figure 4. Example offset and local Moran-like correlation maps for **HII** spaxels of NGC 4192. The top panels show the offsets $\Delta \log_{10} \Sigma_{\text{SFR}}$ (left) and $\Delta[\text{O}/\text{H}]_{\text{O3N2-M13}}$ (right) after removing the mean trends with Σ_* (see text). The bottom-left panel presents the local bivariate Moran-like product I_i constructed from these two offset fields, highlighting spatially local correlation (red) and anti-correlation (blue). The bottom-right panel shows $\log_{10} \Sigma_*$ for the **HII** spaxels used in the other panels, divided into 12 equal-width bins.

$\langle [\text{O}/\text{H}]_{\text{O3N2-M13}} \rangle_m$, and define spaxel-level offsets

$$\begin{cases} \Delta \log_{10} \Sigma_{\text{SFR}}(i) \equiv \log_{10} \Sigma_{\text{SFR}}(i) - \langle \log_{10} \Sigma_{\text{SFR}} \rangle_m, \\ \Delta[\text{O}/\text{H}](i) \equiv [\text{O}/\text{H}](i) - \langle [\text{O}/\text{H}] \rangle_m, \end{cases} \quad (15)$$

where spaxel i belongs to bin m . By construction, these offsets remove the mean rMZR and resolve star forming main sequence (rSFMS) trends within each Σ_* interval, leaving only the relative fluctuations of Σ_{SFR} and Z_{gas} at fixed local stellar mass density. The resulting maps therefore trace spatial variations that are not driven by the underlying Σ_* distribution and its associated large-scale gradients (e.g., see top panels of Figure 4).

Using these mass-normalised quantities, we also measure the residual $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling by computing, in each Σ_* bin of each galaxy, the Spearman rank correlation coefficient between $\Delta[\text{O}/\text{H}]_{\text{O3N2-M13}}$ and $\Delta \log_{10} \Sigma_{\text{SFR}}$ after removing the 3σ outliers. The resulting $\rho(\Sigma_*)$ trends are shown in Figure 5. For the stacked sample (black curve), ρ increases smoothly from negative values at low Σ_* to positive values at high Σ_* : rising from $\rho \approx -0.35$ at $\log_{10} \Sigma_* \sim 7$ to $\rho \approx +0.3$ at $\log_{10} \Sigma_* \gtrsim 8$, with a zero-crossing within $\log_{10}(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.5-8.0$. Individual galaxies follow the same qualitative ordering: systems that predominantly sample the low- Σ_* regime (e.g. NGC 4396, NGC 4457, and part of NGC 4192)

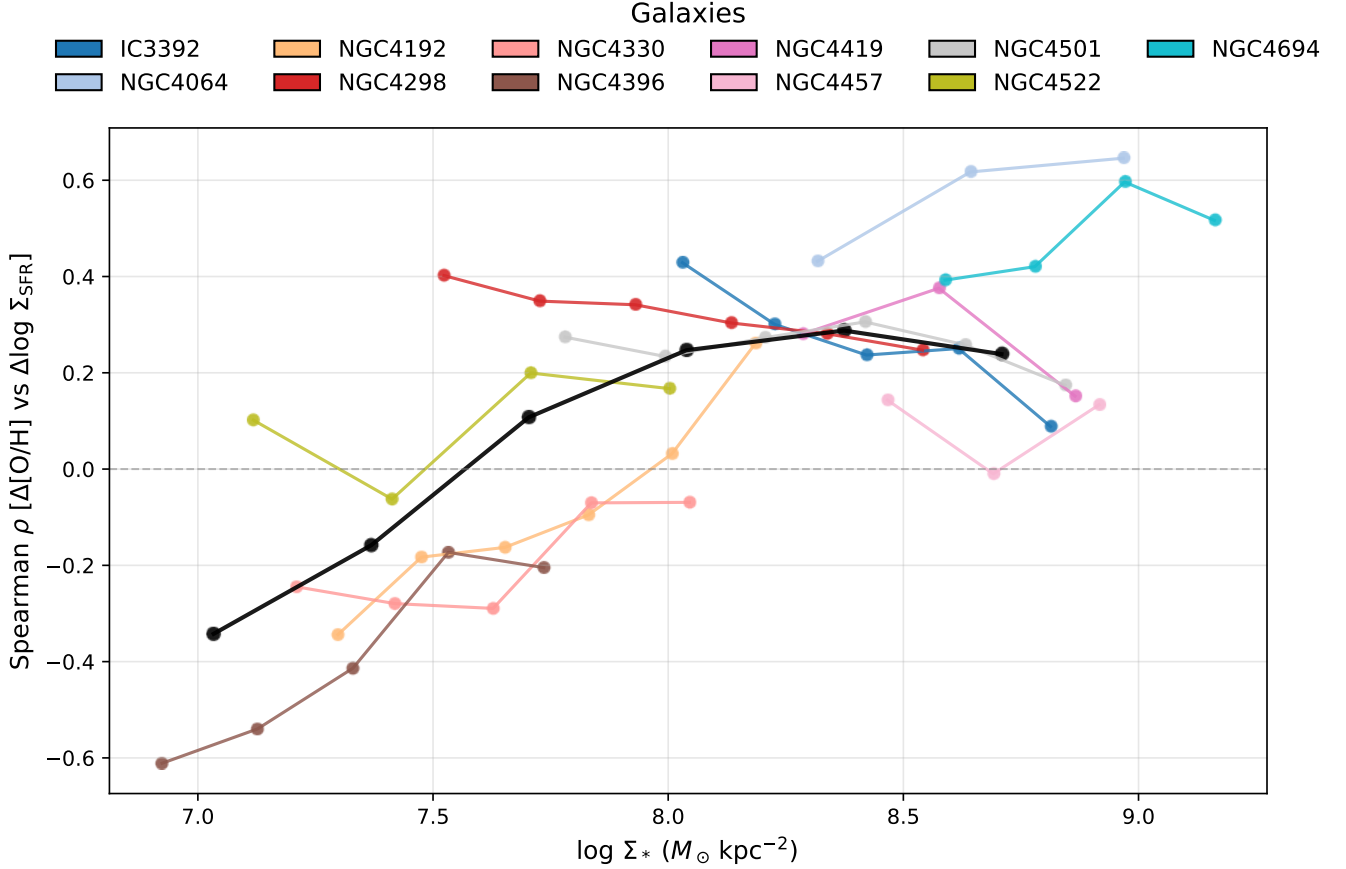


Figure 5. Spearman rank correlation coefficient ρ between $\Delta[\text{O}/\text{H}]_{\text{O3N2-M13}}$ and $\Delta\log_{10} \Sigma_{\text{SFR}}$ as a function of $\log_{10} \Sigma_*$, measured in discrete Σ_* bins for 13 MAUVE–MUSE galaxies (excluding NGC 4383). Each coloured curve corresponds to one galaxy (colour scheme as in Figure 1); the Spearman trends of NGC 4293 and NGC 4698 are omitted due to insufficient statistics. The black curve shows the stacked result obtained by combining all **HII** spaxels from the 13 galaxies.

contribute to the anti-correlated branch, whereas galaxies whose **HII** spaxels lie mainly at high Σ_* populate the positively correlated regime. Notably, NGC 4192 spans a broad Σ_* range that straddles the inversion point and exhibits a clear internal sign reversal, indicating that the transition is intrinsic to the subtle physics within galaxies rather than an artefact of combining different galaxy populations.

We further assess the spatial locality of this residual coupling by constructing the local bivariate Moran-like product I_i defined in Section 2.5. As an illustrative example, the bottom-left panel of Figure 4 shows the I_i map for NGC 4192, derived from the corresponding $\Delta\log_{10} \Sigma_{\text{SFR}}$ and $\Delta[\text{O}/\text{H}]$ fields. As indicated by Equation 15, these two fields are defined as residuals at fixed Σ_* , i.e. after subtracting, in each Σ_* bin, the trends of mean rMZR and rSFMS. In practice, this removes the dominant radial (mass-driven) profiles, so that the I_i map traces purely local spatial correspondence between star formation and metallicity fluctuations. By construction, I_i is the product of two standardised quantities, $z_{\Delta\log_{10} \Sigma_{\text{SFR}}}(i)$ and $z_{\text{lag}, \Delta[\text{O}/\text{H}]}(i)$, each expressed in units of each Σ_* bin’s standard deviation of the corresponding field. Thus $|I_i| \sim 1$ corresponds to the joint product of two $\sim 1\sigma$ deviations (one at the spaxel position, one in its local environment), whereas $|I_i| \ll 1$ indicates that at least one of the two is close to the global mean. Therefore, while the choice is inherently arbitrary, we adopt $|I_i| = 1$ as a convenient, dimensionless threshold to

distinguish weak or typical patterns from stronger local correspondence.

For NGC 4192, most spaxels in Figure 4 have $|I_i| \leq 1$ (grey), implying weak or ambiguous local correspondence, while a non-negligible fraction show significant local correlation ($I_i > 1$, red: higher (lower) Σ_{SFR} with higher (lower) Z_{gas}) or anti-correlation ($I_i < -1$, blue: higher (lower) Σ_{SFR} with lower (higher) Z_{gas}). These two tails are of comparable magnitude, with 7.47% of spaxels having $I_i > 1$ and 11.99% having $I_i < -1$, while the majority (80.54%; $N=117534$) remain in the weak/ambiguous regime. This near-balance between correlated and anti-correlated **HII** spaxels is consistent with NGC 4192 being the clearest “inversion” case in our sample: depending on Σ_* , the net $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling can change sign in Figure 5. Interpreting high- Σ_{SFR} peaks as tracing the cores of individual **HII** regions, the coexistence of red and blue structures within the same Σ_* bins suggests that the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling is spatially heterogeneous: once mass trends of these two properties are removed, neighbouring regions of comparable Σ_* can display either concordant or discordant metallicity responses to star formation. Moreover, the highest $|I_i|$ values are spatially coincident with bright **HII**/high- Σ_{SFR} clumps that satisfy our **HII** BPT selection, whereas diffuse low- Σ_{SFR} regions tend to show $|I_i| \lesssim 1$, supporting that the observed signal is associated with genuine **HII** regions rather than being driven by diffuse ionised gas (DIG) contamination. These

maps show that 1) the sign of the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling is set by spatially varying local physics, so correlated and anti-correlated H II regions can coexist even at fixed Σ_* ; and 2) the observed mass-dependent behaviour arises only as an emergent population trend when one aggregates enough spaxels to sample both sides of the transition near $\log_{10}(\Sigma_*/M_{\odot}, \text{kpc}^{-2}) \sim 7.5-8.0$.

4 Discussion: The Origin of $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (Anti-)Correlation in a Gas-regulator Framework

To interpret the varying sign of the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation, we adopt a spatially-resolved gas-regulator (“bathtub”) framework formulated entirely in spaxel-by-spaxel variations. Throughout this section, all quantities are written as surface densities to reflect the fact that our observational constraints are local (~ 100 pc resolution) rather than global. The model is intended to describe the evolution of the gas regulator in the local Universe (present day), over timescales that are short compared to the ~ 10 Gyr past history of star formation and chemical enrichment. In practice, the variations that matter for our observed $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (anti-)correlation are closer to the lifetimes of individual H II regions and local dynamical or mixing timescales (i.e. a few tens of Myr), as suggested by the fact that the strongest signal is associated with bright H α /high- Σ_{SFR} clumps. Over such intervals, the accumulated growth of the stellar mass surface density Σ_* is negligible compared to its present-day value (or at most comparable to observational uncertainties), so we neglect the time evolution of Σ_* and treat it as a fixed control parameter. We therefore do not attempt to reconstruct the full past star formation or metal enrichment history of each region. Instead, we assume that different H II spaxels within a galaxy began their recent evolution from broadly similar initial conditions and have since evolved under the same set of governing equations.

Within this framework, the time variable, t , plays the role of an evolution parameter that controls the instantaneous state of the local regulator (gas mass, SFR, metallicity, gas supply properties) at a given location. Mathematically, even though we observe different spaxels at the same cosmic time, each with its own set of properties, P (here $P \in \{\Sigma_*, \Sigma_{\text{SFR}}, Z_{\text{gas}}, \text{etc.}\}$), it is useful to think of these spatial variations as sampling different points along a common family of time-dependent solutions. Equivalently, given any pair of spaxels within the same Σ_* bin, the differences in their local properties can be interpreted as the result of having evolved for different effective “phases” or having experienced different recent perturbations under the same functional form of the regulator equations. In this sense, the time coordinate, t , may be regarded as a parameter that indexes the instantaneous value of each property and allows us to trace how fluctuations in one property, P , induce corresponding responses in another. What follows is therefore a local, time-dependent description that we use as a physical language for organizing the spatially-resolved fluctuations measured in the MAUVE–MUSE data.

We first write the star-formation rate surface density as

$$\Sigma_{\text{SFR}}(t) \equiv \frac{\Sigma_g(t)}{\tau_{\text{dep}}(t)}, \quad (16)$$

where $\Sigma_g(t)$ is the gas surface density and $\tau_{\text{dep}}(t)$ the local depletion time. Then, we use the following two differential equations to describe our spatially-resolved framework: the gas mass surface density obeys

$$\frac{d\Sigma_g(t)}{dt} = \Sigma_{\Phi}(t) - [1 - R + \lambda] \Sigma_{\text{SFR}}(t), \quad (17)$$

where $\Sigma_{\Phi}(t)$ is the *gas supply rate (GSR) surface density*[‡], i.e. the rate at which the region gains fresh gas per unit area, R is the returned fraction, and λ is the mass-loading factor; while the surface density of metals (Σ_Z) follows

$$\frac{d\Sigma_Z(t)}{dt} = y(1 - R) \Sigma_{\text{SFR}}(t) - [1 - R + \lambda] \Sigma_{\text{SFR}}(t) Z(t) + \Sigma_{\Phi}(t) Z_{\Phi}, \quad (18)$$

with metal yield per stellar, y , and metallicity of the supplied gas, Z_{Φ} (i.e., background metallicity).

Defining the gas-phase metallicity as

$$Z(t) \equiv \frac{\Sigma_Z(t)}{\Sigma_g(t)}, \quad (19)$$

we obtain

$$\frac{dZ(t)}{dt} = \frac{1}{\Sigma_g(t)} [y(1 - R) \Sigma_{\text{SFR}}(t) + (Z_{\Phi} - Z(t)) \Sigma_{\Phi}(t)], \quad (20)$$

so that at any location, the metallicity responds to enrichment from ongoing star formation and dilution by the supplied gas.

It is convenient to introduce the gas supply timescale

$$\tau_{\Phi}(t) \equiv \frac{\Sigma_g(t)}{\Sigma_{\Phi}(t)} \leq \tau_{\text{dep}}(t), \quad (21)$$

which measures how quickly the local gas reservoir would be replenished by the current GSR. Using $\Sigma_{\text{SFR}} = \Sigma_g/\tau_{\text{dep}}$ and $\Sigma_{\Phi} = \Sigma_g/\tau_{\Phi}$ in Equation 20 yields

$$\frac{dZ(t)}{dt} = \frac{1}{\tau_{\Phi}(t)} \left[Z_{\Phi} + y(1 - R) \frac{\Sigma_{\text{SFR}}(t)}{\Sigma_{\Phi}(t)} - Z(t) \right]. \quad (22)$$

This motivates the definition of the *instantaneous target metallicity*

$$Z^{\dagger}(t) \equiv Z_{\Phi} + y(1 - R) \frac{\Sigma_{\text{SFR}}(t)}{\Sigma_{\Phi}(t)}, \quad (23)$$

so that Equation 22 becomes

$$\frac{dZ(t)}{dt} = \frac{1}{\tau_{\Phi}(t)} [Z^{\dagger}(t) - Z(t)] \iff Z^{\dagger}(t) = Z(t) + \frac{dZ(t)}{dt} \tau_{\Phi}(t). \quad (24)$$

This explicitly shows that $Z^{\dagger}(t)$ acts as the *driver*: it is the metallicity that the system would reach if the current rate of change dZ/dt were sustained over one gas supply timescale. Only when $Z^{\dagger}(t) = Z(t)$ does Equation 24 imply $dZ/dt = 0$, corresponding to a *local minimum or maximum* of the real-time metallicity $Z(t)$ rather than a long-lived equilibrium.

With these definitions, the observed spaxels can be separated conceptually into two complementary cases: either reach (Section 4.1) or evolve toward/away from (Section 4.2) the local local extremum stages.

4.1 Case I: Correlation at $Z = Z^{\dagger}$

For spaxels observed at a series of local extremum time $t = t_m$, the real-time metallicity satisfies $dZ/dt|_{t_m} = 0$, hence $Z(t_m) = Z^{\dagger}(t_m)$,

[‡] We avoid the term “inflow rate surface density” because inflow usually denotes pristine, galaxy-scale accretion from circumgalactic medium (CGM), whereas on local scales it is more appropriate to describe a region’s or giant molecular cloud (GMC)’s ability to gain gas supply from its ambient ISM; moreover, for our observed Virgo members, the global inflow is likely suppressed or truncated.

with

$$Z^\dagger(t_m) = Z_\Phi + \gamma(1-R) \frac{\Sigma_{\text{SFR}}(t_m)}{\Sigma_\Phi(t_m)}. \quad (25)$$

Taking a first-order variation of Equation 25 yields

$$\delta Z^\dagger(t_m) = [Z^\dagger(t_m) - Z_\Phi] [\delta \log_{10} \Sigma_{\text{SFR}}(t_m) - \delta \log_{10} \Sigma_\Phi(t_m)], \quad (26)$$

so the sign of the metallicity fluctuation at extrema is controlled by the competition between spatial variations in SFR and GSR. If Σ_{SFR} varies more strongly than Σ_Φ across neighbouring spaxels, then $\delta Z^\dagger$ follows $\delta \log_{10} \Sigma_{\text{SFR}}$ and a positive $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation is expected. If instead GSR variations dominate while Σ_{SFR} and Σ_Φ co-vary, dilution drives $\delta Z^\dagger$ opposite to $\delta \log_{10} \Sigma_{\text{SFR}}$, yielding an anti-correlation. Where the two contributions are comparable, $\delta Z^\dagger$ is small and little correlation is produced.

Using $\Sigma_{\text{SFR}} \equiv \Sigma_g/\tau_{\text{dep}}$ and $\tau_\Phi \equiv \Sigma_g/\Sigma_\Phi$, Equation 26 is equivalently

$$\delta Z^\dagger(t_m) \propto \delta \log_{10} \tau_\Phi(t_m) - \delta \log_{10} \tau_{\text{dep}}(t_m), \quad (27)$$

highlighting that, at metallicity extrema, the correlation sign is set by whether variations in the gas supply timescale outpace those in depletion time. Furthermore, if we generalize (i.e., substitute) t_m with t , which in fact describes the change of instantaneous target metallicity rather than the real-time metallicity at extrema, we still have the same functional forms of Equation 26 and Equation 27. This tells us a broader context that the change of instantaneous target metallicity is also determined by the comparison of the relative change magnitude between the local SFR and GSR. We will return to this relation later in our discussion.

4.2 Case II: Correlation at $Z \neq Z^\dagger$

In reality, most spaxels are not observed exactly at a local metallicity extremum. To relate their real-time metallicity to the driver $Z^\dagger$, we expand around the nearest extremum time t_m by rewriting $t = t_m + \zeta$ and, for any quantity P ($P \in \{\Sigma_{\text{SFR}}, Z, Z^\dagger, \text{etc.}\}$), defining

$$\Delta_P(\zeta) \equiv P(t_m + \zeta) - P(t_m). \quad (28)$$

Linearising Equation 24 about t_m yields a first-order response,

$$\frac{d\Delta_Z(\zeta)}{d\zeta} + \frac{\Delta_Z(\zeta)}{\tau_\Phi(t_m)} = \frac{\Delta_{Z^\dagger}(\zeta)}{\tau_\Phi(t_m)}, \quad (29)$$

whose solution is

$$\Delta_Z(\zeta) = \frac{1}{\tau_\Phi(t_m)} \int_0^\zeta \exp\left[-\frac{\zeta - \nu}{\tau_\Phi(t_m)}\right] \Delta_{Z^\dagger}(\nu) d\nu. \quad (30)$$

Equation 30 shows that $\Delta_Z(\zeta)$ is a low-pass-filtered version of the driver $\Delta_{Z^\dagger}(\zeta)$: the real-time metallicity responds as an exponential moving average of the target metallicity, smoothed over the gas supply timescale $\tau_\Phi(t_m)$. Physically, this “memory” arises because $Z(t)$ cannot respond instantaneously to changes in $Z^\dagger(t)$, due to finite delays in metal injection from recent star formation and in turbulent mixing (metal diffusion) through the ISM. In practice, because of detection procedures and measurement uncertainties, our observations predominantly probe the regime $\zeta \gtrsim \tau_\Phi(t_m)$, i.e. the time-averaged response rather than an instantaneous snapshot of $Z^\dagger$.

For such observed spaxels, $\delta Z(t)$ follows the recent behaviour of $\delta Z^\dagger(t)$ and therefore inherits the same sign set by star formation versus gas supply variability. Combining Equation 23 with Equation 30 gives the consistent prediction as Equation 26 and Equation 27

$$\begin{aligned} \delta Z(t) &\propto \delta \log_{10}(\Sigma_{\text{SFR}}(t)) - \delta \log_{10}(\Sigma_\Phi(t)) \\ &\propto \delta \log_{10}(\tau_\Phi(t)) - \delta \log_{10}(\tau_{\text{dep}}(t)). \end{aligned} \quad (31)$$

Thus, away from extrema, the observed local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (anti-)correlation is still governed by whether spatial variations are driven primarily by changes in star-formation efficiency (through τ_{dep}) or by changes in the gas supply rate (through τ_Φ), with the finite response time encoded in τ_Φ introducing additional scatter and spatially varying phase lags.

4.3 Interpretation of the Observed (Anti-)Correlation

Within this framework, our resolved MAUVE–MUSE trends in Figure 5 can be interpreted as the outcome of a competition between GSR-driven and SFR-driven variability. Observationally, in high- Σ_* regions (typically inner discs), dense gas and short depletion times make spatial variations in Σ_{SFR} or τ_{dep} relatively strong, while the ability to gain fresh gas (i.e., the GSR) is comparatively steady from spaxel to spaxel. In this regime, Equation 31 favours a positive $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation: fluctuations in SFR dominate over variations in GSR, and oxygen abundance increases with Σ_{SFR} at fixed Σ_* .

On the other hand, at low Σ_* (outer discs), especially in the Virgo environment where the local gas supply can fluctuate strongly due to environmental processes, spatial variations in τ_Φ become more important. When GSR fluctuations dominate while Σ_{SFR} and Σ_Φ co-vary, the dilution associated with enhanced supply tends to lower Z in regions of relatively higher SFR, driving the correlation toward zero or negative values. In between these extremes, where variations in τ_Φ and τ_{dep} are comparable, the net effect on metallicity is small and the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation becomes weak or flat.

WL21 also explain this (anti-)correlation from a theoretical point of view. Their model attributes the positive case (in ~ 100 pc scales) to time-variable SFE acting against a relatively steady gas inflow rate, and the negative case (in integrated galactic scales) to strongly time-variable inflow acting against nearly constant SFE. Our MAUVE–MUSE galaxies, however, show the coexistence of both positive and negative correlations already at ~ 100 pc scales, and our regulator analysis indicates that the sign is set locally by the *relative amplitude* of variations in Σ_{SFR} and the gas–supply–rate surface density Σ_Φ , rather than by scale alone. If the spaxel-to-spaxel changes in Σ_{SFR} are interpreted as being driven primarily by fluctuations in the SFE of individual GMCs, while Σ_Φ is taken as the local analogue of an inflow rate, then our expression in Equation 31 is fully consistent with the WL21 framework: positive (negative) correlations emerge where SFE- or SFR-driven (GSR-driven) variability dominates, and both regimes can naturally coexist within the same galaxy at fixed spatial resolution.

4.4 From Spatially-resolved to Integrated Relations

Although this framework is written entirely in terms of surface densities, the underlying regulator equations have exactly the same functional form if one replaces $(\Sigma_g, \Sigma_*, \Sigma_{\text{SFR}}, \Sigma_\Phi)$ by their galaxy-integrated counterparts $(M_g, M_*, \text{SFR}, \Phi)$ and the associated timescales by effective global depletion and inflow timescales. In this sense, Equation 31 can be read as a scale-free statement: at fixed M_* , the sign of the $Z_{\text{gas}}-\text{SFR}$ (anti-)correlation is determined by the comparison of the relative fluctuation amplitudes in SFR and in the inflow rate, or equivalently by whether variations in depletion or inflow timescale dominate. If global fluctuations in SFR (or SFE) are larger than those in inflow, one expects a positive $Z_{\text{gas}}-\text{SFR}$ correlation at fixed M_* ; if stochasticity in inflow dominates, dilution produces an anti-correlation. This picture provides a natural physical interpretation for the integrated FMR found in large samples, where the SFR

dependence of metallicity is negative at low stellar mass and commonly observed but weakens and may even change sign at the high-mass end (e.g. Mannucci et al. 2010; Yates et al. 2012; Curti et al. 2020). In our view, these global trends reflect the same basic competition encapsulated in Equation 31, with low-mass galaxies residing in a regime where inflow-driven variability is more significant, and massive systems shifting toward a regime where SFR- or SFE-driven variability becomes comparatively stronger.

Within this picture, the apparent prevalence of anti-correlations in both resolved and integrated studies can be understood, at least in part, as a consequence of how different surveys sample the M_* and/or Σ_* parameter space. Global FMR analyses are dominated by galaxies with $M_* \lesssim 10^{10.5} M_\odot$, while more massive systems with $M_* \gtrsim 10^{10.5} M_\odot$ are rarer and span a narrower range of specific SFR, so the low-mass, inflow-dominated regime contributes a larger fraction of the statistical weight. An analogous effect arises in spatially-resolved work: at kpc-scale resolution, many surveys preferentially sample spaxels with $\log(\Sigma_*/M_\odot, \text{kpc}^{-2}) \lesssim 7.8$ and therefore see predominantly anti-correlated or flat $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ behaviour (e.g. Baker et al. 2023; Koller et al. 2024), whereas at ~ 100 pc resolution our MAUVE-MUSE, and possibly MAD data in WL21, maps contain a much larger fraction of spaxels with $\log(\Sigma_*/M_\odot, \text{kpc}^{-2}) \gtrsim 8$, so the positive branch dominates the stacked trends. In both the global and resolved cases, the sign flip emerges only when one samples across the transition region (around $\log \Sigma_* \sim 7.5-8.0$ locally and $\log M_* \sim 10.0-10.5$ globally), and the balance between correlated and anti-correlated regimes is then controlled by the relative number of galaxies/spaxels on either side of this inversion. Furthermore, the more frequent detection of anti-correlation, as well as the weaker and less clear (anti-)correlation signal on global scales than on local scales, may simply raise from an averaging effect, whereby integrating over many spaxels blends regions together with opposite local behaviours. However, what sets the precise location of these transition scales, whether the local (Σ_*) and global (M_*) inversions are causally linked or merely coincident, and which physical conditions change across them (e.g. gas fraction, disc stability, outflow efficiency, or environmental processes), remain open questions that we defer to future work.

5 Robustness of Our Findings

In this section, we assess how robust our observed $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (anti-)correlation is to three key sources of systematics with Figure 6 as a compact visual guide. We first examine the effect of adopting different strong-line abundance prescriptions in Section 5.1, by comparing the four rows of Columns 1 and 3 of Figure 6 (D16, Scal-PG16, O3N2-M13, and O3N2-C20). We then test the sensitivity to star-forming spaxel selection in Section 5.2, by contrasting the conservative **HII** results (Columns 1 and 3) with the more inclusive **HII+Comp** results (Columns 2 and 4). Finally, we investigate the special case of the metallicity outlier NGC 4383 in Section 5.3, highlighted by dotted contours or curves in each panel of Figure 6 relative to the solid loci of the remaining galaxies, noting that the stacked Spearman trend in black continues to exclude this object. The broader purpose of these checks is to emphasise that, although calibration choice, excitation mixing, environment, and mass-dependent sampling can modulate the apparent scatter and the detailed slope in individual regimes, they do not alter the underlying physical interpretation implied by Equation 31, namely that the sign of the local residual correlation is set by the competition between the relative fluctuations

of star formation and gas supply, $\delta \log_{10} \Sigma_{\text{SFR}}$ versus $\delta \log_{10} \Sigma_\Phi$, or equivalently by the balance between $\delta \log_{10} \tau_\Phi$ and $\delta \log_{10} \tau_{\text{dep}}$.

5.1 Sensitivity to Metallicity Calibration

A well-known limitation of any resolved metallicity analysis is that strong-line prescriptions are not anchored to a unique absolute scale. Different calibrations rely on different reference samples and methodologies—direct electron-temperature (T_e) measurements, photoionization model grids, or hybrid/stacked approaches—and they respond differently to variations in excitation, ionization parameter, diffuse ionized gas (DIG), and N/O abundance. Consequently, systematic offsets of order $\sim 0.1-0.3$ dex (and in extreme cases larger up to ~ 0.5 dex) between calibrations applied to the same spectra are well documented, and can alter both the normalization and the detailed slopes of resolved relations (e.g. Kewley & Ellison 2008; Kewley et al. 2019; Maiolino & Mannucci 2019). These long-standing uncertainties motivate our use of four widely adopted diagnostics spanning empirical T_e -calibrated and model-based families: D16, Scal-PG16, O3N2-M13, and O3N2-C20.

The first column of Figure 6 shows that, despite calibration-dependent zero-points, the overall rMZR morphology is broadly stable across all prescriptions: Z_{gas} increases steeply with Σ_* at low surface densities and flattens toward a saturation-like regime at high Σ_* , in line with previous resolved surveys. However, the secondary ordering with Σ_{SFR} is not equally coherent for every diagnostic. For both O3N2-based calibrations, the Σ_{SFR} -stratified contours display a clear and systematic reversal with Σ_* : at low Σ_* , higher Σ_{SFR} preferentially occupies lower Z_{gas} , while at high Σ_* the ordering flips such that higher Σ_{SFR} corresponds to higher Z_{gas} . This behaviour is reflected in the stacked Spearman trends (third column), which show a coherent transition from $\rho < 0$ at low Σ_* to $\rho > 0$ at intermediate/high Σ_* for both O3N2 prescriptions (with galaxy-to-galaxy scatter but a consistent global trend). By contrast, the D16 and Scal-PG16 prescriptions exhibit substantially weaker and more heterogeneous Σ_{SFR} stratification in the H II-only sample. The D16 stacked ρ trend remains mildly positive over most of the Σ_* range; while Scal-PG16 shows a stronger dependence on Σ_* in the stacked statistic, but with opposite-sign behaviour compared to O3N2-M13 and O3N2-C20. Therefore, while the rMZR curvature itself is stable, the inferred $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling (and any apparent sign change) is diagnostic-dependent.

The comparison also reveals systematic differences between calibration families. In our sample, indicators tied more closely to T_e -based anchors (e.g. Scal-PG16 and the O3N2 calibrations) show tighter internal sequences with metallicity dispersions of $\sim 0.1-0.15$ dex at fixed Σ_* and Σ_{SFR} , whereas the model-based D16 prescription produces visibly larger scatter ($\sim 0.2-0.25$ dex) and spans a broader metallicity range ($\sim 0.4-0.5$ dex wider) across the same parameter space. This behaviour is consistent with PHANGS-MUSE (Kreckel et al. 2019), which shows that calibrations tied to photoionization models or to a small number of line ratios are more susceptible to local variations in ionization parameter and abundance ratios, leading to increased spaxel-to-spaxel dispersion and systematic offsets relative to T_e -anchored scales. Within the O3N2 family itself, the two prescriptions are highly consistent in their trends; the main difference is a nearly uniform offset, with O3N2-C20 yielding metallicities systematically higher than O3N2-M13 by $\sim 0.1-0.2$ dex.

While all calibrations recover an inversion, they do not yield identical qualitative behaviour in detail. The D16 Spearman trends are weighted toward positive ρ over most of the Σ_* range, mirroring the predominantly positive resolved correlation reported by WL21 on ~ 100 pc scales. This suggests that the *choice of calibration*

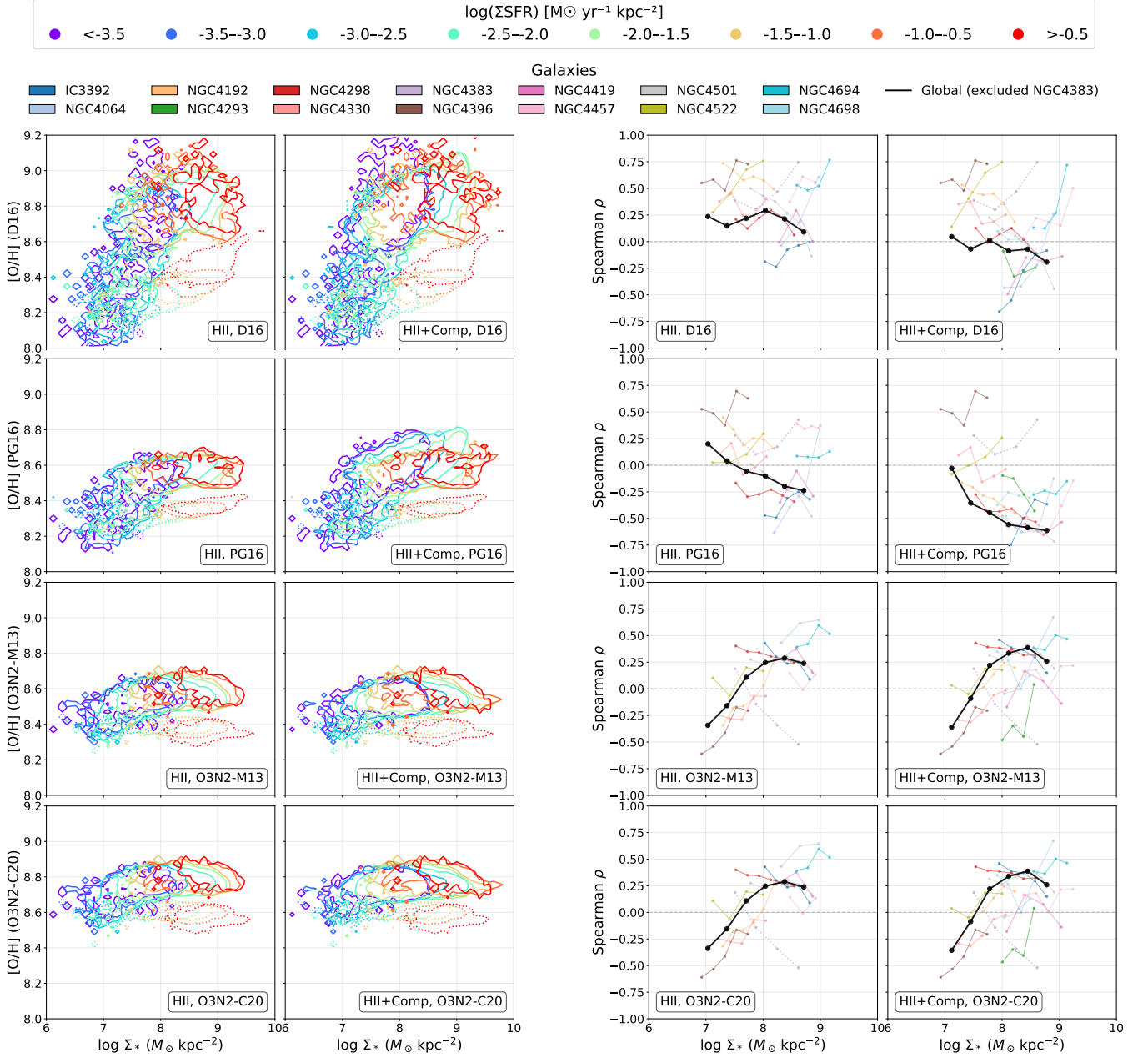


Figure 6. Comparison of the rMZR (left two columns) and the mass-dependent Spearman rank correlation coefficient ρ between $\Delta[\text{O}/\text{H}]$ and $\Delta \log_{10} \Sigma_{\text{SFR}}$ (right two columns) for four gas-phase metallicity calibrations: D16 (top row), Scal-PG16 (second row), O3N2-M13 (third row), and O3N2-C20 (bottom row). Columns 1 and 3 use only **HII**-classified star-forming spaxels, whereas Columns 2 and 4 use the more inclusive **HII+Comp** selection (see the BPT demarcation in Figure 1). The rMZR panels are colour-coded by $\log_{10} \Sigma_{\text{SFR}}$ as in Figure 3. The Spearman panels show $\rho(\Sigma_*)$ measured in discrete Σ_* bins for individual galaxies (coloured curves) and for the stacked sample (black curve; excluding NGC 4383), following Figure 5. The parameter spaces occupied by NGC 4383 are indicated by dotted contours/curves in each corresponding panel.

can influence whether a given data set appears dominated by correlated or anti-correlated regimes, even when the underlying spaxel population is the same. Moreover, at the high- Σ_* end the O3N2 prescriptions favour a positive residual coupling, whereas D16 and Scal-PG16 lean toward weaker or even negative behaviour. Similar calibration-dependent qualitative differences have been noted in other MUSE-based surveys, where O3N2- and N2S2-based prescriptions can yield different resolved slopes from S- or N2S2-based prescriptions (e.g. Kreckel et al. 2019). We therefore caution that the detailed

mass dependence of the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation can diminish or even reverse under different metallicity calibrations.

We adopt the O3N2 indicator as our primary metallicity diagnostic for several reasons. First, both the O3N2-M13 and O3N2-C20 calibrations are derived from large IFS datasets (CALIFA and MaNGA, respectively), making them particularly well-suited for spatially-resolved studies like ours. Second, these calibrations are anchored to T_e -based abundance measurements, providing a more direct physical foundation. Third, O3N2 prescriptions show tighter internal sequences in our sample compared to some alternative di-

agnostics (see Figure 6). Importantly, results by O3N2 indicators clearly delineates a key observational feature: a negative $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation at the low-mass end transitioning to a positive correlation at the high-mass end. In our resolved sample, this inversion occurs at $\log_{10}(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.5 - 8.0$. This resolved trend mirrors the global, integrated picture established by Yates et al. (2012), whose observations derived by both strong line diagnostics and grids of photoionization models (see their Figure 1) reveal a similar reversal, negative at low masses and positive at high masses, with a transition point around a characteristic mass of $\log_{10}(M_*/M_\odot) \approx 10.0 - 10.5$. Similarly, Curti et al. (2020) also present such reversal of $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ correlation at the high mass end in their Figure 6 and 7.

5.2 Sensitivity to the Star-forming Spaxel Definition

Our baseline analysis relies on a conservative **HII**-only mask, in which spaxels are required to fall within the **HII** locus of both the [N II] and [S II] BPT diagrams with robust line detections and uncertainties that remain fully inside the same excitation class. This stringent selection is designed to minimise contamination from shocks, AGN-like excitation, and diffuse ionised gas (DIG), all of which can bias strong-line metallicity indicators and $H\alpha$ -based Σ_{SFR} estimates. However, MAUVE–MUSE also provides a more inclusive “star-forming” mask that merges **HII** and composite regions (**HII+Comp**), motivated by the expectation that composite spaxels are still largely powered by O/B stars’ photonization but may include a non-negligible contribution from DIG or harder radiation fields. We therefore repeat the full rMZR and offset-correlation analysis using **HII+Comp** spaxels to assess the sensitivity of our conclusions to excitation masking.

The comparison is summarised in Figure 6 as well, where Columns 1 and 3 use the conservative **HII** selection, and Columns 2 and 4 adopt **HII+Comp**. Including composite spaxels expands the sampled parameter space, especially toward high Σ_* where inner discs and circumnuclear regions are more likely to host mixed excitation and more complex physics. For the two O3N2-based prescriptions (O3N2-M13 and O3N2-C20), both the rMZR colour ordering and the mass-dependent Spearman trends are essentially unchanged when composites are added. The stacked $\rho(\Sigma_*)$ curves retain the same negative-to-positive transition at $\log_{10}(\Sigma_*/M_\odot \text{ kpc}^{-2}) \sim 7.5-8.0$, and the scatter around the media rMZR locus increases only marginally. This stability indicates that our main result is not driven by a narrow excitation-class definition, and that O3N2 indicators are comparatively resilient to moderate DIG/composite admixture in the MAUVE–MUSE data.

By contrast, the N2S2- and S-calibrations show a stronger response to including composites. In particular, the Scal-PG16 rMZR exhibits a noticeable increase in scatter at the high- Σ_* end under the **HII+Comp** mask, consistent with the known sensitivity of S2- and N2-based ratios to harder ionizing spectra and DIG-enhanced low-ionization lines. The corresponding $\rho(\Sigma_*)$ trends also shift: for D16 the stacked Spearman curve becomes dominated by weak or negative correlations over a broader Σ_* range, and some individual galaxies show reduced significance or a delayed zero-crossing; while for Scal-PG16 the stacked Spearman trend become completely negative without no inversion point anymore. These changes underline that the N2S2- and S-calibrations (i.e., D16 and Scal-PG16 here) leveraging [N II] and [S II] can be more susceptible to excitation mixing, and they reinforce the need to treat excitation masking and calibration choice as coupled systematics when interpreting resolved $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relations.

Crucially, despite these quantitative shifts, the qualitative inversion

of the residual $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling persists in the stacked sample for both O3N2-M13 and O3N2-C20. The low- Σ_* anti-correlation branch and the high- Σ_* positive branch remain present, demonstrating that the sign flip is not an artefact of excluding composite spaxels when using O3N2 prescriptions.

5.3 NGC4383 – the Metallicity Outlier

NGC 4383 is a clear metallicity outlier within the MAUVE–MUSE sample. To illustrate its behaviour without allowing it to bias ensemble trends, we reintroduce its spaxels in Figure 6 as dotted contours (rMZR panels) and dotted light-mauve curves (Spearman panels), while keeping the stacked Spearman trend (black curve) computed from the other 13 galaxies only. From the rMZR comparison (first two columns), NGC 4383 remains systematically offset to lower Z_{gas} at fixed Σ_* by at least 0.2 dex under all four metallicity calibrations. The persistence of this depression across prescriptions and excitation selections indicates that it is not produced by calibration-specific systematics or by the choice of star-forming mask, motivating our designation of NGC 4383 as a metallicity outlier.

The Spearman trend of NGC 4383 adds a second, more striking peculiarity. Unlike the other galaxies, NGC 4383 shows a mass-dependent correlation pattern that is nearly the mirror image of the stacked relation: wherever the combined MAUVE–MUSE behaviour is positive (negative), the NGC 4383 curve tends to be negative (positive), especially in O3N2 indicators. In other words, the sign of the residual $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling in NGC 4383 appears to reverse relative to the dominant local regime traced by the rest of the sample. If included in the stack, this anti-aligned behaviour would disproportionately populate the low- Z_{gas} tail and partially cancel the mass-ordered trend, artificially steepening the low- Σ_* rMZR branch and shifting $\rho(\Sigma_*)$ toward the non-correlation case. We therefore exclude NGC 4383 from all stacked rMZR and correlation measurements in the main text, and investigate it separately here.

A plausible physical explanation for both its depressed metallicity scale and its opposite-sign residual coupling is unusually efficient removal of enriched gas. NGC 4383 has been identified as a system with strong feedback activity in the MAUVE–MUSE context, and intense metal-rich outflows could lower Z_{gas} at fixed Σ_* while also perturbing the local balance between enrichment and gas supply that sets the sign of the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation. Alternatively (or in addition), the anomaly could reflect a recent gas-accretion or re-accretion event: Chung et al. (2009) reported kinematic anomalies on the eastern side of the H I disc and suggested recent accretion. In our framework, the inflow of relatively metal-poor gas would naturally dilute Z_{gas} at fixed Σ_* and can drive an opposite-sign residual coupling by enhancing gas-supply variability relative to enrichment. In this sense, NGC 4383 provides a valuable pathological case for testing the limits of the resolved regulator picture and for isolating environment- or feedback-driven departures from the population-average behaviour. A detailed discussion of its ionized-gas properties and environmental context is presented by Watts et al. (2024), and we defer a full physical interpretation to future work.

6 Conclusions

In this paper, after removing the metallicity outlier (NGC 4383), we have used 13 Virgo spiral galaxies from the MAUVE–MUSE internal release to revisit the resolved mass–metallicity relation and its secondary dependence on star formation rate surface density on

~ 100 pc scales. By combining high-quality IFS data with a spatially-resolved regulator framework, we have characterised where and why the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation (and also its residual coupling) changes sign. Our main conclusions are as follows.

(i) Our MAUVE–MUSE galaxies follow a well-defined resolved mass–metallicity relation that is consistent with previous IFS surveys. Using the **O3N2-M13** calibration as our fiducial metallicity scale, the rMZR rises steeply at low Σ_* and flattens toward high Σ_* , in line with earlier work on kpc-scale data. Colour-coding by Σ_{SFR} reveals a clear secondary dependence: at fixed Σ_* , low- Σ_* regions tend to show lower Z_{gas} at higher Σ_{SFR} , whereas high- Σ_* regions show the opposite ordering (see [Figure 3](#)).

(ii) After removing the primary rMZR and rSFMS trends, the residual coupling between metallicity and SFR surface density exhibits a robust mass-dependent inversion. The Spearman rank correlation coefficient between $\Delta[\text{O}/\text{H}]$ and $\Delta \log_{10} \Sigma_{\text{SFR}}$ transitions smoothly from $\rho < 0$ at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \lesssim 7.5$ to $\rho > 0$ at $\log(\Sigma_*/M_\odot \text{ kpc}^{-2}) \gtrsim 8.0$, with a zero-crossing in between (see [Figure 5](#)). Notably, NGC 4192 spans a broad Σ_* range that straddles the inversion point and exhibits a clear internal sign reversal, rather than simply populating one branch or the other. This suggests that the transition is driven by an intrinsic physical mechanism operating within individual galaxies, rather than arising as an artifact of stacking different galaxy populations. Local Moran-like correlation maps further show that, once Σ_* trends are removed, correlated and anti-correlated H II regions can coexist within the same galaxy, and the familiar mass-dependent behaviour emerges only as an ensemble property.

(iii) To interpret these patterns, we formulated a spatially-resolved gas–regulator model in surface densities and identified an “instantaneous target metallicity” $Z^\dagger$ that drives the evolution of the real-time metallicity Z . Analysis around metallicity extrema and toward/away from them leads to a simple condition (see [Equation 31](#)): at fixed Σ_* , the sign of the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ (anti-)correlation is controlled by the relative fluctuation amplitudes of star formation and gas supply. Positive correlations arise where SFR- or SFE-driven variability dominates (short, strongly varying depletion times at nearly steady gas supply), while negative correlations arise where gas-supply variability dominates and dilution drives Z_{gas} opposite to Σ_{SFR} . Our MAUVE–MUSE trends are therefore naturally understood as the outcome of a competition between SFR-driven and GSR-driven fluctuations that varies systematically with Σ_* .

(iv) We argued that the same regulator physics can be extrapolated from spatially-resolved to integrated galaxy-wide relations ([Section 4.4](#)). Replacing surface densities by their global analogues, [Equation 31](#) implies that the sign of the integrated $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ dependence at fixed M_* is likewise set by whether fluctuations in SFR (or SFE) outweigh those in the inflow rate. In this view, the negative SFR dependence of metallicity commonly seen at low stellar masses and the weaker or reversed dependence at the high-mass end in global FMR analyses are two manifestations of the same underlying competition between depletion and inflow timescales. The apparent prevalence of anti-correlation, and the weaker and less clear direct correlation signal on global scales than on local scales, can be understood as a consequence of parameter-space sampling and spatial averaging: many surveys predominantly probe galaxies or spaxels on the low-mass, inflow-dominated side of the inversion and then integrate over regions with mixed local behaviour.

(v) Finally, we tested the robustness of these conclusions against three key systematics using [Figure 6](#). First, adopting four widely used metallicity diagnostics (**D16**, **Scal-PG16**, **O3N2-M13**, and **O3N2-**

C20), we find that the overall rMZR curvature is stable across all calibrations. However, the mass-dependent sign flip in the residual $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling remains tension according to the choice of diagnostic. The O3N2-based indicators (**O3N2–O3N2-M13** and **O3N2–O3N2-C20**) show the clearest and most internally consistent transition from anti-correlation at low Σ_* to correlation at high Σ_* . By contrast, the **Scal-PG16** prescription exhibits a completely opposite trend, while the **D16** calibration remains predominantly positively correlated over the full Σ_* range. We therefore conclude that, although the resolved metallicity scaling relations are qualitatively similar across diagnostics, the presence and sign of the inferred $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ inversion are calibration-dependent. Second, relaxing the star-forming mask from a conservative **HII-only** selection to an inclusive **HII+Comp** mask increases the scatter and modulates the detailed $\rho(\Sigma_*)$ trends, with the strongest changes occurring for the N2S2- and S-based diagnostics (**D16** and **Scal-PG16**). For the O3N2-based calibrations (**O3N2-M13** and **O3N2-C20**), the mass-dependent inversion remains identifiable. In contrast, under the **Scal-PG16** prescription the coupling becomes predominantly negative across the full Σ_* range, while **D16** shows no clear monotonic correlation, with the stacking Spearman trend fluctuating around zero. Third, we showed that NGC 4383 is a genuine metallicity outlier: it lies ~ 0.1 – 0.2 dex below the rMZR locus and exhibits an almost mirror-image residual correlation pattern relative to the stack ([Section 5.3](#)).

In general, these results suggest that the spatially resolved $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation is not governed by a single universal trend, but by a balance between star-formation-driven and gas-supply-driven variability that depends on local mass surface density, environment, and feedback. Our MAUVE–MUSE data show that both correlated and anti-correlated regimes naturally coexist within the same discs at ~ 100 pc resolution, and that the familiar mass-dependent inversion emerges as a population-level imprint of this underlying competition. Future work combining MAUVE–MUSE data with resolved H I and CO observations, as well as higher-redshift IFS surveys will be crucial for constraining Σ_ϕ and its timescale more directly, and for testing whether the inversion scales identified here are universal or themselves evolve with redshift, environment, and galaxy assembly history.

Software

This research made use of `astropy` ([Astropy Collaboration et al. 2022](#), <https://www.astropy.org/>), `numpy` ([Harris et al. 2020](#), <https://numpy.org/>), `matplotlib` ([Hunter 2007](#), <https://matplotlib.org/>), `scipy` ([Virtanen et al. 2020](#), <https://scipy.org/>), `spectlite` ([Kirkby et al. 2023](#), <https://spectlite.readthedocs.io/en/latest/>) and `nGIST` ([Fraser-McKelvie et al. 2025](#), <https://geckos-survey.github.io/gist-documentation/>).

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Data Availability

The full set of MAUVE–MUSE datacubes and associated nGIST value-added products will be released as part of the main public data release once the survey is completed (anticipated in 2027). Requests for access prior to the public release should be directed to the corresponding author. Readers interested in joining the MAUVE collaboration and accessing the full dataset are encouraged to consult the [team](#) page on the MAUVE survey website. The analysis pipeline used in this work is publicly available at https://github.com/Rongjun-ANU/MAUVE_MUSE_Zgas_SigmaSFR.

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